OM-273473L 2024-05

Processes

MIG (GMAW) Welding

Air Carbon Arc (CAC-A) Cutting and Gouging

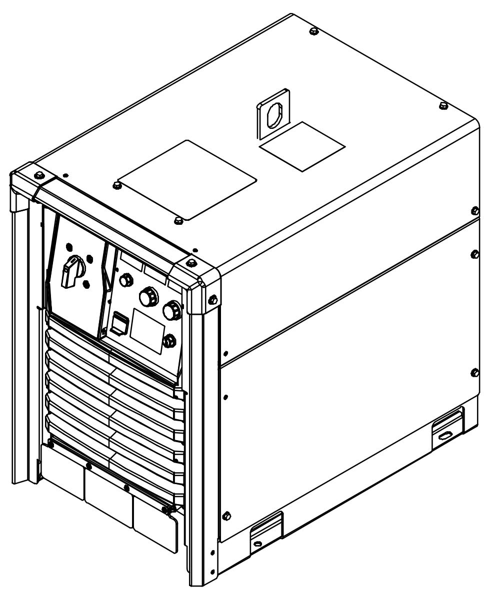
Description

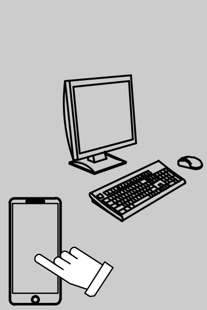
CV DC 3 Phase

Arc Welding Power Source

Auto-Continuum® 350 And 500

w/ Insight Core





For product information, Owner's Manual translations, and more, visit

www.MillerWelds.com

OWNER'S MANUAL

From Miller to You

Thank you and *congratulations* on choosing Miller. Now you can get the job done and get it done right. We know you don't have time to do it any other way.

That's why when Niels Miller first started building arc welders in 1929, he made sure his products offered long-lasting value and superior quality. Like you, his customers couldn't afford anything less. Miller products had to be more than the best they could be. They had to be the best you could buy.

Today, the people that build and sell Miller products continue the tradition. They're just as committed to providing equipment and service that meets the high standards of quality and value established in 1929.

This Owner's Manual is designed to help you get the most out of your Miller products. Please take time to read the Safety Precautions. They will help you protect yourself against potential hazards on the worksite. We've made installation and operation quick and easy. With Miller, you can count on years of reliable service with proper maintenance. And if for some reason the unit needs repair, there's a Troubleshooting section that will help you figure out what the problem is, and our extensive service network is there to help fix the problem. Warranty and maintenance information for your particular model are also provided.

Miller Electric manufactures a full line of welders and welding-related equipment. For information on other quality Miller products, contact your local Miller distributor to receive the latest full line catalog or individual specification sheets. **To locate your nearest distributor or service agency call 1-800-4-A-Miller, or visit us at www.MillerWelds.com on the web.**

Working as hard as you do – every power source from Miller is backed by the most hassle-free warranty in the business.

ISO 9001
Quality

Miller is the first welding equipment manufacturer in the U.S.A. to be registered to the ISO 9001 Quality System Standard.

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SECTION 1 – SAFETY PRECAUTIONS – READ BEFORE USING

Protect yourself and others from injury—read, follow, and save these important safety precautions and operating instructions.

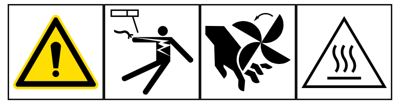
1-1. Symbol Usage

DANGER! – Indicates a hazardous situation which, if not avoided, will result in death or serious injury. The possible hazards are shown in the adjoining symbols or explained in the text.

Indicates a hazardous situation which, if not avoided, could result in death or serious injury. The possible hazards are shown in the adjoining symbols or explained in the text.

***NOTICE** – Indicates statements not related to personal injury.*

☞ *Indicates special instructions.*



This group of symbols means Warning! Watch Out! ELECTRIC SHOCK, MOVING PARTS, and HOT PARTS hazards. Consult symbols and related instructions below for necessary actions to avoid these hazards.

1-2. Arc Welding Hazards

The symbols shown below are used throughout this manual to call attention to and identify possible hazards. When you see the symbol, watch out, and follow the related instructions to avoid the hazard. The safety information given below is only a summary of the more complete safety information found in the Principal Safety Standards. Read and follow all Safety Standards.

Only qualified persons should install, operate, maintain, and repair this equipment. A qualified person is defined as one who, by possession of a recognized degree, certificate, or professional standing, or who by extensive knowledge, training and experience, has successfully demonstrated the ability to solve or resolve problems relating to the subject matter, the work, or the project and has received safety training to recognize and avoid the hazards involved.

During operation, keep everybody, especially children, away.

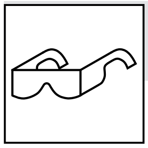
ELECTRIC SHOCK can kill.

Touching live electrical parts can cause fatal shocks or severe burns. The electrode and work circuit is electrically live whenever the output is on. The input power circuit and machine internal circuits are also live when power is on. In semiautomatic or automatic wire welding, the wire, wire reel, drive roll housing, and all metal parts touching the welding wire are electrically live. Incorrectly installed or improperly grounded equipment is a hazard.

- Do not touch live electrical parts.
- Wear dry, hole-free insulating gloves and body protection.
- Insulate yourself from work and ground using dry insulating mats or covers big enough to prevent any physical contact with the work or ground.
- Do not use AC weld output in damp, wet, or confined spaces, or if there is a danger of falling.
- Do not store or use equipment in standing water.
- Use AC output ONLY if required for the welding process.
- If AC output is required, use remote output control if present on unit.
- Additional safety precautions are required when any of the following electrically hazardous conditions are present: in damp locations or while wearing wet clothing; on metal structures such as floors, gratings, or scaffolds; when in cramped positions such as sitting, kneeling, or lying; or when there is a high risk of unavoidable or accidental contact with the workpiece or ground. For these conditions, use the following equipment in order presented: 1) a semiautomatic DC constant voltage (wire) welder, 2) a DC manual (stick) welder, or 3) an AC welder with reduced open-circuit voltage. In most situations, use of a DC, constant voltage wire welder is recommended. And, do not work alone!
- Disconnect input power or stop engine before installing or servicing this equipment. Lockout/tagout input power according to OSHA 29 CFR 1910.147 (see Safety Standards).
- Properly install, ground, and operate this equipment according to its Owner's Manual and national, state, and local codes.
- Always verify the supply ground — check and be sure that input power cord ground wire is properly connected to ground terminal in disconnect box or that cord plug is connected to a properly grounded receptacle outlet.
- When making input connections, attach proper grounding conductor first—double-check connections.
- Keep cords dry, free of oil and grease, and protected from hot metal and sparks.
- Frequently inspect input power cord and ground conductor for damage or bare wiring — replace immediately if damaged — bare wiring can kill.
- Turn off all equipment when not in use.
- Do not use worn, damaged, undersized, or repaired cables.
- Do not drape cables over your body.
- If earth grounding of the workpiece is required, ground it directly with a separate cable.
- Do not touch electrode if you are in contact with the work, ground, or another electrode from a different machine.
- Use only well-maintained equipment. Repair or replace damaged parts at once. Maintain unit according to manual.
- Do not touch electrode holders connected to two welding machines at the same time since double open-circuit voltage will be present.
- Wear a safety harness if working above floor level.
- Keep all panels and covers securely in place.
- Clamp work cable with good metal-to-metal contact to workpiece or worktable as near the weld as practical.
- Insulate work clamp when not connected to workpiece to prevent contact with any metal object.
- Do not connect more than one electrode or work cable to any single weld output terminal. Disconnect cable for process not in use.
- Use GFCI protection when operating auxiliary equipment in damp or wet locations.

HOT PARTS can burn.

- Do not touch hot parts bare handed.
- Allow cooling period before working on equipment.
- To handle hot parts, use proper tools and/or wear heavy, insulated welding gloves and clothing to prevent burns.



FLYING METAL OR DIRT can injure eyes.

- Welding, chipping, wire brushing, and grinding cause sparks and flying metal. As welds cool, they can throw off slag.
- Wear approved safety glasses with side shields even under your welding helmet.

FUMES AND GASES can be hazardous.

Welding produces fumes and gases. Breathing these fumes and gases can be hazardous to your health.

- Keep your head out of the fumes. Do not breathe the fumes.
- Ventilate the work area and/or use local forced ventilation at the arc to remove welding fumes and gases. The recommended way to determine adequate ventilation is to sample for the composition and quantity of fumes and gases to which personnel are exposed.
- If ventilation is poor, wear an approved air-supplied respirator.
- Read and understand the Safety Data Sheets (SDSs) and the manufacturer's instructions for adhesives, coatings, cleaners, consumables, coolants, degreasers, fluxes, and metals.
- Work in a confined space only if it is well ventilated, or while wearing an air-supplied respirator. Always have a trained watchperson nearby. Welding fumes and gases can displace air and lower the oxygen level causing injury or death. Be sure the breathing air is safe.
- Do not weld in locations near degreasing, cleaning, or spraying operations. The heat and rays of the arc can react with vapors to form highly toxic and irritating gases.
- Do not weld on coated metals, such as galvanized, lead, or cadmium plated steel, unless the coating is removed from the weld area, the area is well ventilated, and while wearing an air-supplied respirator. The coatings and any metals containing these elements can give off toxic fumes if welded.

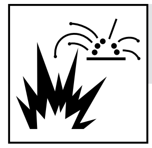
BUILDUP OF GAS can injure or kill.

- Shut off compressed gas supply when not in use.
- Always ventilate confined spaces or use approved air-supplied respirator.

ARC RAYS can burn eyes and skin.

Arc rays from the welding process produce intense visible and invisible (ultraviolet and infrared) rays that can burn eyes and skin. Sparks fly off from the weld.

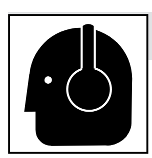
- Wear an approved welding helmet fitted with a proper shade of filter lenses to protect your face and eyes from arc rays and sparks when welding or watching (see ANSI Z49.1 and Z87.1 listed in Safety Standards).
- Wear approved safety glasses with side shields under your helmet.
- Use protective screens or barriers to protect others from flash, glare, and sparks; warn others not to watch the arc.
- Wear body protection made from leather or flame-resistant clothing (FRC). Body protection includes oil-free clothing such as leather gloves, heavy shirt, cuffless trousers, high shoes, and a cap.



WELDING can cause fire or explosion.

Welding on closed containers, such as tanks, drums, or pipes, can cause them to blow up. Sparks can fly off from the welding arc. The flying sparks, hot workpiece, and hot equipment can cause fires and burns. Accidental contact of electrode to metal objects can cause sparks, explosion, overheating, or fire. Check and be sure the area is safe before doing any welding.

- Remove all flammables within 35 ft (10.7 m) of the welding arc. If this is not possible, tightly cover them with approved covers.
- Do not weld where flying sparks can strike flammable material.
- Protect yourself and others from flying sparks and hot metal.
- Be alert that welding sparks and hot materials from welding can easily go through small cracks and openings to adjacent areas.
- Watch for fire, and keep a fire extinguisher nearby.
- Be aware that welding on a ceiling, floor, bulkhead, or partition can cause fire on the hidden side.
- Do not cut or weld on tire rims or wheels. Tires can explode if heated. Repaired rims and wheels can fail. See OSHA 29 CFR 1910.177 listed in Safety Standards.
- Do not weld on containers that have held combustibles, or on closed containers such as tanks, drums, or pipes unless they are properly prepared according to AWS F4.1 (see Safety Standards).
- Do not weld where the atmosphere can contain flammable dust, gas, or liquid vapors (such as gasoline).
- Connect work cable to the work as close to the welding area as practical to prevent welding current from traveling long, possibly unknown paths and causing electric shock, sparks, and fire hazards.
- Do not use welder to thaw frozen pipes.
- Remove stick electrode from holder or cut off welding wire at contact tip when not in use.
- Wear body protection made from leather or flame-resistant clothing (FRC). Body protection includes oil-free clothing such as leather gloves, heavy shirt, cuffless trousers, high shoes, and a cap.
- Remove any combustibles, such as a butane lighter or matches, from your person before doing any welding.
- After completion of work, inspect area to ensure it is free of sparks, glowing embers, and flames.
- Use only correct fuses or circuit breakers. Do not oversize or bypass them.
- Follow requirements in OSHA 1910.252 (a) (2) (iv) and NFPA 51B for hot work and have a fire watcher and extinguisher nearby.
- Read and understand the Safety Data Sheets (SDSs) and the manufacturer's instructions for adhesives, coatings, cleaners, consumables, coolants, degreasers, fluxes, and metals.



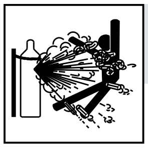
NOISE can damage hearing.

Noise from some processes or equipment can damage hearing.

- Wear approved ear protection if noise level is high.

ELECTRIC AND MAGNETIC FIELDS (EMF) can affect Implanted Medical Devices.

- Wearers of Pacemakers and other Implanted Medical Devices should keep away.
- Implanted Medical Device wearers should consult their doctor and the device manufacturer before going near arc welding, spot welding, gouging, plasma arc cutting, or induction heating operations.

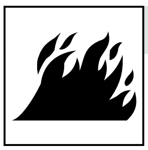


CYLINDERS can explode if damaged.

Compressed gas cylinders contain gas under high pressure. If damaged, a cylinder can explode. Since gas cylinders are normally part of the welding process, be sure to treat them carefully.

- Protect compressed gas cylinders from excessive heat, mechanical shocks, physical damage, slag, open flames, sparks, and arcs.
- Install cylinders in an upright position by securing to a stationary support or cylinder rack to prevent falling or tipping.
- Keep cylinders away from any welding or other electrical circuits.
- Never drape a welding torch over a gas cylinder.
- Never allow a welding electrode to touch any cylinder.
- Never weld on a pressurized cylinder—explosion will result.
- Use only correct compressed gas cylinders, regulators, hoses, and fittings designed for the specific application; maintain them and associated parts in good condition.
- Turn face away from valve outlet when opening cylinder valve. Do not stand in front of or behind the regulator when opening the valve.
- Keep protective cap in place over valve except when cylinder is in use or connected for use.
- Use the proper equipment, correct procedures, and sufficient number of persons to lift, move, and transport cylinders.
- Read and follow instructions on compressed gas cylinders, associated equipment, and Compressed Gas Association (CGA) publication P-1 listed in Safety Standards.

1-3. Additional Hazards For Installation, Operation, And Maintenance

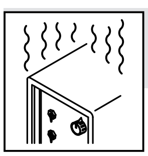


FIRE OR EXPLOSION hazard.

- Do not install or place unit on, over, or near combustible surfaces.
- Do not install unit near flammables.
- Do not overload building wiring — be sure power supply system is properly sized, rated, and protected to handle this unit.

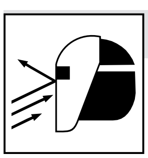
FALLING EQUIPMENT can injure.

- Use lifting eye to lift unit only, NOT running gear, gas cylinders, or any other accessories.
- Use correct procedures and equipment of adequate capacity to lift and support unit.
- If using lift forks to move unit, be sure forks are long enough to extend beyond opposite side of unit.
- Keep equipment (cables and cords) away from moving vehicles when working from an aerial location.
- Follow the guidelines in the Applications Manual for the Revised NIOSH Lifting Equation (Publication No. 94-110) when manually lifting heavy parts or equipment.



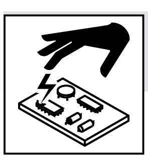
OVERUSE can cause OVERHEATING.

- Allow cooling period; follow rated duty cycle.
- Reduce current or reduce duty cycle before starting to weld again.
- Do not block or filter airflow to unit.



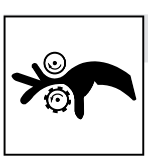
FLYING SPARKS can injure.

- Wear a face shield to protect eyes and face.
- Shape tungsten electrode only on grinder with proper guards in a safe location wearing proper face, hand, and body protection.
- Sparks can cause fires—keep flammables away.



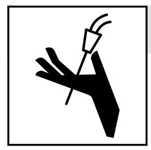
STATIC (ESD) can damage PC boards.

- Put on grounded wrist strap BEFORE handling boards or parts.
- Use proper static-proof bags and boxes to store, move, or ship PC boards.



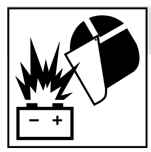
MOVING PARTS can injure.

- Keep away from moving parts.
- Keep away from pinch points such as drive rolls.



WELDING WIRE can injure.

- Do not press gun trigger until instructed to do so.
- Do not point gun toward any part of the body, other people, or any metal when threading welding wire.

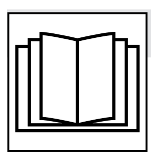


BATTERY EXPLOSION can injure.

- Do not use welder to charge batteries or jump start vehicles unless it has a battery charging feature designed for this purpose.

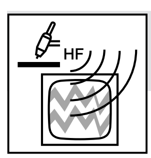
MOVING PARTS can injure.

- Keep away from moving parts such as fans.
- Keep all doors, panels, covers, and guards closed and securely in place.
- Have only qualified persons remove doors, panels, covers, or guards for maintenance and troubleshooting as necessary.
- Reinstall doors, panels, covers, or guards when maintenance is finished and before reconnecting input power.



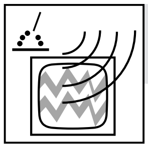
READ INSTRUCTIONS.

- Read and follow all labels and the Owner's Manual carefully before installing, operating, or servicing unit. Read the safety information at the beginning of the manual and in each section.
- Use only genuine replacement parts from the manufacturer.
- Perform installation, maintenance, and service according to the Owner's Manuals, industry standards, and national, state, and local codes.



H.F. RADIATION can cause interference.

- High-frequency (H.F.) can interfere with radio navigation, safety services, computers, and communications equipment.
- Have only qualified persons familiar with electronic equipment perform this installation.
- The user is responsible for having a qualified electrician promptly correct any interference problem resulting from the installation.
- If notified by the FCC about interference, stop using the equipment at once.
- Have the installation regularly checked and maintained.
- Keep high-frequency source doors and panels tightly shut, keep spark gaps at correct setting, and use grounding and shielding to minimize the possibility of interference.



ARC WELDING can cause interference.

- Electromagnetic energy can interfere with sensitive electronic equipment such as microprocessors, computers, and computer-driven equipment such as robots.
- Be sure all equipment in the welding area is electromagnetically compatible.
- To reduce possible interference, keep weld cables as short as possible, close together, and down low, such as on the floor.
- Locate welding operation 100 meters from any sensitive electronic equipment.
- Be sure this welding machine is installed and grounded according to this manual.
- If interference still occurs, the user must take extra measures such as moving the welding machine, using shielded cables, using line filters, or shielding the work area.

1-4. California Proposition 65 Warnings

⚠ **WARNING – This product can expose you to chemicals including lead, which are known to the state of California to cause cancer and birth defects or other reproductive harm.**

For more information, go to www.P65Warnings.ca.gov.

1-5. Principal Safety Standards

Safety in Welding, Cutting, and Allied Processes, American Welding Society standard ANSI Standard Z49.1. Website: www.aws.org.

Safe Practice For Occupational And Educational Eye And Face Protection, ANSI Standard Z87.1, from American National Standards Institute. Website: safetyequipment.org.

Safe Practices for the Preparation of Containers and Piping for Welding and Cutting, American Welding Society Standard AWS F4.1. Website: www.aws.org.

National Electrical Code, NFPA Standard 70 from National Fire Protection Association. Website: www.nfpa.org.

Safe Handling of Compressed Gases in Cylinders, CGA Pamphlet P-1 from Compressed Gas Association. Website: www.cganet.com.

Safety in Welding, Cutting, and Allied Processes, CSA Standard W117.2 from Canadian Standards Association. Website: www.csagroup.org.

Standard for Fire Prevention During Welding, Cutting, and Other Hot Work, NFPA Standard 51B from National Fire Protection Association. Website: www.nfpa.org.

OSHA, Occupational Safety and Health Standards for General Industry, Title 29, Code of Federal Regulations (CFR), Part 1910.177 Subpart N, Part 1910 Subpart Q, and Part 1926, Subpart J. Website: www.osha.gov.

OSHA *Important Note Regarding the ACGIH TLV, Policy Statement on the Uses of TLVs and BEIs.* Website: www.osha.gov.

Applications Manual for the Revised NIOSH Lifting Equation from the National Institute for Occupational Safety and Health (NIOSH). Website: www.cdc.gov/NIOSH.

SOM 2024–01

1-6. EMF Information

Electric current flowing through any conductor causes localized electric and magnetic fields (EMF). The current from arc welding (and allied processes including spot welding, gouging, plasma arc cutting, and induction heating operations) creates an EMF field around the welding circuit. EMF fields can interfere with some medical implants, e.g. pacemakers. Protective measures for persons wearing medical implants have to be taken. For example, restrict access for passers−by or conduct individual risk assessment for welders. All welders should use the following procedures in order to minimize exposure to EMF fields from the welding circuit:

1. Keep cables close together by twisting or taping them, or using a cable cover.
2. Do not place your body between welding cables. Arrange cables to one side and away from the operator.
3. Do not coil or drape cables around your body.
4. Keep head and trunk as far away from the equipment in the welding circuit as possible.
5. Connect work clamp to workpiece as close to the weld as possible.
6. Do not work next to, sit or lean on the welding power source.
7. Do not weld whilst carrying the welding power source or wire feeder.

About Implanted Medical Devices:

Implanted Medical Device wearers should consult their doctor and the device manufacturer before performing or going near arc welding, spot welding, gouging, plasma arc cutting, or induction heating operations. If cleared by your doctor, then following the above procedures is recommended.

SECTION 2 – CONSIGNES DE SÉCURITÉ - LIRE AVANT UTILISATION

Pour écarter les risques de blessure pour vous-même et pour autrui — lire, appliquer et ranger en lieu sûr ces consignes relatives aux précautions de sécurité et au mode opératoire.

2-1. Symboles utilisés

DANGER! – Indique une situation dangereuse qui si on l'évite pas peut donner la mort ou des blessures graves. Les dangers possibles sont montrés par les symboles joints ou sont expliqués dans le texte.

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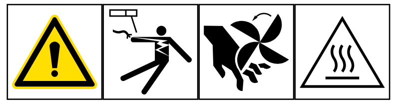
AVIS – *Indique des déclarations pas en relation avec des blessures personnelles.*

☞ *Indique des instructions spécifiques.*



Ce groupe de symboles veut dire Avertissement! Attention! DANGER DE CHOC ELECTRIQUE, PIECES EN MOUVEMENT, et PIECES CHAUDES. Reportez-vous aux symboles et aux directives ci-dessous afin de connaître les mesures à prendre pour éviter tout danger.

2-2. Dangers relatifs au soudage à l'arc

Les symboles représentés ci-dessous sont utilisés dans ce manuel pour attirer l'attention et identifier les dangers possibles. En présence de ce symbole, prendre garde et suivre les instructions afférentes pour éviter tout risque. Les consignes de sécurité présentées ci-après ne font que résumer l'information contenue dans les Normes de sécurité principales. Lire et suivre toutes les Normes de sécurité.

L'installation, l'utilisation, l'entretien et les réparations ne doivent être confiés qu'à des personnes qualifiées. Une personne qualifiée est définie comme celle qui, par la possession d'un diplôme reconnu, d'un certificat ou d'un statut professionnel, ou qui, par une connaissance, une formation et une expérience approfondies, a démontré avec succès sa capacité à résoudre les problèmes liés à la tâche, le travail ou le projet et a reçu une formation en sécurité afin de reconnaître et d'éviter les risques inhérents.

Au cours de l'utilisation, tenir toute personne à l'écart et plus particulièrement les enfants.

UNE DÉCHARGE ÉLECTRIQUE peut entraîner la mort.

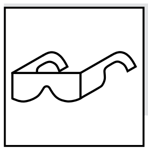
Le contact d'organes électriques sous tension peut provoquer des accidents mortels ou des brûlures graves. Le circuit de l'électrode et de la pièce est sous tension lorsque le courant est délivré à la sortie. Le circuit d'alimentation et les circuits internes de la machine sont également sous tension lorsque l'alimentation est sur Marche. Dans le mode de soudage avec du fil, le fil, le dérouleur, le bloc de commande du rouleau et toutes les parties métalliques en contact avec le fil sont sous tension électrique. Un équipement installé ou mis à la terre de manière incorrecte ou impropre constitue un danger.

- Ne pas toucher aux pièces électriques sous tension.
- Porter des gants isolants et des vêtements de protection secs et sans trous.
- S'isoler de la pièce à couper et du sol en utilisant des housses ou des tapis assez grands afin d'éviter tout contact physique avec la pièce à couper ou le sol.
- Ne pas utiliser de sortie de soudage CA dans des zones humides ou confinées ou s'il y a un risque de chute.
- Ne stockez pas et n'utilisez pas l'équipement dans de l'eau stagnante.
- Se servir d'une source électrique à courant électrique UNIQUEMENT si le procédé de soudage le demande.
- Si l'utilisation d'une source électrique à courant électrique s'avère nécessaire, se servir de la fonction de télécommande si l'appareil en est équipé.
- D'autres consignes de sécurité sont nécessaires dans les conditions suivantes : risques électriques dans un environnement humide ou si l'on porte des vêtements mouillés ; sur des structures métalliques telles que sols, grilles ou échafaudages ; en position coincée comme assise, à genoux ou couchée ; ou s'il y a un risque élevé de contact inévitable ou accidentel avec la pièce à souder ou le sol. Dans ces conditions, utiliser les équipements suivants, dans l'ordre indiqué : 1) un poste à souder DC à tension constante (à fil), 2) un poste à souder DC manuel (électrode) ou 3) un poste à souder AC à tension à vide réduite. Dans la plupart des situations, l'utilisation d'un poste à souder DC à fil à tension constante est recommandée. En outre, ne pas travailler seul !
- Couper l'alimentation ou arrêter le moteur avant de procéder à l'installation, à la réparation ou à l'entretien de l'appareil. Déverrouiller l'alimentation selon la norme OSHA 29 CFR 1910.147 (voir normes de sécurité).
- Brancher correctement la mise à la terre et utiliser cet appareil conformément à son manuel d'utilisateur et aux codes nationaux, provinciaux et municipaux.
- Toujours vérifier la mise à la terre — vérifier et assurez-vous que le conducteur de mise à la terre du cordon d'alimentation est bien raccordé à la borne de mise à la terre dans le boîtier de déconnexion ou que la fiche du cordon est raccordée à une prise correctement mise à la terre.
- En effectuant les raccordements d'entrée, fixer d'abord le conducteur de mise à la terre approprié et contre-vérifier les connexions.
- Les câbles doivent être exempts d'humidité, d'huile et de graisse; protégez-les contre les étincelles et les pièces métalliques chaudes.
- Vérifier fréquemment le cordon d'alimentation et le conducteur de mise à la terre afin de s'assurer qu'il n'est pas altéré ou dénudé -, le remplacer immédiatement s'il l'est -. Un fil dénudé peut entraîner la mort.
- L'équipement doit être hors tension lorsqu'il n'est pas utilisé.
- Ne pas utiliser des câbles usés, endommagés, de grosseur insuffisante ou mal épissés.
- Ne pas enrouler les câbles autour du corps.
- Si la pièce soudée doit être mise à la terre, le faire directement avec un câble distinct.
- Ne pas toucher l'électrode quand on est en contact avec la pièce, la terre ou une électrode provenant d'une autre machine.
- Ne pas toucher des porte électrodes connectés à deux machines en même temps à cause de la présence d'une tension à vide doublée.
- N'utiliser qu'un matériel en bon état. Réparer ou remplacer sur-le-champ les pièces endommagées. Entretenir l'appareil conformément à ce manuel.

- Porter un harnais de sécurité si l’on doit travailler au-dessus du sol.
- S’assurer que tous les panneaux et couvercles sont correctement en place.
- Fixer le câble de retour de façon à obtenir un bon contact métal-métal avec la pièce à souder ou la table de travail, le plus près possible de la soudure.
- Isoler la pince de masse quand pas mis à la pièce pour éviter le contact avec tout objet métallique.
- Ne pas raccorder plus d’une électrode ou plus d’un câble de masse à une même borne de sortie de soudage. Débrancher le câble pour le procédé non utilisé.
- Utiliser une protection différentielle lors de l’utilisation d’un équipement auxiliaire dans des endroits humides ou mouillés.

LES PIÈCES CHAUDES peuvent provoquer des brûlures.

- Ne pas toucher des parties chaudes à mains nues.
- Prévoir une période de refroidissement avant de travailler à l’équipement.
- Ne pas toucher aux pièces chaudes, utiliser les outils recommandés et porter des gants de soudage et des vêtements épais pour éviter les brûlures.



DES PIECES DE METAL ou DES SALETES peuvent provoquer des blessures dans les yeux.

- Le soudage, l’écaillement, le passage de la pièce à la brosse en fil de fer, et le meulage génèrent des étincelles et des particules métalliques volantes. Pendant la période de refroidissement des soudures, elles risquent de projeter du laitier.
- Porter des lunettes de sécurité avec écrans latéraux ou un écran facial.

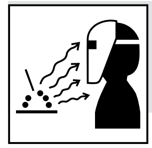
LES FUMÉES ET LES GAZ peuvent être dangereux.

Le soudage génère des fumées et des gaz. Leur inhalation peut être dangereux pour votre santé.

- Eloigner votre tête des fumées. Ne pas respirer les fumées.
- À l’intérieur, ventiler la zone et/ou utiliser une ventilation forcée au niveau de l’arc pour l’évacuation des fumées et des gaz de soudage. Pour déterminer la bonne ventilation, il est recommandé de procéder à un prélèvement pour la composition et la quantité de fumées et de gaz auxquelles est exposé le personnel.
- Si la ventilation est médiocre, porter un respirateur anti-vapeurs approuvé.
- Lire et comprendre les fiches de données de sécurité et les instructions du fabricant concernant les adhésifs, les revêtements, les nettoyants, les consommables, les produits de refroidissement, les dégraisseurs, les flux et les métaux.
- Travailler dans un espace fermé seulement s’il est bien ventilé ou en portant un respirateur à alimentation d’air. Demander toujours à un surveillant dûment formé de se tenir à proximité. Des fumées et des gaz de soudage peuvent déplacer l’air et abaisser le niveau d’oxygène provoquant des blessures ou des accidents mortels. S’assurer que l’air de respiration ne présente aucun danger.
- Ne pas souder dans des endroits situés à proximité d’opérations de dégraissage, de nettoyage ou de pulvérisation. La chaleur et les rayons de l’arc peuvent réagir en présence de vapeurs et former des gaz hautement toxiques et irritants.
- Ne pas souder des métaux munis d’un revêtement, tels que l’acier galvanisé, plaqué en plomb ou au cadmium à moins que le revêtement n’ait été enlevé dans la zone de soudure, que l’endroit soit bien ventilé, et en portant un respirateur à alimentation d’air. Les revêtements et tous les métaux renfermant ces éléments peuvent dégager des fumées toxiques en cas de soudage.

LES ACCUMULATIONS DE GAZ risquent de provoquer des blessures ou même la mort.

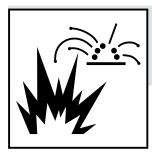
- Fermer l’alimentation du gaz comprimé en cas de non utilisation.
- Veiller toujours à bien aérer les espaces confinés ou se servir d’un respirateur d’adduction d’air homologué.



LES RAYONS DE L’ARC peuvent provoquer des brûlures dans les yeux et sur la peau.

Le rayonnement de l’arc du procédé de soudage génère des rayons visibles et invisibles intenses (ultraviolets et infrarouges) susceptibles de provoquer des brûlures dans les yeux et sur la peau. Des étincelles sont projetées pendant le soudage.

- Porter un casque de soudage approuvé muni de verres filtrants approprié pour protéger visage et yeux pendant le soudage (voir ANSI Z49.1 et Z87.1 énuméré dans les normes de sécurité).
- Porter des lunettes de sécurité avec écrans latéraux même sous votre casque.
- Avoir recours à des écrans protecteurs ou à des rideaux pour protéger les autres contre les rayonnements les éblouissements et les étincelles ; prévenir toute personne sur les lieux de ne pas regarder l’arc.
- Porter une protection corporelle en cuir ou des vêtements ignifuges (FRC). La protection du corps comporte des vêtements sans huile, comme des gants de cuir, une chemise solide, des pantalons sans revers, des chaussures hautes et une casquette.

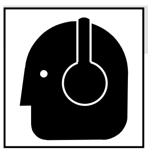


LE SOUDAGE peut provoquer un incendie ou une explosion.

Le soudage effectué sur des conteneurs fermés tels que des réservoirs, tambours ou des conduites peut provoquer leur éclatement. Des étincelles peuvent être projetées de l’arc de soudure. La projection d’étincelles, des pièces chaudes et des équipements chauds peut provoquer des incendies et des brûlures. Le contact accidentel de l’électrode avec des objets métalliques peut provoquer des étincelles, une explosion, un surchauffement ou un incendie. Avant de commencer le soudage, vérifier et s’assurer que l’endroit ne présente pas de danger.

- Déplacer toutes les substances inflammables à une distance de 10,7 m de l’arc de soudage. En cas d’impossibilité les recouvrir soigneusement avec des protections homologués.
- Ne pas souder dans un endroit là où des étincelles peuvent tomber sur des substances inflammables.
- Se protéger et d’autres personnes de la projection d’étincelles et de métal chaud.
- Des étincelles et des matériaux chauds du soudage peuvent facilement passer dans d’autres zones en traversant de petites fissures et des ouvertures.
- Surveiller tout déclenchement d’incendie et tenir un extincteur à proximité.
- Le soudage effectué sur un plafond, plancher, paroi ou séparation peut déclencher un incendie de l’autre côté.
- Ne pas couper ou souder des jantes ou des roues. Les pneus peuvent exploser s’ils sont chauffés. Les jantes et les roues réparées peuvent défaillir. Voir OSHA 29 CFR 1910.177 énuméré dans les normes de sécurité.
- Ne pas effectuer le soudage sur des conteneurs fermés tels que des réservoirs, tambours, ou conduites, à moins qu’ils n’aient été préparés correctement conformément à AWS F4.1 (voir les Normes de Sécurité).
- Ne pas souder là où l’air ambiant pourrait contenir des poussières, gaz ou émanations inflammables (vapeur d’essence, par exemple).

- Brancher le câble de masse sur la pièce le plus près possible de la zone de soudage pour éviter le transport du courant sur une longue distance par des chemins inconnus éventuels en provoquant des risques d'électrocution, d'étincelles et d'incendie.
- Ne pas utiliser le poste de soudage pour dégeler des conduites gelées.
- En cas de non utilisation, enlever la baguette d'électrode du porte-électrode ou couper le fil à la pointe de contact.
- Porter une protection corporelle en cuir ou des vêtements ignifuges (FRC). La protection du corps comporte des vêtements sans huile, comme des gants de cuir, une chemise solide, des pantalons sans revers, des chaussures hautes et une casquette.
- Avant de souder, retirer toute substance combustible de vos poches telles qu'un allumeur au butane ou des allumettes.
- Une fois le travail achevé, assurez-vous qu'il ne reste aucune trace d'étincelles incandescentes ni de flammes.
- Utiliser exclusivement des fusibles ou coupe-circuits appropriés. Ne pas augmenter leur puissance; ne pas les ponter.
- Suivre les recommandations dans OSHA 1910.252 (a) (2) (iv) et NFPA 51B pour les travaux à chaud et avoir de la surveillance et un extincteur à proximité.
- Lire et comprendre les fiches de données de sécurité et les instructions du fabricant concernant les adhésifs, les revêtements, les nettoyants, les consommables, les produits de refroidissement, les dégraisseurs, les flux et les métaux.



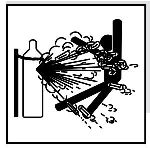
Le BRUIT peut endommager l'ouïe.

Le bruit des processus et des équipements peut affecter l'ouïe.

- Porter des protections approuvées pour les oreilles si le niveau sonore est trop élevé.

Les CHAMPS ÉLECTROMAGNÉTIQUES (CEM) peuvent affecter les implants médicaux.

- Les porteurs de stimulateurs cardiaques et autres implants médicaux doivent rester à distance.
- Les porteurs d'implants médicaux doivent consulter leur médecin et le fabricant du dispositif avant de s'approcher de la zone où se déroule du soudage à l'arc, du soudage par points, du gougeage, de la découpe plasma ou une opération de chauffage par induction.

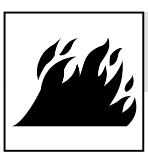


Si des BOUTEILLES sont endommagées, elles pourront exploser.

Des bouteilles de gaz comprimé protecteur contiennent du gaz sous haute pression. Si une bouteille est endommagée, elle peut exploser. Du fait que les bouteilles de gaz font normalement partie du procédé de soudage, les manipuler avec précaution.

- Protéger les bouteilles de gaz comprimé d'une chaleur excessive, des chocs mécaniques, des dommages physiques, du laitier, des flammes ouvertes, des étincelles et des arcs.
- Placer les bouteilles debout en les fixant dans un support stationnaire ou dans un porte-bouteilles pour les empêcher de tomber ou de se renverser.
- Tenir les bouteilles éloignées des circuits de soudage ou autres circuits électriques.
- Ne jamais placer une torche de soudage sur une bouteille à gaz.
- Une électrode de soudage ne doit jamais entrer en contact avec une bouteille.
- Ne jamais souder une bouteille pressurisée - risque d'explosion.
- Utiliser seulement des bouteilles de gaz comprimé, régulateurs, tuyaux et raccords convenables pour cette application spécifique; les maintenir ainsi que les éléments associés en bon état.
- Tourner le dos à la sortie de vanne lors de l'ouverture de la vanne de la bouteille. Ne pas se tenir devant ou derrière lerégulateur lors de l'ouverture de la vanne.
- Maintenir le chapeau de protection sur la soupape, sauf en cas d'utilisation ou de branchement de la bouteille.
- Utilisez les équipements corrects, les bonnes procédures et suffisamment de personnes pour soulever, déplacer et transporter les bouteilles.
- Lire et suivre les instructions sur les bouteilles de gaz comprimé, l'équipement connexe et le dépliant P-1 de la CGA (Compressed Gas Association) mentionné dans les principales normes de sécurité.

2-3. Symboles de dangers supplémentaires en relation avec l'installation, le fonctionnement et la maintenance

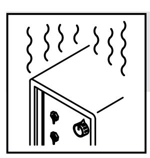


Risque D'INCENDIE OU D'EXPLOSION.

- Ne pas placer l'appareil sur, au-dessus ou à proximité de surfaces inflammables.
- Ne pas installer l'appareil à proximité de produits inflammables
- Ne pas surcharger l'installation électrique – s'assurer que l'alimentation est correctement dimensionné et protégé avant de mettre l'appareil en service.

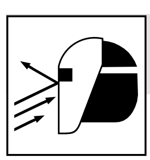
LA CHUTE DE L'ÉQUIPEMENT peut provoquer des blessures.

- Utiliser l'anneau de levage uniquement pour soulever l'appareil, NON PAS les organes de roulement, les bouteilles de gaz ou tout autre accessoire.
- Utilisez les procédures correctes et des équipements d'une capacité appropriée pour soulever et supporter l'appareil.
- En utilisant des fourches de levage pour déplacer l'unité, s'assurer que les fourches sont suffisamment longues pour dépasser du côté opposé de l'appareil.
- Tenir l'équipement (câbles et cordons) à distance des véhicules mobiles lors de toute opération en hauteur.
- Suivre les consignes du Manuel des applications pour l'équation de levage NIOSH révisée (Publication Nº94– 110) lors du levage manuelle de pièces ou équipements lourds.



L'EMPLOI EXCESSIF peut SURCHAUFFER L'ÉQUIPEMENT.

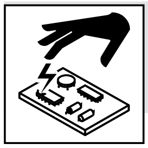
- Laisser l'équipement refroidir ; respecter le facteur de marche nominal.
- Réduire le courant ou le cycle opératoire avant de recommancer le soudage.
- Ne pas obstruer les passages d'air du poste.



LES ÉTINCELLES PROJETÉES peuvent provoquer des blessures.

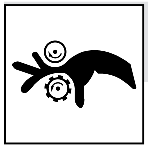
- Porter un écran facial pour protéger le visage et les yeux.

- Affûter l'électrode au tungstène uniquement à la meuleuse dotée de protecteurs. Cette manœuvre est à exécuter dans un endroit sûr lorsque l'on porte l'équipement homologué de protection du visage, des mains et du corps.
- Les étincelles risquent de causer un incendie - éloigner toute substance inflammable.



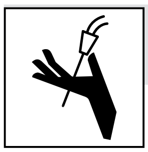
LES CHARGES ÉLECTROSTATIQUES peuvent endommager les circuits imprimés.

- Établir la connexion avec la barrette de terre AVANT de manipuler des cartes ou des pièces.
- Utiliser des pochettes et des boîtes antistatiques pour stocker, déplacer ou expédier des cartes de circuits imprimes.



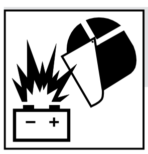
Les PIÈCES MOBILES peuvent causer des blessures.

- Ne pas s'approcher des organes mobiles.
- Ne pas s'approcher des points de coincement tels que des rouleaux de commande.



LES FILS DE SOUDAGE peuvent provoquer des blessures.

- Ne pas appuyer sur la gachette avant d'en avoir reçu l'instruction.
- Ne pas diriger le pistolet vers soi, d'autres personnes ou toute pièce mécanique en engageant le fil de soudage.

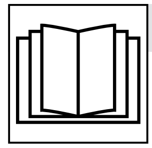


L'EXPLOSION DE LA BATTERIE peut provoquer des blessures.

- Ne pas utiliser l'appareil de soudage pour charger des batteries ou faire démarrer des véhicules à l'aide de câbles de démarrage, sauf si l'appareil dispose d'une fonctionnalité de charge de batterie destinée à cet usage.

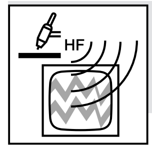
Les PIÈCES MOBILES peuvent causer des blessures.

- S'abstenir de toucher des organes mobiles tels que des ventilateurs.
- Maintenir fermés et verrouillés les portes, panneaux, recouvrements et dispositifs de protection.
- Lorsque cela est nécessaire pour des travaux d'entretien et de dépannage, faire retirer les portes, panneaux, recouvrements ou dispositifs de protection uniquement par du personnel qualifié.
- Remettre les portes, panneaux, recouvrements ou dispositifs de protection quand l'entretien est terminé et avant de rebrancher l'alimentation électrique.



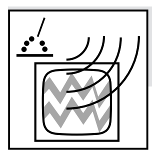
LIRE LES INSTRUCTIONS.

- Lire et appliquer les instructions sur les étiquettes et le Mode d'emploi avant l'installation, l'utilisation ou l'entretien de l'appareil. Lire les informations de sécurité au début du manuel et dans chaque section.
- N'utiliser que des pièces de remplacement provenant du fabricant.
- Effectuer l'installation, l'entretien et toute intervention selon les manuels d'utilisateurs, les normes nationales, provinciales et de l'industrie, ainsi que les codes municipaux.



LE RAYONNEMENT HAUTE FRÉQUENCE (H.F.) risque de provoquer des interférences.

- Le rayonnement haute fréquence (H. F.) peut provoquer des interférences avec les équipements de radio-navigation et de communication, les services de sécurité et les ordinateurs.
- Demander seulement à des personnes qualifiées familiarisées avec des équipements électroniques de faire fonctionner l'installation.
- L'utilisateur est tenu de faire corriger rapidement par un électricien qualifié les interférences résultant de l'installation.
- Si le FCC signale des interférences, arrêter immédiatement l'appareil.
- Effectuer régulièrement le contrôle et l'entretien de l'installation.
- Maintenir soigneusement fermés les portes et les panneaux des sources de haute fréquence, maintenir les éclateurs à une distance correcte et utiliser une terre et un blindage pour réduire les interférences éventuelles.



LE SOUDAGE À L'ARC risque de provoquer des interférences.

- L'énergie électromagnétique risque de provoquer des interférences pour l'équipement électronique sensible tel que les ordinateurs et l'équipement commandé par ordinateur tel que les robots.
- Veiller à ce que tout l'équipement de la zone de soudage soit compatible électromagnétiquement.
- Pour réduire la possibilité d'interférence, maintenir les câbles de soudage aussi courts que possible, les grouper, et les poser aussi bas que possible (ex. par terre).
- Veiller à souder à une distance de 100 mètres de tout équipement électronique sensible.
- Veiller à ce que ce poste de soudage soit posé et mis à la terre conformément à ce mode d'emploi.
- En cas d'interférences après avoir pris les mesures précédentes, il incombe à l'utilisateur de prendre des mesures supplémentaires telles que le déplacement du poste, l'utilisation de câbles blindés, l'utilisation de filtres de ligne ou la pose de protecteurs dans la zone de travail.

2-4. Proposition californienne 65 Avertissements

⚠ **AVERTISSEMENT – Ce produit peut vous exposer à des produits chimiques tels que le plomb, reconnus par l'État de Californie comme cancérigènes et sources de malformations ou d'autres troubles de la reproduction.**

Pour plus d'informations, consulter www.P65Warnings.ca.gov.

2-5. Principales normes de sécurité

Safety in Welding, Cutting, and Allied Processes, American Welding Society standard ANSI Standard Z49.1. Website: www.aws.org.

Safe Practice For Occupational And Educational Eye And Face Protection, ANSI Standard Z87.1, from American National Standards Institute. Website: safetyequipment.org.

Safe Practices for the Preparation of Containers and Piping for Welding and Cutting, American Welding Society Standard AWS F4.1. Website: www.aws.org.

National Electrical Code, NFPA Standard 70 from National Fire Protection Association. Website: www.nfpa.org.

Safe Handling of Compressed Gases in Cylinders, CGA Pamphlet P-1 from Compressed Gas Association. Website: www.cganet.com.

Safety in Welding, Cutting, and Allied Processes, CSA Standard W117.2 from Canadian Standards Association. Website: www. csa-group.org.

Standard for Fire Prevention During Welding, Cutting, and Other Hot Work, NFPA Standard 51B from National Fire Protection Association. Website: www.nfpa.org.

OSHA, Occupational Safety and Health Standards for General Industry, Title 29, Code of Federal Regulations (CFR), Part 1910.177 Subpart N, Part 1910 Subpart Q, and Part 1926, Subpart J. Website: www.osha.gov.

OSHA *Important Note Regarding the ACGIH TLV, Policy Statement on the Uses of TLVs and BEIs.* Website: www.osha.gov.

Applications Manual for the Revised NIOSH Lifting Equation from the National Institute for Occupational Safety and Health (NIOSH). Website: www.cdc.gov/NIOSH.

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2-6. Informations relatives aux CEM

Le courant électrique qui traverse tout conducteur génère des champs électromagnétiques (CEM) à certains endroits. Le courant issu d'un soudage à l'arc (et de procédés connexes, y compris le soudage par points, le gougeage, le découpage plasma et les opérations de chauffage par induction) crée un champ électromagnétique (CEM) autour du circuit de soudage. Les champs électromagnétiques produits peuvent causer interférence à certains implants médicaux, p. ex. les stimulateurs cardiaques. Des mesures de protection pour les porteurs d'implants médicaux doivent être prises: par exemple, des restrictions d'accès pour les passants ou une évaluation individuelle des risques pour les soudeurs. Tous les soudeurs doivent appliquer les procédures suivantes pour minimiser l'exposition aux CEM provenant du circuit de soudage:

1. Rassembler les câbles en les torsadant ou en les attachant avec du ruban adhésif ou avec une housse.
2. Ne pas se tenir au milieu des câbles de soudage. Disposer les câbles d'un côté et à distance de l'opérateur.
3. Ne pas courber et ne pas entourer les câbles autour de votre corps.
4. Maintenir la tête et le torse aussi loin que possible du matériel du circuit de soudage.
5. Connecter la pince sur la pièce aussi près que possible de la soudure.
6. Ne pas travailler à proximité d'une source de soudage, ni s'asseoir ou se pencher dessus.
7. Ne pas souder tout en portant la source de soudage ou le dévidoir.

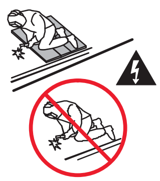
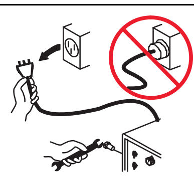
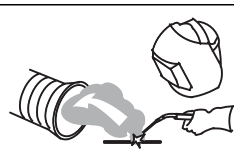
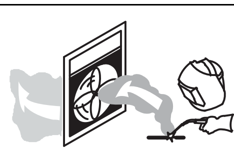
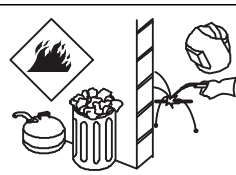
En ce qui concerne les implants médicaux :

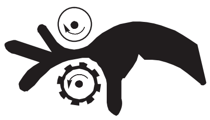
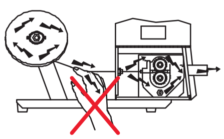
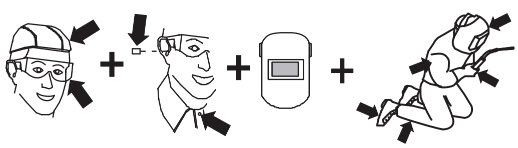
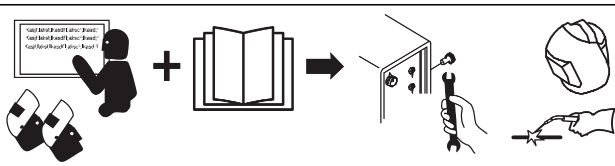
Les porteurs d'implants doivent d'abord consulter leur médecin avant de s'approcher des opérations de soudage à l'arc, de soudage par points, de gougeage, du coupage plasma ou de chauffage par induction. Si le médecin approuve, il est recommandé de suivre les procédures précédentes.

SECTION 3 – DEFINITIONS

3-1. Additional Safety Symbol Definitions

☞ *Some symbols are found only on CE products.*

	Warning! Watch Out! There are possible hazards as shown by the symbols.
	Do not discard product (where applicable) with general waste. Reuse or recycle Waste Electrical and Electronic Equipment (WEEE) by disposing at a designated collection facility. Contact your local recycling office or your local distributor for further information.
	Wear dry insulating gloves. Do not touch electrode with bare hand. Do not wear wet or damaged gloves.
	Protect yourself from electric shock by insulating yourself from work and ground.
	Disconnect input plug or power before working on machine.
	Keep your head out of the fumes.
	Use forced ventilation or local exhaust to remove the fumes.
	Use ventilating fan to remove fumes.
	Keep flammables away from welding. Do not weld near flammables.

	Welding sparks can cause fires. Have a fire extinguisher nearby, and have a watchperson ready to use it.
	Do not weld on drums or any closed containers.
	Do not remove or paint over (cover) the label.
	Drive rolls can injure fingers.
	Welding wire and drive parts are at welding voltage during operation - keep hands and metal objects away.
	Pinch points can injure.
	Wear hat and safety glasses. Use ear protection and button shirt collar. Use welding helmet with correct shade of filter. Wear complete body protection made from leather or flame-resistant clothing (FRC). Body protection includes oil-free clothing such as leather gloves, heavy shirt, cuffless trousers, high shoes, and a cap.
	Become trained and read the instructions before working on the machine or welding.

3-2. Miscellaneous Symbols And Definitions

A	Amperage
V	Voltage
U_0	Rated No-Load Voltage
	Direct Current (DC)
~	Alternating Current
U_1	Rated Supply Voltage
U_2	Conventional Load Voltage
I_2	Rated Welding Current
V	Input Power Or Input Voltage
	Gas Input
IP	Input Protection Rating
	Purge By Gas
∅ MM/IN	Wire Diameter
	Remote

3~	Three Phase
+	Positive
−	Negative
X	Duty Cycle
	Line Connection
S	Suitable for Welding in an Environment with Increased Risk of Electric Shock
	Increase
Hz	Hertz
	Three Phase Static Frequency Converter-Transformer-Rectifier
	Wire Type
	Gas Metal Arc Welding (GMAW)
	Gas Metal Arc Welding (GMAW) MIG/Gun Control
	Self Shielded Flux Cored Arc Welding (FCAW)
	Wire Feed Spool Gun

t2	Gas Postflow
t1	Gas Preflow
	Cold Jog (Inch) Toward Workpiecel
	Pulse
%	Percent
	Variable Inductance
	Read Instructions
	Constant Voltage
	Wire Feed
V	Voltage Output
	Trigger Hold Off
	Off

SECTION 4 – SPECIFICATIONS

4-1. Serial Number And Rating Label Location

The serial number and rating information for this product is located on the back. Use rating label to determine input power requirements and/or rated output. For future reference, write serial number in space provided on back cover of this manual.

4-2. Software Licensing Agreement

The End User License Agreement and any third-party notices and terms and conditions pertaining to third-party software can be found at https://www.millerwelds.com/eula and are incorporated by reference herein.

4-3. Information About Default Weld Parameters And Settings

***NOTICE –** Each welding application is unique. Although certain Miller Electric products are designed to determine and default to certain typical welding parameters and settings based upon specific and relatively limited application variables input by the end user, such default settings are for reference purposes only; and final weld results can be affected by other variables and application-specific circumstances. The appropriateness of all parameters and settings should be evaluated and modified by the end user as necessary based upon application-specific requirements. The end user is solely responsible for selection and coordination of appropriate equipment, adoption or adjustment of default weld parameters and settings, and ultimate quality and durability of all resultant welds. Miller Electric expressly disclaims any and all implied warranties including any implied warranty of fitness for a particular purpose.*

4-4. 350 Model Specifications

☞ *Do not use information in unit specifications table to determine electrical service requirements. See Sections 5-10 thru 5-11 for information on connecting input power.*

☞ *This equipment will deliver rated output at an ambient air temperature up to 104°F (40°C).*

Input Power	Rated Output	Voltage Range in CV Mode	Amperage Range In CC Mode	Max Open-Circuit Voltage	RMS Amps Input at Rated Load Output, 50/60 Hz 3-Phase at NEMA Load Voltages and Class I Rating					KVA	KW
					230 V	380 V	400 V	460 V	575 V		
3-Phase	350 A at 34 VDC, 100% Duty Cycle*	10-44 V	20-400 A	75 VDC	36.7 (0–1A**)	21.8 (0–1A**)	20.8 (0–1A**)	18.8 (0–1A**)	14.6 (0–1A**)	14.4 (0.8**)	13.8 (0.17-**)
*See Section 4-9 for Duty Cycle Rating.											
** While idling; Input amperage fluctuates while idling and is always less than one ampere. Use one ampere for power efficiency calculations.											

4-5. 500 Model Specifications

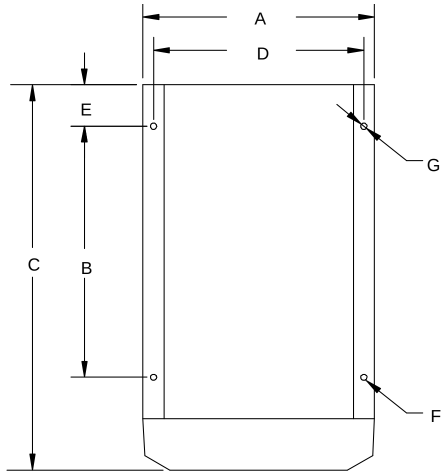
☞ *Do not use information in unit specifications table to determine electrical service requirements. See Sections 5-10 thru 5-11 for information on connecting input power.*

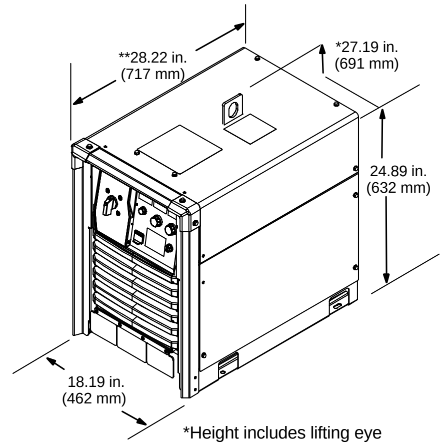
☞ *This equipment will deliver rated output at an ambient air temperature up to 104°F (40°C).*

Input Power	Rated Output	Voltage Range in CV Mode	Amperage Range In CC Mode	Max Open-Circuit Voltage	RMS Amps Input at Rated Load Output, 50/60 Hz 3-Phase at NEMA Load Voltages and Class I Rating					KVA	KW
					230 V	380 V	400 V	460 V	575 V		
3-Phase	500 A at 40 VDC, 100% Duty Cycle*	10-44 V	20-600 A	75 VDC	58.7 (0–1A**)	34.9 (0–1A**)	33.2 (0–1A**)	28.9 (0–1A**)	23.3 (0–1A**)	23.1 (0.8**)	21.9 (0.17-**)
*See Section 4-10 for Duty Cycle Rating.											
** While idling; Input amperage fluctuates while idling and is always less than one ampere. Use one ampere for power efficiency calculations.											

4-6. Dimensions And Weight

Hole Layout Dimensions	
A	17.52 in. (445 mm)
B	17.33 in. (441 mm)
C	26.17 in. (665 mm)
D	16.09 in. (409 mm)
E	2.28 in. (58 mm)
F	.47 in. (12 mm)
G	.47 in. x 1 in. (12 mm x 25 mm)
Weight	
350 Model: 130 lb (59.4 kg) **350 Model w/Aux Power:** 139 lb (63.5 kg) **500 Model:** 151 lb (69 kg) **500 Model w/Aux Power:** 160 lb (73.1 kg)	
Lifting Eye Weight Rating: 1000 lb (453 KG) Maximum	





*Height includes lifting eye

**Length includes E−Stop cover

278352-C

4-7. Environmental Specifications

A. IP Rating

IP Rating
IP23S This equipment is designed for outdoor use. It may be stored, but is not intended to be used for welding outside during precipitation unless sheltered.

B. Temperature Specifications

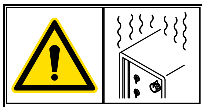
Operating Temperature Range*	Storage/Transportation Temperature Range
14 to 104°F (-10 to 40°C)	-4 to 131°F (-20 to 55°C)

*Output is derated at temperatures above 104°F (40°C).

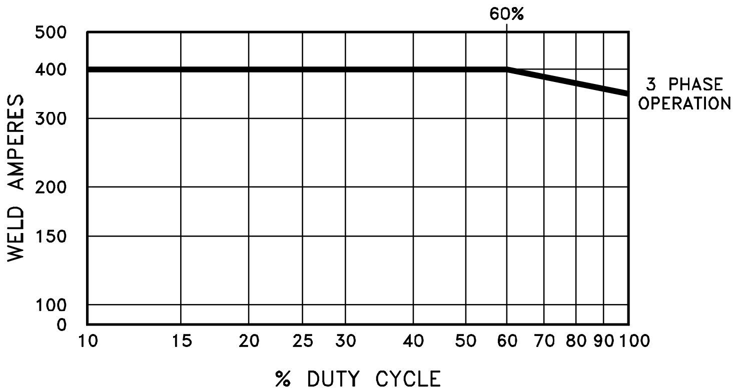
4-8. Static Characteristics

The static (output) characteristics of the welding power source can be described as *flat* during the GMAW process and *drooping* during the SMAW and GTAW processes. Static characteristics are also affected by control settings (including software), electrode, shielding gas, weldment material, and other factors. Contact the factory for specific information on the static characteristics of the welding power source.

4-9. Duty Cycle And Overheating For 350 Model



350 Model

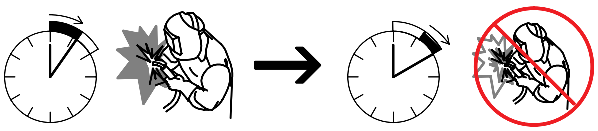


Duty Cycle is percentage of 10 minutes that unit can weld at rated load without overheating.

If unit overheats, output stops. Wait fifteen minutes for unit to cool. Reduce amperage or duty cycle before starting to weld again.

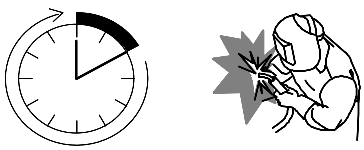
***NOTICE** – Exceeding duty cycle can damage unit and void warranty.*

60% Duty Cycle At 400A



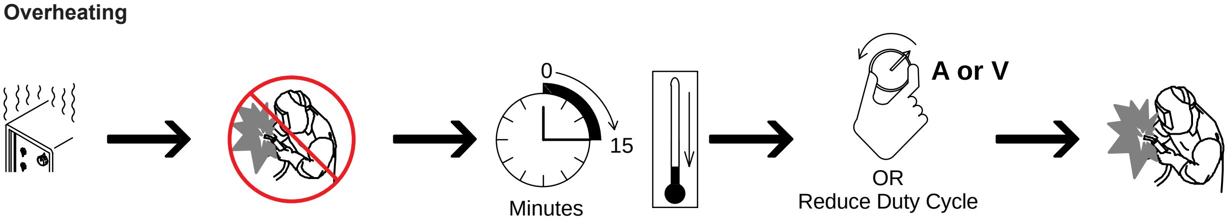
6 Minutes Welding 4 Minutes Resting

100% Duty Cycle At 350A

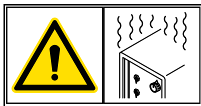


Continuous Welding

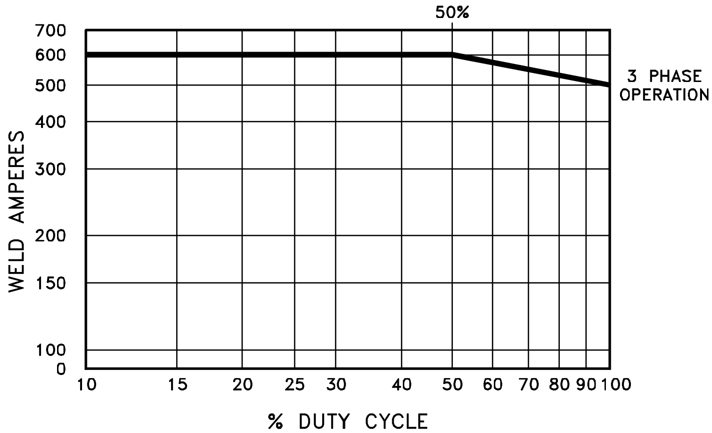
Overheating



4-10. Duty Cycle And Overheating For 500 Model



500 Model

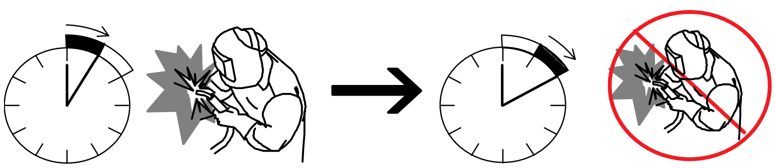


Duty Cycle is percentage of 10 minutes that unit can weld at rated load without overheating.

If unit overheats, output stops. Wait fifteen minutes for unit to cool. Reduce amperage or duty cycle before starting to weld again.

***NOTICE** – Exceeding duty cycle can damage unit and void warranty.*

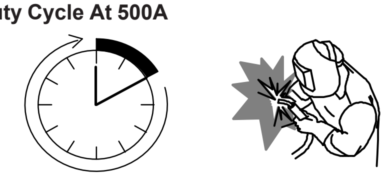
50% Duty Cycle At 600A



5 Minutes Welding

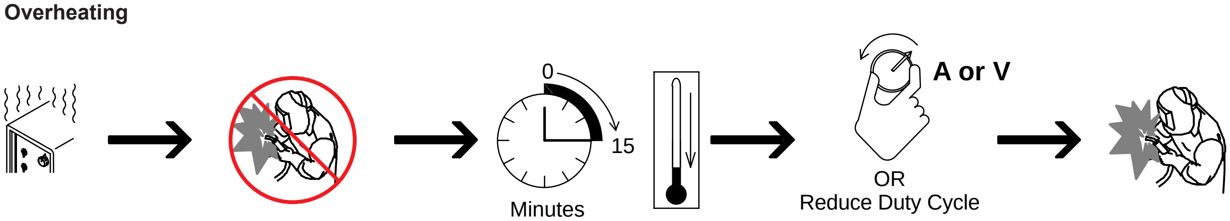
5 Minutes Resting

100% Duty Cycle At 500A



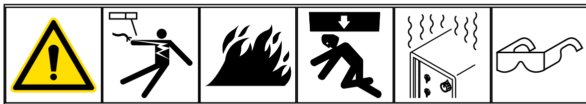
Continuous Welding

Overheating

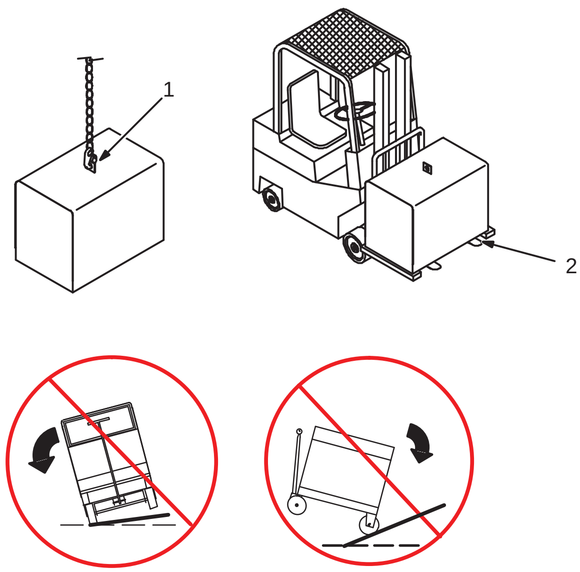


SECTION 5 – INSTALLATION

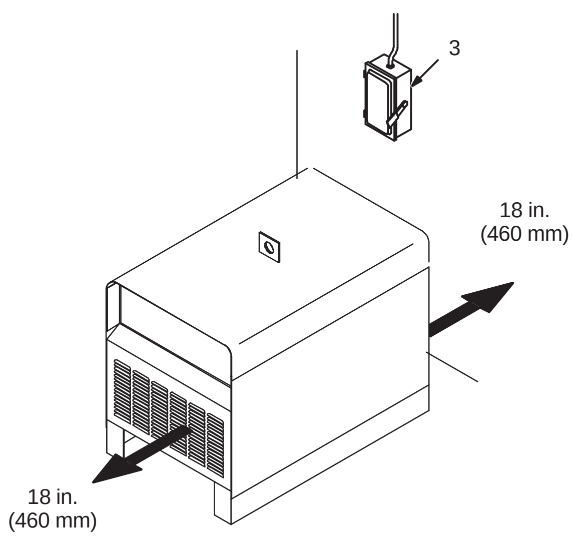
5-1. Selecting A Location



Movement



Location And Airflow



⚠ **Do not move or operate unit where it could tip.**

⚠ **Special installation may be required where gasoline or volatile liquids are present - see NEC Article 511 or CEC Section 20.**

1 Lifting Eye

2 Lifting Forks

Use lifting eye or lifting forks to move unit.

If using lifting forks, extend forks beyond opposite side of unit.

3 Line Disconnect Device

Locate unit near correct input power supply.

5-2. Selecting Cable Sizes*

***NOTICE –** The Total Cable Length in Weld Circuit (see table below) is the combined length of both weld cables. For example, if the power source is 100 ft (30 m) from the workpiece, the total cable length in the weld circuit is 200 ft (2 cables x 100 ft). Use the 200 ft (60 m) column to determine cable size.*

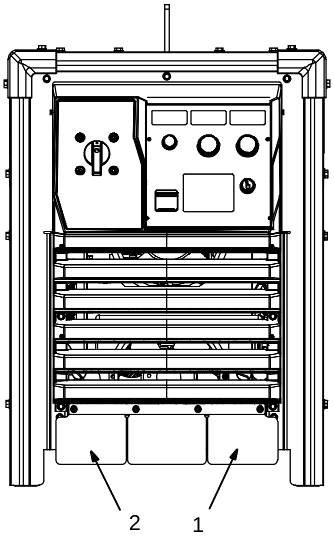
	Weld Cable Size and Total Cable (Copper) Length in Weld Circuit Not Exceeding*****							
	100 ft (30 m) or Less		**150 ft (45 m)**	**200 ft (60 m)**	**250 ft (70 m)**	**300 ft (90 m)**	**350 ft (105 m)**	**400 ft (120 m)**
Welding Amperes	**10 - 60% Duty Cycle AWG (mm²)**	**60 - 100% Duty Cycle AWG (mm²)**	**10 - 100% Duty Cycle AWG (mm²)**					
100	4 (20)	4 (20)	4 (20)	3 (30)	2 (35)	1 (50)	1/0 (60)	1/0 (60)
150	3 (30)	3 (30)	2 (35)	1 (50)	1/0 (60)	2/0 (70)	3/0 (95)	3/0 (95)
200	3 (30)	2 (35)	1 (50)	1/0 (60)	2/0 (70)	3/0 (95)	4/0 (120)	4/0 (120)
250	2 (35)	1 (50)	1/0 (60)	2/0 (70)	3/0 (95)	4/0 (120)	2x2/0 (2x70)	2x2/0 (2x70)
300	1 (50)	1/0 (60)	2/0 (70)	3/0 (95)	4/0 (120)	2x2/0 (2x70)	2x3/0 (2x95)	2x3/0 (2x95)
350	1/0 (60)	2/0 (70)	3/0 (95)	4/0 (120)	2x2/0 (2x70)	2x3/0 (2x95)	2x3/0 (2x95)	2x4/0 (2x120)
400	1/0 (60)	2/0 (70)	3/0 (95)	4/0 (120)	2x2/0 (2x70)	2x3/0 (2x95)	2x4/0 (2x120)	2x4/0 (2x120)
500	2/0 (70)	3/0 (95)	4/0 (120)	2x2/0 (2x70)	2x3/0 (2x95)	2x4/0 (2x120)	3x3/0 (3x95)	3x3/0 (3x95)
600	3/0 (95)	4/0 (120)	2x2/0 (2x70)	2x3/0 (2x95)	2x4/0 (2x120)	3x3/0 (3x95)	3x4/0 (3x120)	3x4/0 (3x120)

* This chart is a general guideline and may not suit all applications. If cable overheats, use next size larger cable.

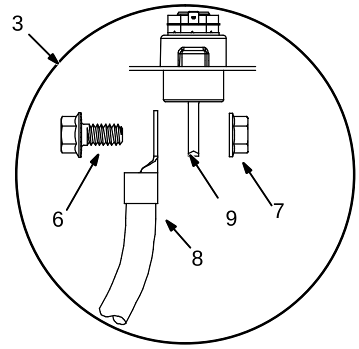
**Weld cable size (AWG) is based on either a 4 volts or less drop or a current density of at least 300 circular mils per ampere.
( ) = mm² for metric use.

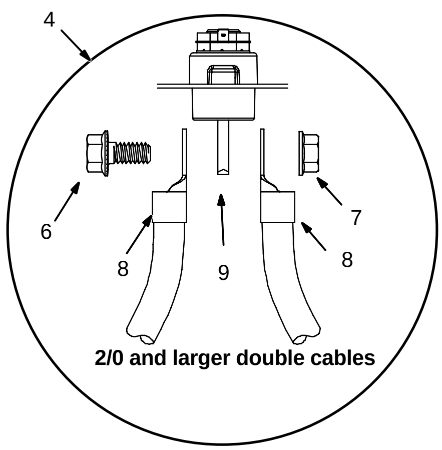
***For distances longer than those shown in this guide, see AWS Fact Sheet No. 39, Welding Cables, available from the American Welding Society at http://www.aws.org.

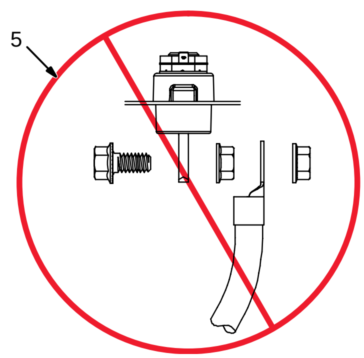
5-3. Connecting Weld Output Cables



Front View







⚠ **Turn off power before connecting to weld output terminals.**

⚠ **Do not use worn, damaged, undersized, or repaired cables.**

Ensure all connections are tight.

☞ *Do not place anything between weld cable terminal and output terminal. Make sure that the surfaces of the weld cable terminal and output terminal are clean.*

1 Negative (-) Terminal, located under stud cover

2 Positive (+) Terminal, located under stud cover

3 Correct Weld Cable Connection for single feeder

4 Correct Weld Cable Connection for 2/0 and larger double cables

5 Incorrect Weld Cable Connection

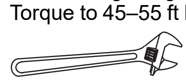
6 Weld Output Terminal Bolt

7 Nut

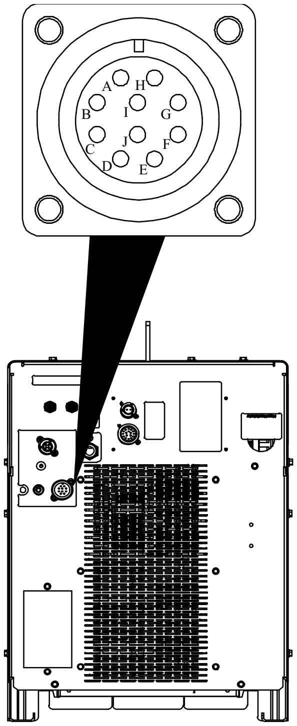
8 Weld Cable Terminal

9 Output Terminal

Remove supplied nut and bolt from weld output terminal. Insert bolt through hole in weld cable terminal and hole in weld output terminal. Screw nut onto bolt until weld cable terminal is tight against the output terminal. Torque to 45–55 ft lb (61–75 N m)



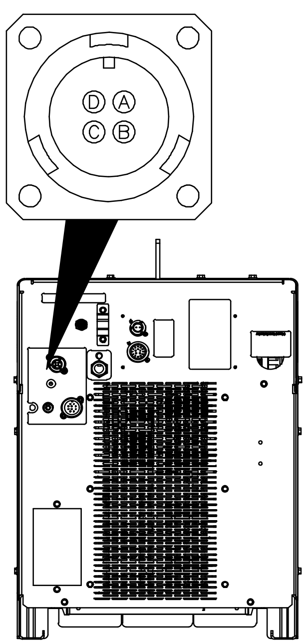
5-4. 10–Pin Receptacle Information



Ref. 278352-C

Socket	Socket Information
A	+50 Volts DC Common
B	+50 Volts DC Common
C	Voltage Sense Positive, Originates from Feeder Wire Drive.
D	+50 Volts DC Power
E	+50 Volts DC Power
F	ENET Rx —
G	ENET Tx —
H	Drain
I	ENET Tx +
J	ENET Rx +

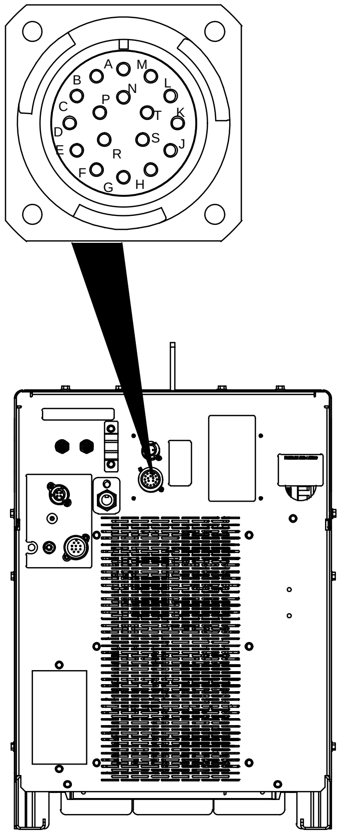
5-5. Volt Sense Receptacle Information



Ref. 278352C

Socket	Socket Information
A	Not Used
B	Volt Sense Negative
C	Not Used
D	Not Used

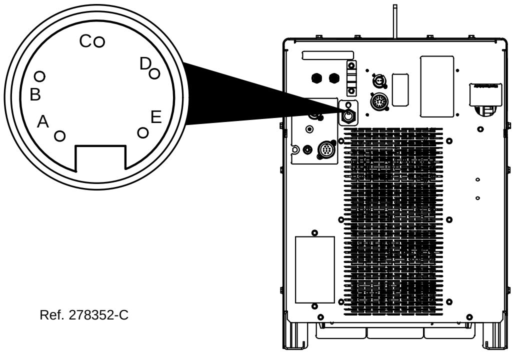
5-6. Peripheral Receptacle RC–22 Functions



278352-C

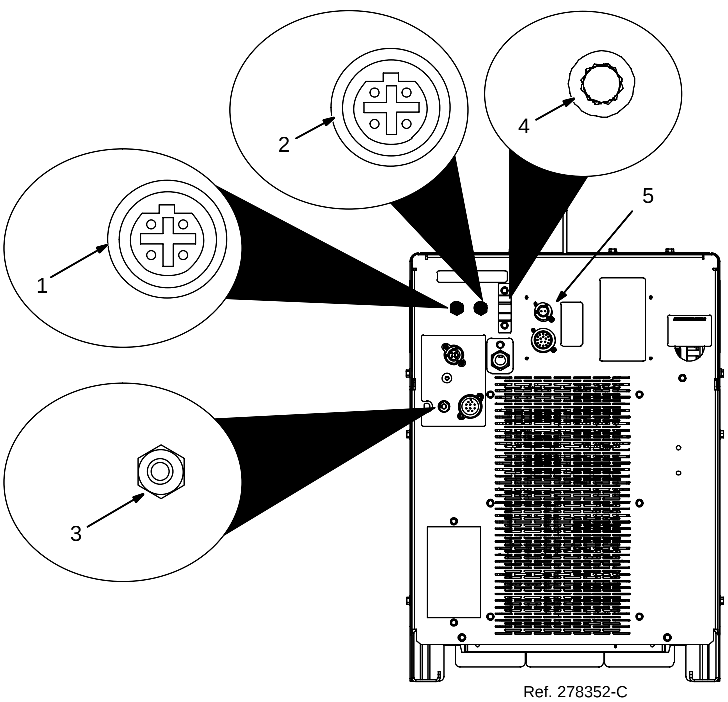
Socket		Ref. Com	Socket Information
A			Output Common
B	Closure when weld contactor on	A	Digital Output 1 (DO1)
C	Not Defined		Digital Output 2 (DO2)
D	Closure to signal when touched	A	Digital Output 3 (DO3)
E			Not Used
F	Circuit Common for AIM Board		Chassis Ground
G	Input Common		Input Common
H	Turn on Touch Sense with +24 VDC	G	Request to turn on Touch Sense (DI1)
J	24 VDC Flow Switch Input	G	Low flow input, 24VDC Input 24VDC = coolant flowing 0V = No Flow, Flow switch is open. (Externally Sourced)
K	Not Defined		Digital Input 3 (DI3)
L			Not Used
M			Not Used
N	+24 VDC output goes high when touched	F	Touched hardware signal 14 ohm impedance
P			Not Used
R			Not Used
S			Not Used
T			Not Used

5-7. Devicenet Receptacle Information



Socket	Socket Information
A	Chassis Ground
B	+24 volts DC; Available Current is 1 Ampere
C	+24 volts DC DC Common
D	CAN H
E	CAN L

5-8. Supplementary Protector CB1, Communication Panel, And E-Stop



1 Ethernet Receptacle

Used for connecting a computer directly to the power source to access configuration web pages.

2 Ethernet Receptacle

Used for connecting a robot directly to the power source.

3 Supplementary Protector CB1

CB1 protects the wire feed motor from overload. If CB1 opens, the wire feeder does not work.

☞ *Press button to reset breaker. If breaker continues to open, contact a Factory Authorized Service Agent.*

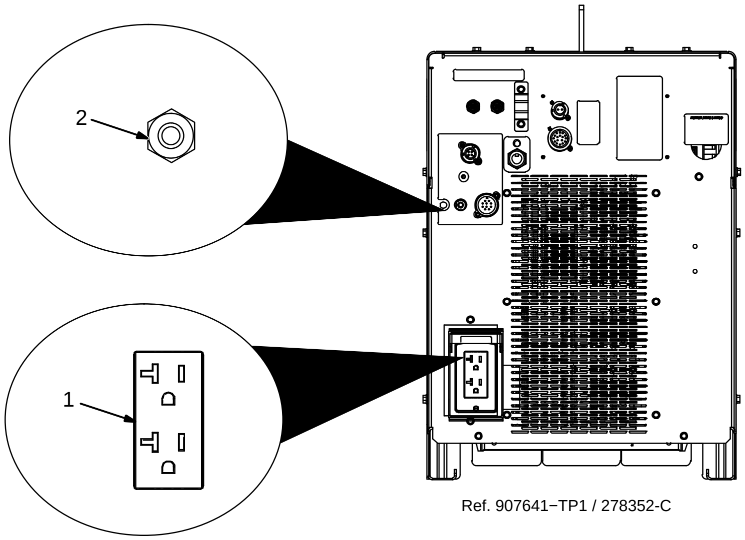
4 Wireless (WiFi) Antenna

Antenna for connecting to the Internet via (WiFi) wireless connection if selected during configuration.

5 E-Stop Receptacle RC24

A short across the two sockets allows unit to weld.

5-9. Supplementary Protector CB2 And Aux. Power (Optional)



Ref. 907641−TP1 / 278352-C

1 115 VAC Receptacle

2 Supplementary Protector CB2

CB2 protects the 115 VAC receptacle from overload. If CB2 opens, the aux power does not work.

Press button to reset breaker. If breaker continues to open, contact a Factory Authorized Service Agent.

5-10. Electrical Service Guide

Failure to follow these electrical service guide recommendations could create an electric shock or fire hazard. These recommendations are for an individual branch circuit sized for the rated output and duty cycle of one welding power source. In individual branch circuit installations, the National Electrical Code (NEC) allows the receptacle or conductor rating to be less than the rating of the circuit protection device. All components of the circuit must be physically compatible. See NEC articles 210.21, 630.11, and 630.12.

NOTICE – *INCORRECT INPUT POWER can damage this welding power source. This welding power source requires a CONTINUOUS supply of input power at rated frequency ±10%) and voltage ±10%). Phase to ground voltage shall not exceed +10% of rated input voltage. Do not use a generator with automatic idle device (that idles engine when no load is sensed) to supply input power to this welding power source.*

NOTICE – *Actual input voltage should not be 10% less than minimum and/or 10% more than maximum input voltages listed in table. If actual input voltage is outside this range, output may not be available.*

350 Amp Model	**60 Hz 3-Phase**				
Rated Supply Voltage (V)	230	380	400	460	575
Rated Maximum Supply Current I_{1max} (A)	43.7	26.2	24.6	21.4	17.1
Rated Effective Supply Current I_{1eff} (A)	36.7	21.8	28.8	18.8	14.6
Maximum Recommended Standard Fuse Rating In Amperes[1]					
Time Delay Fuses[2]	50	30	30	25	20
Normal Operating Fuses[3]	60	35	35	30	25
Maximum Recommended Supply Conductor Length In Feet (Meters)[4]	119 (36)	215 (66)	241 (74)	193 (59)	196 (60)
Raceway Installation					
Minimum Supply Conductor Size In AWG (mm^2)[5]	8 (10)	10 (6)	10 (6)	12 (4)	14 (2.5)
Minimum Grounding Conductor Size In AWG (mm^2) [5]	8 (10)	10 (6)	10 (6)	12 (4)	14 (2.5)

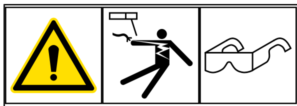
Reference: 2023 National Electrical Code (NEC) (including article 630)

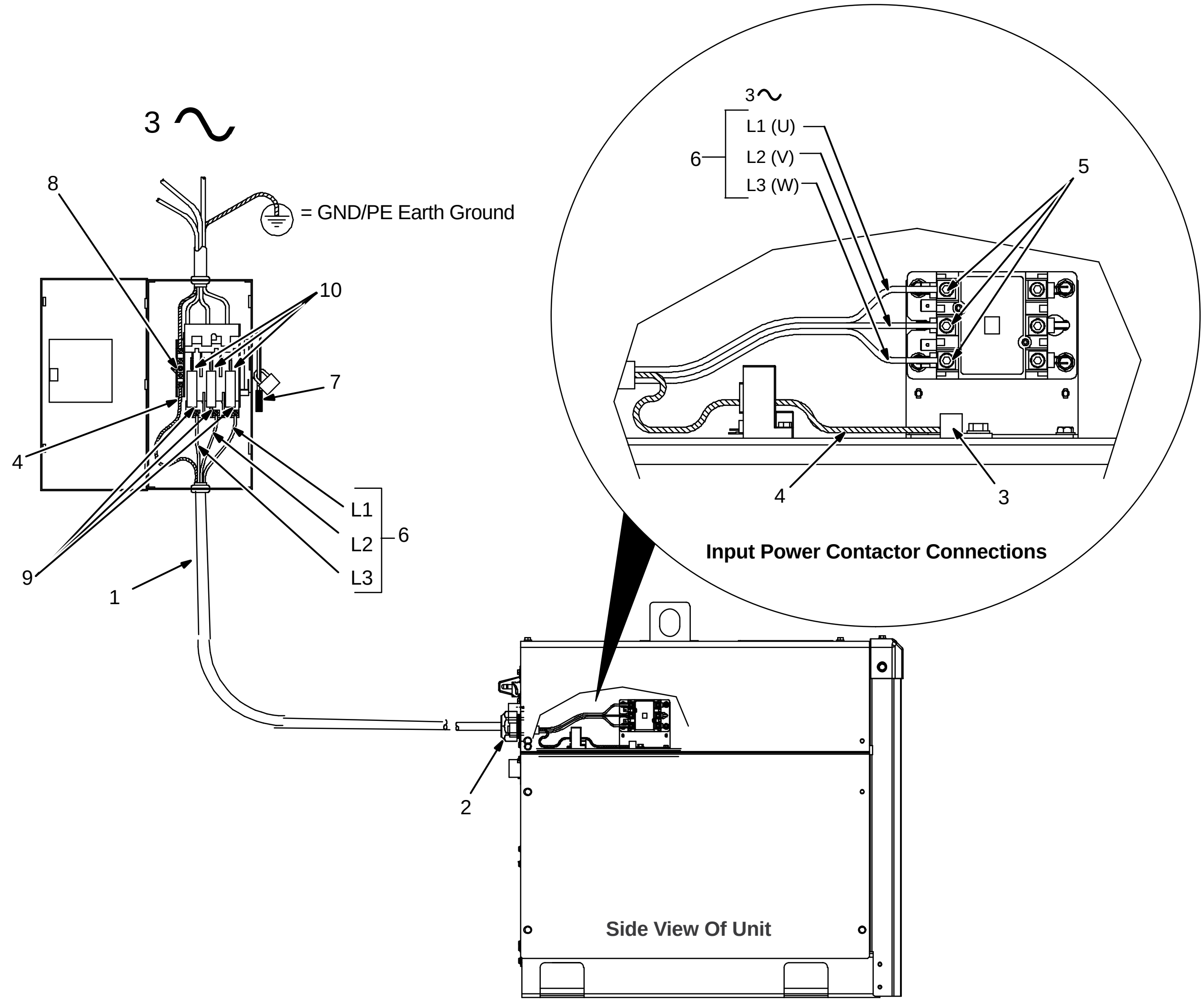
500 Amp Model	**60 Hz 3-Phase**				
Rated Supply Voltage (V)	230	380	400	460	575
Rated Maximum Supply Current I_{1max} (A)	78.8	46.6	44.1	38.6	30.4
Rated Effective Supply Current I_{1eff} (A)	57.6	34.7	33.1	28.7	23.3
Maximum Recommended Standard Fuse Rating In Amperes[1]					
Time Delay Fuses [2]	90	50	50	45	35
Normal Operating Fuses[3]	110	70	60	50	45
Maximum Recommended Supply Conductor Length In Feet (Meters)[4]	104 (32)	184 (56)	206 (63)	177 (54)	282 (86)
Raceway Installation					
Minimum Supply Conductor Size In AWG (mm²)[5]	6(16)	8 (10)	8 (10)	10 (6)	10 (6)
Minimum Grounding Conductor Size In AWG (mm²) [5]	6(16)	8 (10)	8 (10)	10 (6)	10 (6)

Reference: 2023 National Electrical Code (NEC) (including article 630)

1 If a circuit breaker is used in place of a fuse, choose a circuit breaker with time-current curves comparable to the recommended fuse.
2 "Time-Delay" fuses are UL class "RK5" . See UL 248.
3 "Normal Operating" (general purpose - no intentional delay) fuses are UL class "K5" (up to and including 60 amps), and UL class "H" (65 amps and above). See UL 248.
4 Maximum total length of copper supply conductors in entire installation, raceway and/or flexible cord.
5 Raceway conductor data in this section specifies conductor size (excluding flexible cord or cable) between the panelboard and the equipment per NEC Table 310.16 and is based on allowable ampacities of insulated copper conductors having a temperature rating of 75°C (167°F) with not more than three single current-carrying conductors in a raceway.

5-11. Connecting 3-Phase Input Power





⚠ **Turn Off welding power source, and check voltage on input capacitors according to Section before proceeding.**

⚠ **Installation must meet all National and Local Codes—have only qualified persons make this installation.**

⚠ **Disconnect and lockout/tagout input power before connecting input conductors from unit. Follow established procedures regarding the installation and removal of lockout/tagout devices.**

⚠ **Make input power connections to the welding power source first.**

⚠ **Always connect green or green/yellow conductor to supply grounding terminal first, and never to a line terminal.**

See rating label on unit and check input voltage available at site.

1 Input Power Conductors (Customer Supplied Cord)

Select size and length of conductors using Electrical Service Guide. Conductors must comply with national, state, and local electrical codes. If applicable, use lugs of proper amperage capacity and correct hole size.

Welding Power Source Input Power Connections

2 Strain Relief Kit (Supplied With Machine)

Install strain relief as explained in instructions supplied with kit.

Install strain relief of proper size for unit and conductors. Route conductors (cord) through strain relief. Tighten strain relief.

3 Welding Power Source Grounding Terminal

4 Green Or Green/Yellow Grounding Conductor

Connect green or green/yellow grounding conductor to machine grounding terminal first.

5 Welding Power Source Line Terminals

6 Input Conductors (L1, L2, And L3)

Connect input conductors L1, L2, and L3 to welding power source line terminals.

Reinstall side panel on welding power source.

Disconnect Device Input Power Connections

7 Disconnect Device (switch shown in the OFF position)

8 Disconnect Device Grounding Terminal

9 Disconnect Device Line Terminals

Connect green or green/yellow grounding conductor to disconnect device grounding terminal first.

Connect input conductors L1, L2, and L3 to disconnect device line terminals.

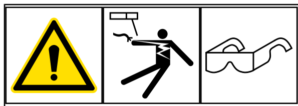
10 Over-Current Protection

Select type and size of over-current protection using Electrical Service Guide (fused disconnect switch shown).

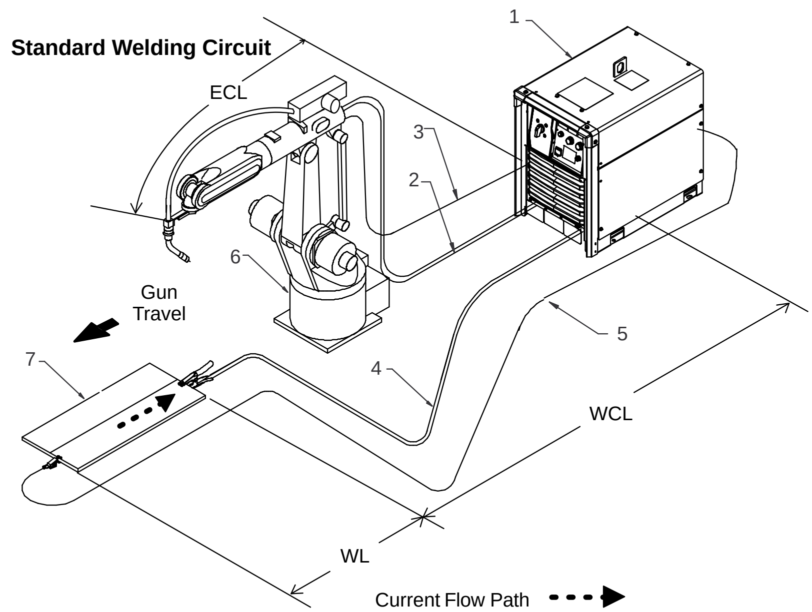
Close and secure door on disconnect device. Follow established lockout/tagout procedures to put unit in service.

SECTION 6 – RECOMMENDED SET-UP PROCEDURES

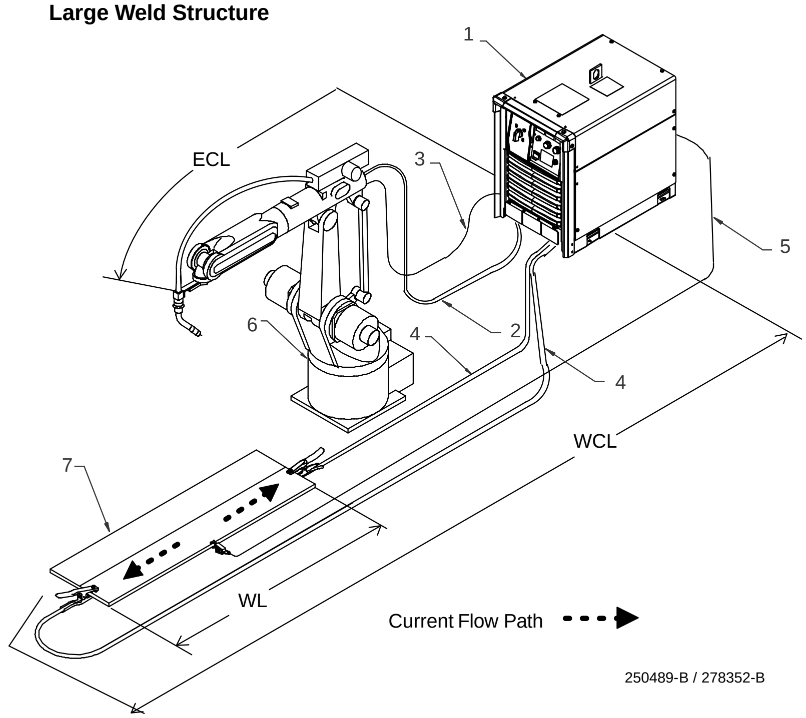
6-1. Welding Circuit



Standard Welding Circuit



Large Weld Structure



250489-B / 278352-B

☞ *Minimizing the welding circuit loop can prevent extreme voltage drops that produce poor welding characteristics.*

1 Welding Power Source

2 Electrode Cable

3 Feeder Cable

4 Work Cable

5 Volt Sense Lead

6 Wire Feeder

7 Work Piece

In pulse welding applications using inverter power sources, cable resistance can result in less than satisfactory performance. In most cases, a welding circuit length of 50 ft (15 m) or less will provide satisfactory performance with a standard welding circuit connection.

The length of a welding circuit is determined as follows:

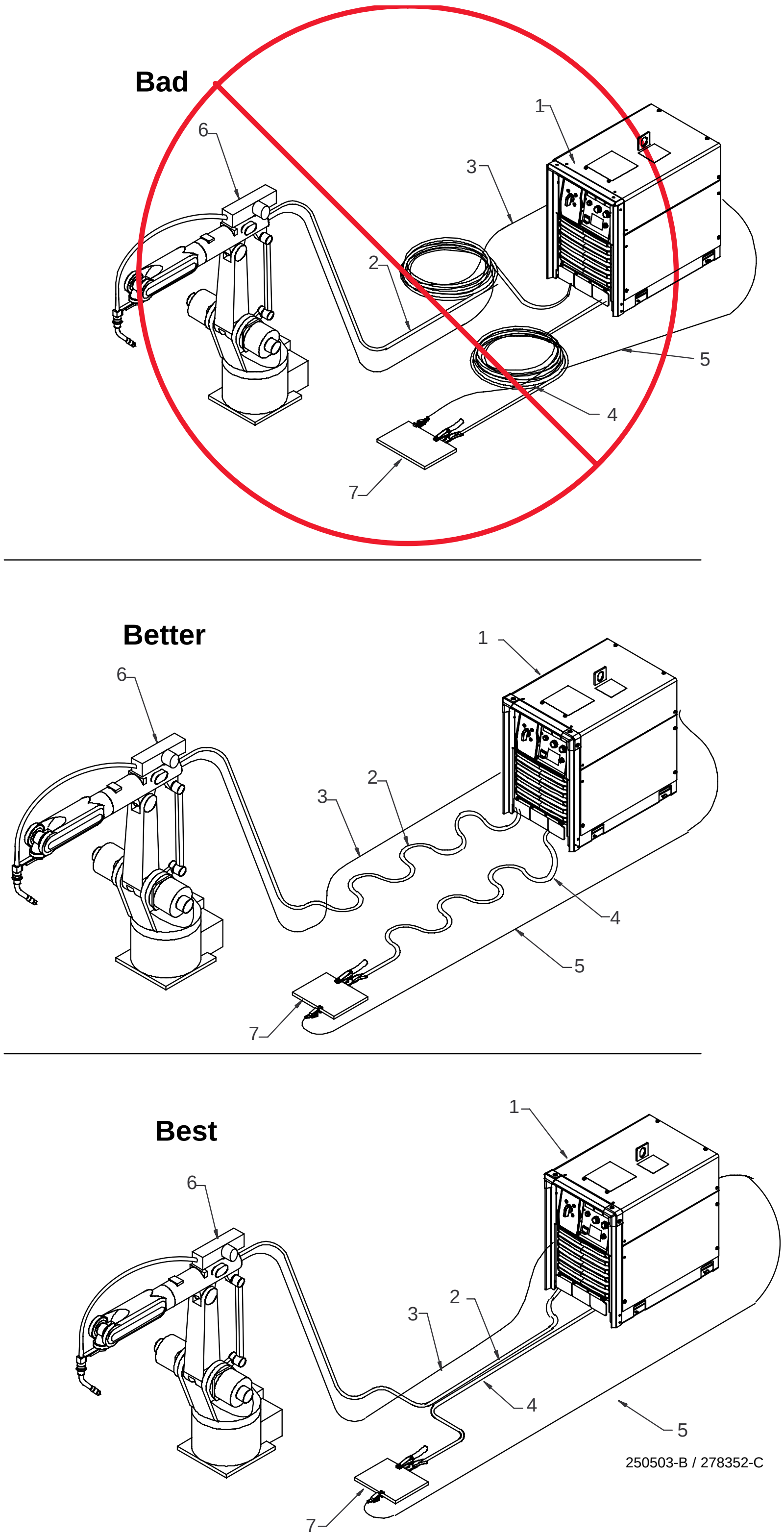
Welding Circuit = Electrode Cable Length (ECL) + Work Cable Length (WCL) + Workpiece Length (WL)

See section 5–3 for weld cable size.

Variations in welding processes and welding circuit resistance can affect apparent voltage at the welding arc. Voltage sensing can improve welding performance by providing accurate feedback to the welding power source.

It is important to connect the voltage sensing lead as near to the weld as possible, but not in the return current path.

6-2. Arranging Welding Cables To Reduce Welding Circuit Inductance



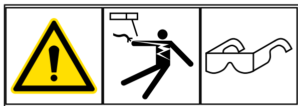
1 Welding Power Source
2 Electrode Cable
3 Feeder Cable
4 Work Cable
5 Volt Sense Lead
6 Robot
7 Work Piece

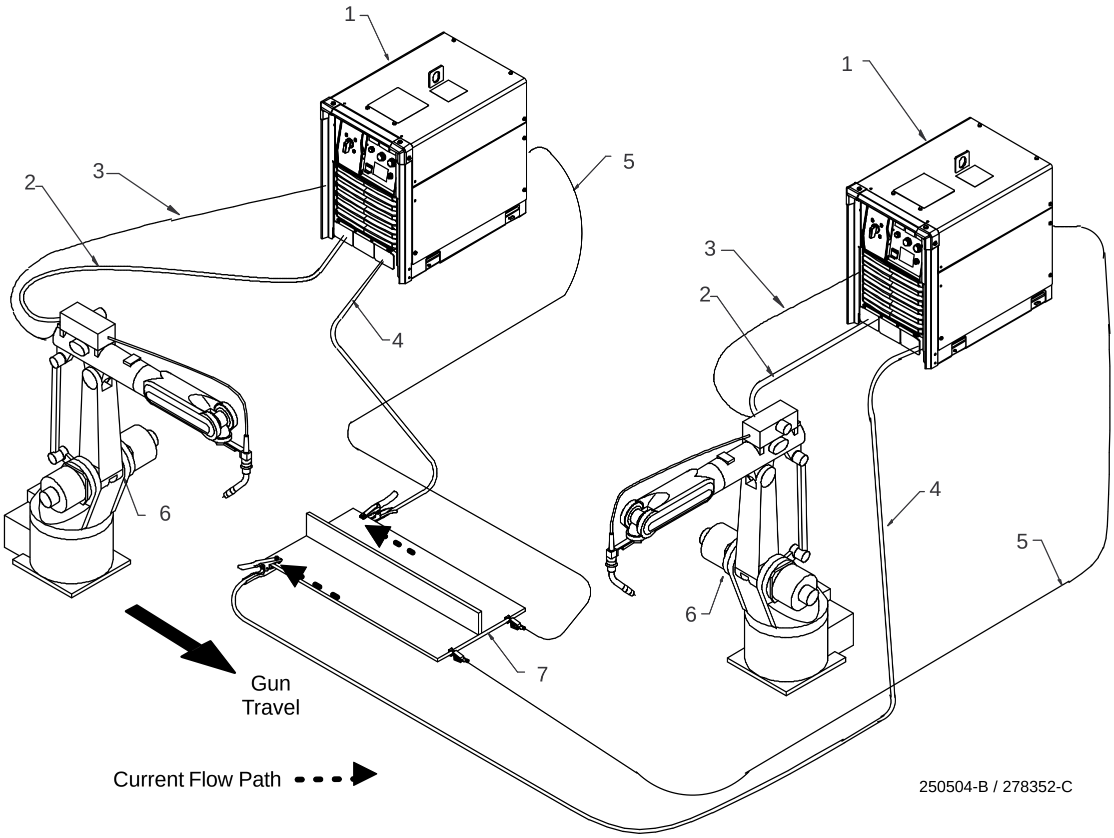
The arrangement of the cables has an effect that is significant to the welding properties. As an example, Accupulse welding process can produce high welding circuit inductance depending on cable length and arrangement. This can result in limited current rise during droplet transfer into the welding puddle.

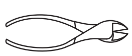
The Vsense option can be turned On or Off in the Setup menu or by using the web pages. The electrode sense lead is contained in the feeder control cable and compensates for voltage drop in all semi-automatic processes when the Vsense option is On . The work sense lead connects to the Axcess welding power source 4-pin connector located above the negative output terminal. This work sense lead compensates for work cable voltage drop when connected to the welding power source when the Vsense option is On.

Do not coil excess cables. Use cables that are the appropriate length for the application. Whenever using long weld cables [longer than 50 ft (15 m)] try to arrange positive and negative weld cables together to reduce the magnetic field surrounding the cables. Avoid coupling the feeder and work sense leads with the weld cables.

6-3. Using Multiple Welding Power Sources







1 Welding Power Source
2 Electrode Cable
3 Feeder Cable
4 Work Cable
5 Voltage Sensing Lead
6 Robot
7 Work Piece

Each welding power source should have a separate work cable connection to the workpiece. Do not stack or join work cables together at the workpiece. This is very important for pulse welding applications.

It is important to connect the voltage sensing lead as near to the weld as possible, but not in the return current path.

Connect voltage sensing lead at the end of the weld joint.

The direction of the welding path should be away from the work cable connections.

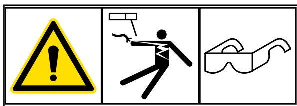
Connect work clamp at the beginning of the weld joint.

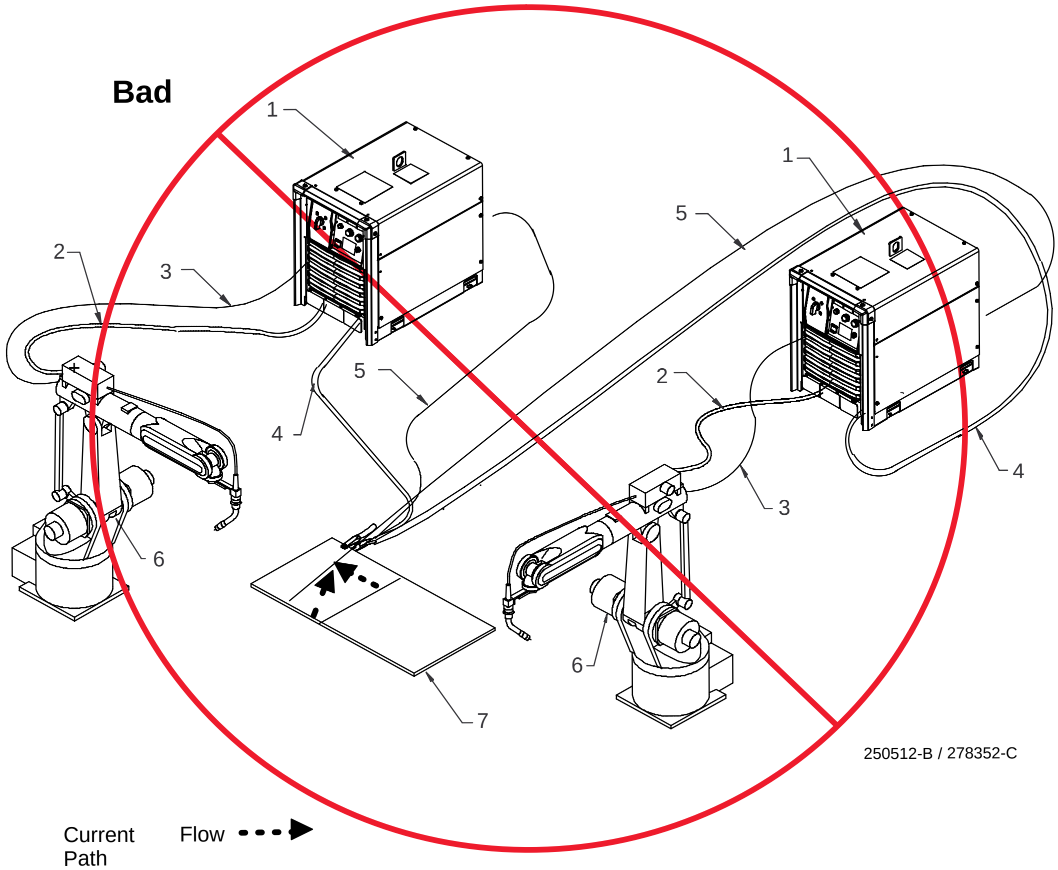
Each welding gun should have its own source of shielding gas. Use a separate shielding gas regulator and shielding gas connection for each welding gun.

Arc blow is the deflection of a welding arc from its normal path due to magnetic forces. It will adversely affect the appearance of a weld, cause excessive spatter, and impair the quality of a weld.

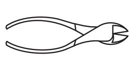
Arc blow occurs primarily during the welding of steel or ferromagnetic metals. Weld current will take the path of least resistance, but not always the most direct path through the workpiece to the work lead connection. The most intense magnet force will be around the arc due to a difference in resistance for the magnetic path in the base metal. The work clamp connection is important and should be placed at the starting point of a weld. It is recommended to have as short of an arc as possible so that there is less of an arc for the magnetic forces to control. Conditions affecting the magnetic force acting on the arc vary so widely that the reference here is only about cabling connections and arc preferences.

6-4. Voltage Sensing Lead And Work Cable Connections For Multiple Welding Arcs





Current Path Flow



1 Welding Power Source

2 Electrode Cable

3 Feeder Cable

4 Work Cable

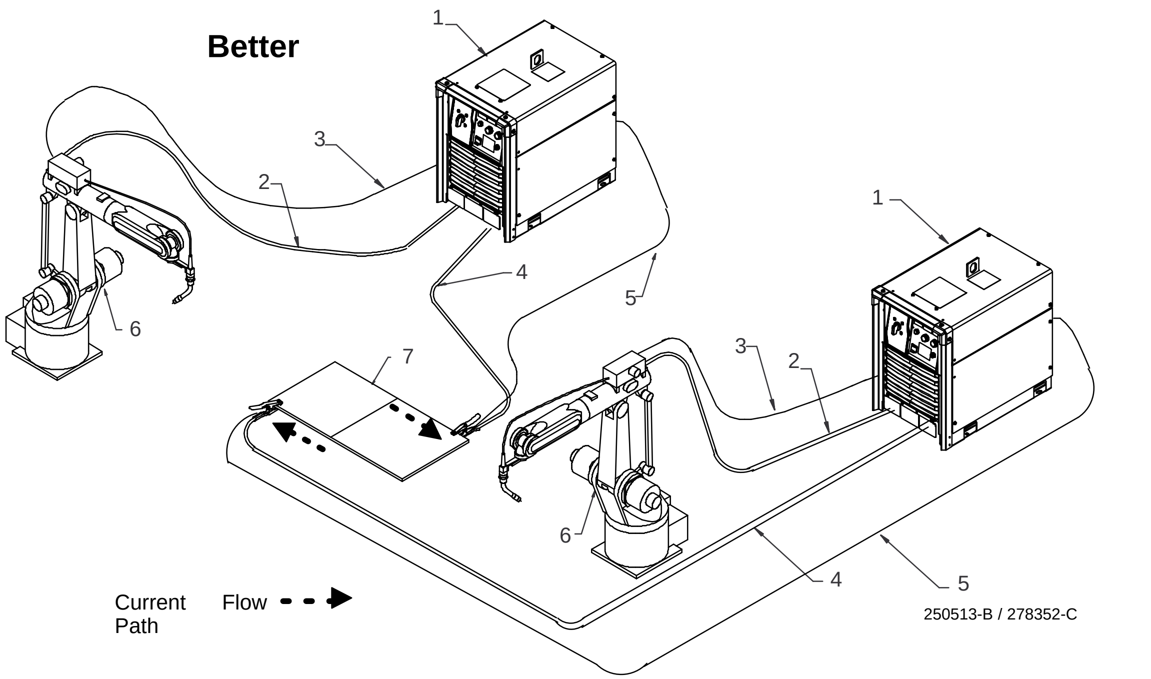
5 Voltage Sensing Lead

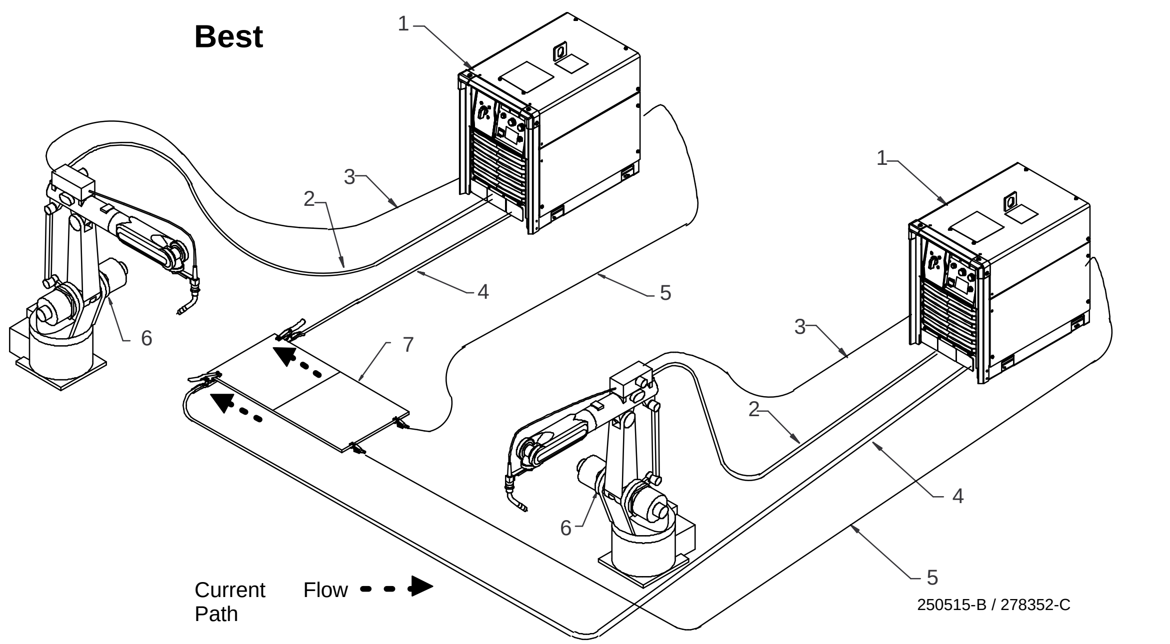
6 Robot

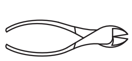
7 Work Piece

This arrangement is a bad setup due to sensing leads being directly in the current flow path of the welding arc. Interaction between welding circuits will affect voltage drop in the workpiece. The voltage drop across the workpiece will not be measured correctly for the voltage feedback signal. Voltage feedback to the welding power sources will not be correct at either sense lead and result in poor arc starts and arc quality.

6-5. Voltage Sensing Lead And Work Cable Connections For Multiple Welding Arcs







1 Welding Power Source

2 Electrode Cable

3 Feeder Cable

4 Work Cable

5 Voltage Sensing Lead

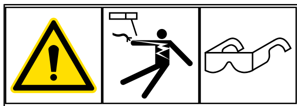
6 Robot

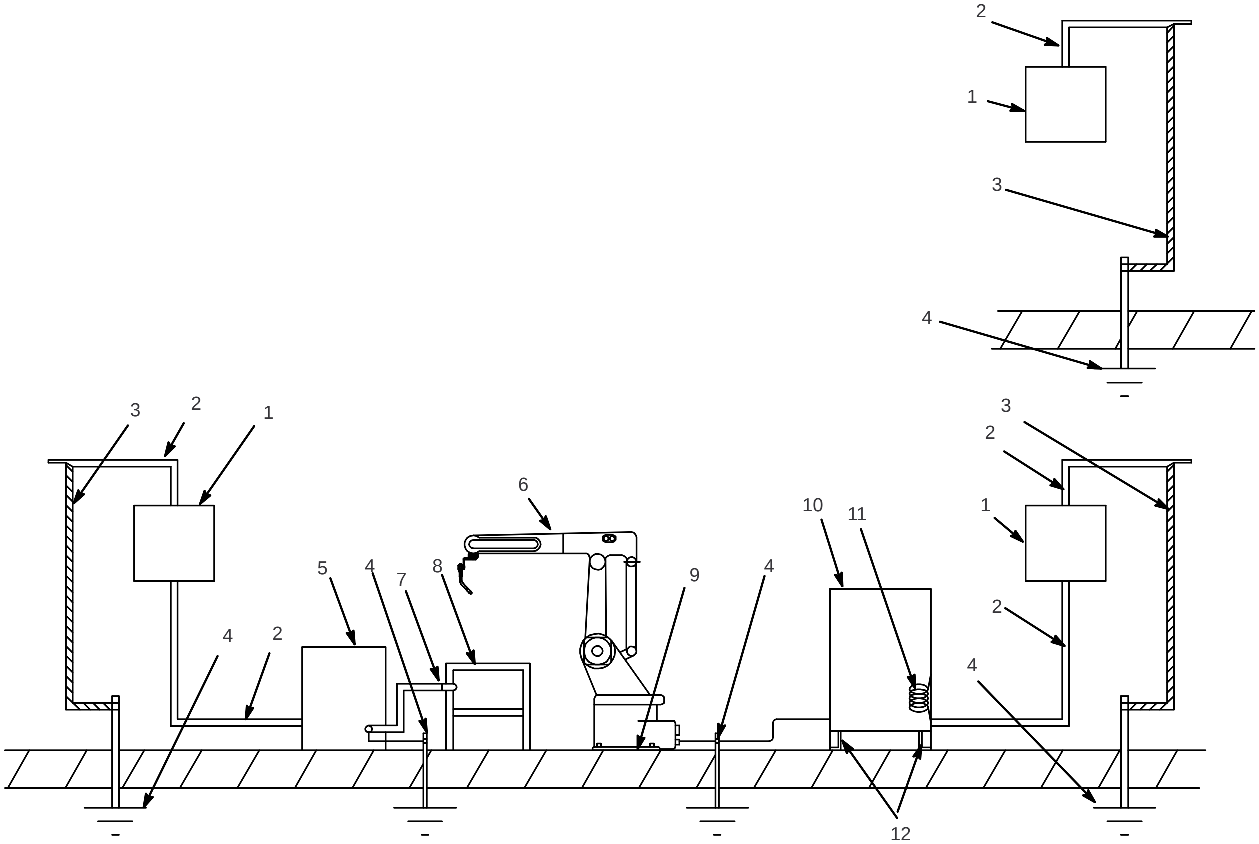
7 Work Piece

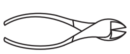
The top arrangement is a better setup for supporting separate voltage feedback to the welding power sources. The most accurate voltage sensing may not be achieved due to voltage drops in the workpiece. This may require compensation in the welding parameters.

The bottom arrangement is the best setup for proper voltage sensing at the work piece. Voltage feedback to the welding power sources will be more accurate and result in reliable arc starts and better arc quality.

6-6. Robot Ground Locations







1 Line Disconnect Switch Boxes
2 Input Power Supply Lines
3 Input Power Supply Ground
4 Earth Ground Location
5 Welding Power Source
6 Robot
7 Work Weld Cable
8 Work Table
9 Insulating Plate Under Robot Control
10 Robot Control
11 Robot Control Ground Lead
12 Insulating Liner Strips

Each welding power source should have a separate work cable connection to the workpiece. Do not stack or join work cables together at the workpiece. This is very important for pulse welding applications.

It is important to connect the voltage sensing lead as near to the weld as possible, but not in the return current path.

Connect voltage sensing lead at the end of the weld joint.

The direction of the welding path should be away from the work cable connections.

Connect work clamp at the beginning of the weld joint.

Each welding gun should have its own source of shielding gas. Use a separate shielding gas regulator and shielding gas connection for each welding gun.

Arc blow is the deflection of a welding arc from its normal path due to magnetic forces. It will adversely affect the appearance of a weld, cause excessive spatter, and impair the quality of a weld.

Arc blow occurs primarily during the welding of steel or ferromagnetic metals. Weld current will take the path of least resistance, but not always the most direct path through the workpiece to the work lead connection. The most intense magnet force will be around the arc due to a difference in resistance for the magnetic path in the base metal. The work clamp connection is important and should be placed at the starting point of a weld. It is recommended to have as short of an arc as possible so that there is less of an arc for the magnetic forces to control. Conditions affecting the magnetic force acting on the arc vary so widely that the reference here is only about cabling connections and arc preferences.

6-7. 30 Points Of Mechanics In MIG Welding

Primary Power

- Check primary power connection at line disconnect switch or receptacle and/or cord plug.
- Check primary power connection at welding power source.

Secondary Power

- Check secondary weld output connections at welding power source.
- Inspect condition and routing of positive weld cable to wire drive motor.
- Check connection of positive weld cable to wire drive motor.
- Inspect condition and routing of negative weld cable to fixture.
- Check connection of negative weld cable to fixture.
- Inspect condition of any rotary grounds, grounding shoes or other auxiliary grounds.
- Check installation, routing and condition of welding gun.

Shielding Gas

- Check gas hose connection to shielding gas supply regulator.
- Check shielding gas flow rate.
- Check gas hose routing.
- Check gas hose connection at wire drive housing.
- Check gun connection at wire drive and be sure O-rings seal at drive housing.
- Check condition of gas diffuser.
- Check condition of nozzle.
- Check O-ring for proper sealing at nozzle.

Welding Wire

- Inspect condition of de-reeler. Check for wear at quick-connect coupling and replace if necessary.
- Check placement of payoff pack or drum for smooth feed path.
- Inspect condition and routing of input conduit.
- Check installation of quick-connect coupling at rear of wire drive so that it does not contact drive rolls. Check for wear and replace if necessary.
- Check drive rolls and replace if worn.
- Check for drive roll tension setting.
- Check intermediate guide for proper size to match wire size and replace if worn.
- Check for proper length of liner at both ends and be sure it is cut without burrs.
- Check liner for proper size to match wire size.
- Check liner for wear and clean out to prevent plugging.
- Check contact tip for proper size to match wire size.
- Check contact tip for wear and change at regular intervals.
- Check contact tip for a tight fit and secure installation at gun.

6-8. Arc Blow

Arc blow is the deflection of the welding arc from its normal path due to magnetic forces. This condition is usually encountered in direct current welding of magnetic materials, such as iron and nickel. Arc blow can happen in alternating current welding under certain conditions, but these cases are rare and the intensity of the arc blow is always less severe. Direct current flowing through the electrode and base metal will set up a magnetic field around the electrode. This magnetic field tends to deflect the arc to the side at times, but usually the arc deflects either forward or backwards along the joint.

Back blow is encountered when welding toward the work cable connection on a workpiece near the end of a joint or into a corner. Forward blow is encountered when welding away from the work cable connection on a workpiece at the start of a joint. In general, arc blow is the result of two basic conditions:

1 The change of current flow direction as it enters the work and is conducted toward the work cable.
2 The asymmetric arrangement of magnetic material around the arc, a condition that normally exists when welding is performed near the end of ferromagnetic materials.

Although arc blow cannot always be completely eliminated, it can be controlled or reduced to an acceptable level through knowledge of the two conditions listed above.

Except in cases where arc blow is unusually severe, certain steps can be taken to eliminate or reduces its severity. Some or all of the following steps may be necessary:

- Place work cable connection as far as possible from joints to be welded.
- If back blow is the problem, place the work cable connection at the start of the joint to be welded and weld toward a heavy tack weld.
- If forward blow is the problem, place the work cable connection at the end of the joint to be welded.
- Position electrode angle so that arc force counteracts arc blow.
- Use the shortest possible arc that maintains good welding practices (this helps arc force to counteract arc blow).
- Reduce welding current if possible.
- Weld toward a heavy tack weld or runoff tab.
- Use the back step sequence of welding.
- Wrap the work cable around the workpiece in the direction that sets up a magnetic field to counteract the magnetic field causing the arc blow.

6-9. Basic Welding Troubleshooting

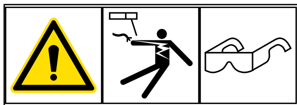
Listed below are some problems, causes and remedies related to welding operations; however, this list does not contain every possible condition that could be encountered in welding.

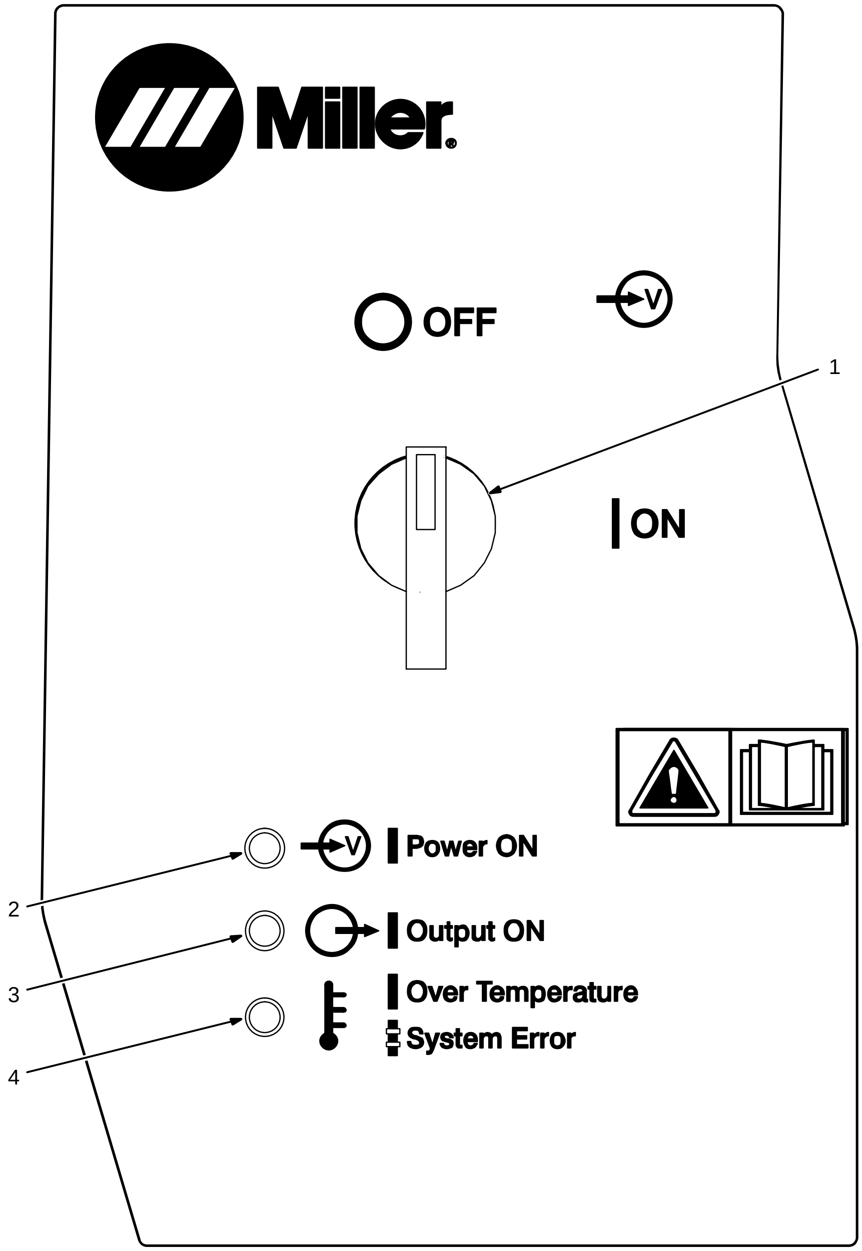
Trouble	Probable Cause	Remedy
No weld output; unit completely inoperative.	Line disconnect switch in Off position.	Place switch in On position.
	Power source switch in Off position.	Place switch in On position.
	Primary power fuse blown or circuit breaker tripped.	Replace fuse or reset circuit breaker and check input voltage.
Weld output is present, but wire stops feeding while welding.	Wire feeder protective fuse blown or circuit breaker tripped.	Replace fuse or reset circuit breaker and find overload condition.
	Wire feeder drive rolls misaligned	Align drive rolls.
	Wrong size drive rolls	Replace with proper size drive rolls.
	Too much or too little drive roll pressure.	Adjust drive roll pressure
	Too much tension set at wire spool.	Reduce wire spool tension.
	Restriction in unspooler or drum adapter.	Replace unspooler or repair restriction.
	Feeder motor burnt out.	Test motor and replace if necessary.
	Gun liner dirty or restricted.	Remove gun liner and clean or replace.
	Wrong type or size of liner.	Install proper size liner.
	Broken or damaged gun or torch.	Replace faulty parts.
	Contact tip opening restricted.	Replace contact tip.
	Wrong size or type of contact tip.	Replace with proper size and type contact tip.
	Sharp bends or kinks in gun cable or liner.	Straighten gun cable and/or replace liner.
	Gun overheating.	Use gun with proper amperage rating.
	Wrong size wire.	Match wire size to liner and contact tip.
	Guides rubbing on drive rolls.	Adjust or position guides properly.
	Drive rolls jammed.	Remove foreign object from gears.
	Motor cable disconnected or damaged.	Connect, repair or replace motor cable.
Porosity in Weld	Dirty base metal, heavy oxides, mill scale, oil etc.	Clean base metal by brushing, grinding or use chemical cleansing before welding.
	Regulator/flowmeter faulty.	Adjust or replace regulator/flowmeter.
	Gas cylinder valve closed.	Open gas cylinder valve.
	Gas regulator diaphragm defective.	Replace regulator.
	Flowmeter cracked or broken.	Repair or replace flowmeter.
	Gas hose disconnected or leaking.	Connect or replace gas hose.
	Too much or too little gas flow.	Adjust for proper gas flow.
	Moisture in shielding gas.	Replace gas cylinder or supply.
	Wrong gas for wire type or transfer mode.	Use correct shielding gas.
	Feeder gas solenoid faulty.	Replace solenoid
	Gun or outlet cable leaking.	Repair or replace faulty parts.
	Wire feed speed setting too high.	Reduce wire feed speed.
	Contact tip extends too far out of nozzle.	Adjust or replace parts (max distance should not exceed 1/8 in (3.2 mm).
	Nozzle to work distance too large.	Reduce nozzle to work distance.
	Incorrect gun or torch angle.	Set proper gun angle (porosity or dirty welds mean gun angle is too large).
	Nozzle Restriction	Clean off spatter or remove restriction.
	Breeze or drafts in weld zone.	Shield weld zone from drafts
	Low shielding gas cylinder pressure.	Replace gas cylinder.
	Gas leak at gun to feeder connection.	Properly install gun or replace O-rings at gun connector.
Excessive Spatter	Voltage set too high.	Reduce voltage setting (reduce trim or arc adjust for pulse welding).
	Incorrect gun or torch angle.	Set proper gun angle.
	Too much or too little gas flow.	Adjust for proper gas flow.
	Wrong gas for wire type or transfer mode.	Use correct shielding gas.
	Wrong electrode wire type or size.	Use proper electrode wire.

Trouble	Probable Cause	Remedy
	Wrong inductance setting.	Adjust inductance.
	Electrode wire dirty or old.	Replace with new electrode wire.
	Oily or dirty base metal.	Clean base metal by brushing, grinding or use chemical cleansing before welding.
	Distance too great for wire stick-out or nozzle to work.	Adjust wire stick-out or reduce nozzle to work distance.
	Wrong transfer mode.	Set proper transfer mode.
	Travel speed too slow.	Increase travel speed so that arc is on leading edge of weld puddle.
Wandering, hunting or erratic arc.	Restriction in unspooler or drum adapter.	Replace unspooler or repair restriction.
	Dirty or worn gun liner or inlet cable.	Remove gun liner or inlet cable and clean or replace.
	Sharp bends or kinks in gun cable or liner.	Straighten gun cable and/or replace liner.
	Loose or worn contact tip.	Tighten or replace contact tip.
	Wrong size or type of contact tip.	Replace with proper size and type contact tip.
	Gun overheating.	Use gun with proper amperage rating.
	Loose power cables or other electrical connections.	Tighten, repair or replace connections or cables, also check all rotary or brush type connections.
	Incorrect gun or torch angle.	Set proper gun angle.
	Too much or too little gas flow.	Adjust for proper gas flow
	Moisture in shielding gas.	Replace gas cylinder or supply.
	Wrong gas for wire type or transfer mode	Use correct shielding gas.
	Wrong program selection for pulse welding.	Make proper program selection.
	Improper or unsteady analog command from robot controller.	Check signal from robot controller (as a troubleshooting aid go to power source control of voltage and wire speed).
	Gun or outlet cable leaking.	Repair or replace faulty parts.
	Incorrect nozzle to work distance.	Set proper distance [3/8 in to 5/8 in (9.5 to 15.9 mm) for short arc, 5/8 in to 1 in (15.9 to 25.4 mm) for pulse welding, and 3/4 in to 1-1/4 in (19.1 to 31.8 mm) for spray welding].
	Voltage sensing leads open or shorted.	Repair or replace voltage sense leads.
	High frequency noise in the area.	Be sure proper grounding methods are followed when TIG or plasma equipment is used in the area.
	Arc blow.	See section 6–4
	Drive motor tachometer or motor cable open or shorted.	Check drive motor tachometer and cables, and repair or replace.
	Wrong size drive rolls.	Replace with proper size drive rolls.
	Too much or too little drive roll pressure	Adjust drive roll pressure.
Welding wire burns back to contact tip at the start of a weld.	Restriction in wire feed system.	Check inlet cable, gun liner and wire guides.
	Worn drive rolls	Replace drive rolls.
	Wrong size drive rolls.	Replace with proper size drive rolls.
	Improper start parameters	Adjust start parameters.
	Worn contact tip.	Replace contact tip.
	Wrong size or type of contact tip	Replace with proper size and type contact tip.
	Not enough cast in welding wire.	Add a wire straightener to put cast in wire.
Welding wire burns back to contact tip during welding.	Restriction in wire feed system.	Check inlet cable, gun liner and wire guides.
	Worn drive rolls	Replace drive rolls.
	Wrong size drive rolls.	Replace with proper size drive rolls.
	Too much or too little drive roll pressure.	Adjust drive roll pressure.
	Worn contact tip.	Replace contact tip.
	Wrong size or type of contact tip	Replace with proper size and type contact tip.
	Not enough cast in welding wire.	Add a wire straightener to put cast in wire.
Welding wire burns back to contact tip at the end of a weld.	Welding power source output not shutting off.	Make sure all switches are in correct position, repair power source if necessary.
	Burnback setting too high or too long.	Adjust burnback setting or turn off completely.

SECTION 7 – OPERATION

7-1. Operator Controls





Turn off power before connecting to weld output terminals.

Do not use worn, damaged, under-sized, or repaired cables.

1 Power Switch

Turns unit On or Off.

2 Power On LED

Power LED illuminates when unit is energized.

3 Output On LED

Output On LED illuminates when weld output is energized.

4 Error LED

On Solid-

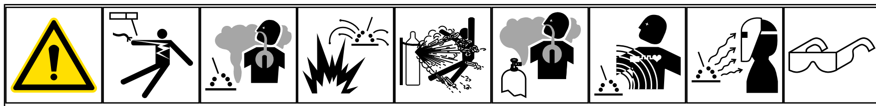
Unit has overheated and weld output has been disabled. Allow unit to cool to operating temperature. Error LED will turn off and welding may resume.

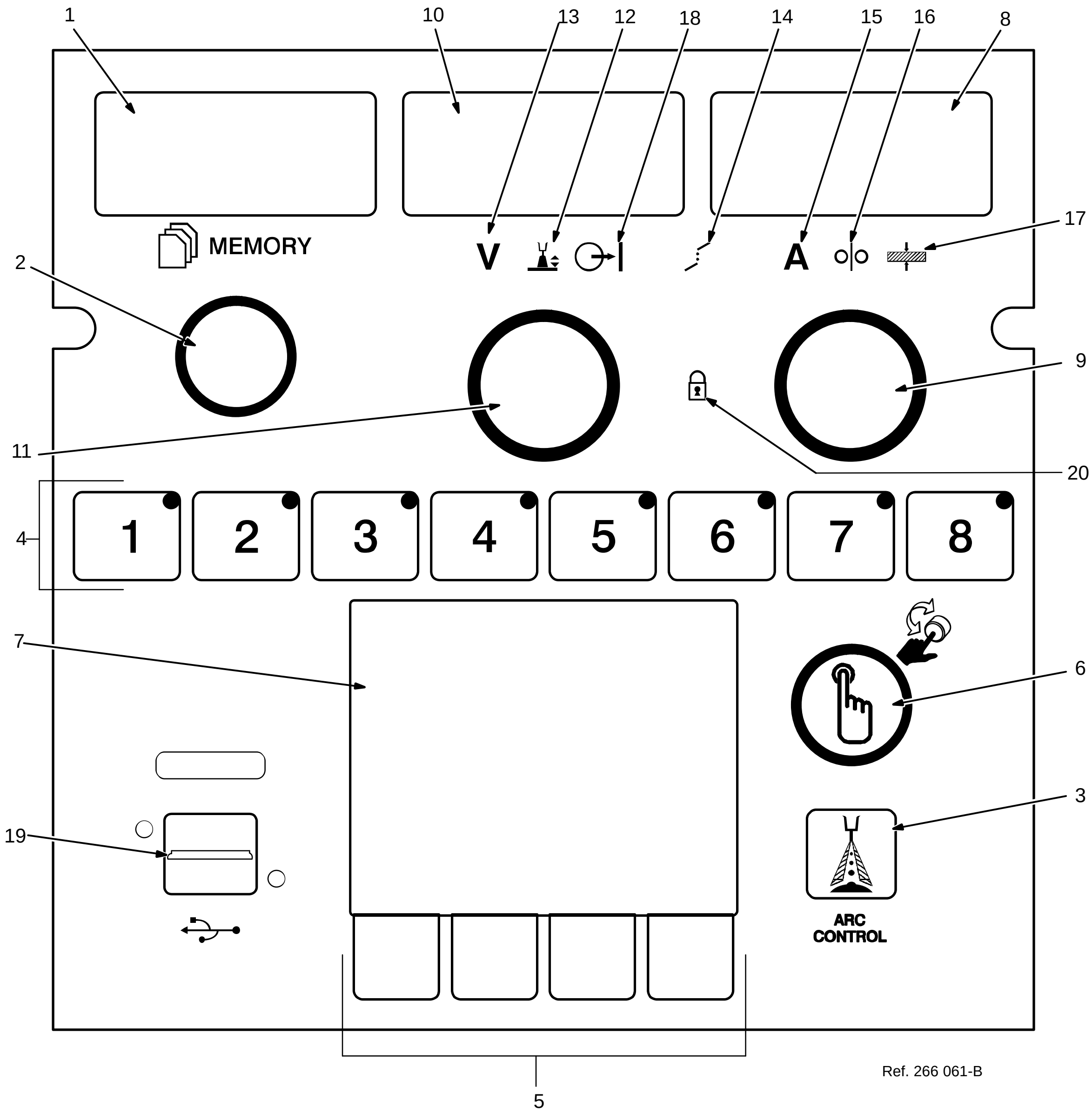
Blinking-

A system error has occurred. Use the User Interface to determine the source of the error as well as possible solutions. Follow on-screen instructions to clear the error. When the error has been cleared the Error LED will stop blinking and welding may resume.

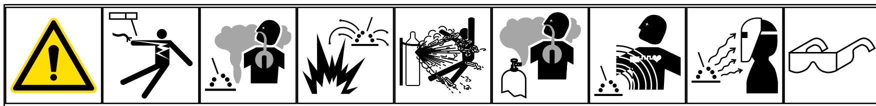
After turning unit off, wait until Power ON LED is off before turning unit back on. Cycling power too quickly can cause software issues on power up.

7-2. Wire Feeder User Interface





7-3. Description Of Front Panel Controls



1 Memory Display

The Memory display shows the active weld program. The range is from 0001 to 9999.

2 Memory/Variable Adjustment Knob

Use this knob to select a program number or increase/decrease the variable shown in the Memory display.

3 Arc Control Button

The Arc Control button is process dependent. It allows user to change the variable that appears on the Values/Parameters display, such as inductance, DIG, RMD (ball size) or trim. The range can be +/- depending on the process.

4 Memory 1-8 Buttons And LEDs

Use these buttons to select a stored program from the weld program library data. The LED will indicate which program is selected and active.

The Variables/Parameters display will show weld program library summary data for the selected program.

5 Soft Key Buttons

Use these buttons to navigate the menus appearing in the LCD display.

6 Scroll Knob And Select/Save Button

Rotate this knob to scroll available options or increase/decrease the variable shown in the LCD display.

Pressing this knob acts as a Select/Save button. Press the button to select options available in the LCD display

7 Values/Parameters LCD Display

This display shows all values and parameters that have been selected using the Soft Key buttons and the Scroll knob and Save/ Select button.

8 Amperage/Wire Feed Speed Display

This display shows amperage, or wire feed speed depending on the process selected

9 Amperage/Wire Feed Speed/Material Thickness Adjustment Knob

Use this knob to increase/decrease amperage, wire feed speed, or material thickness depending on the process selection. These values appear on the Amperage/Wire Feed Speed/Material Thickness display.

10 Voltage/Arc Length/Inductance/Arc Adjust/RMD Display

This display shows voltage, arc length, inductance, arc adjust or RMD depending on the process selected.

11 Voltage/Arc Control Adjustment Knob

Use this knob to increase/decrease voltage or arc length depending on the process selection. These values appear on the Voltage/Arc Length/Inductance/Arc Adjust/RMD display.

12 Arc Length Indicator

LED illuminates to show that arc length function is active.

13 Voltage Indicator

LED illuminates to show that the voltage function is active.

14 Synergic Mode Active Indicator

LED illuminates to show that a synergic weld process is active.

15 Amperage Indicator

LED illuminates to show that a SMAW, GMAW or CAG process is selected. Values appear on the Voltage/Arc Length/Inductance/Arc Adjust/RMD display.

16 Wire Feed Speed Indicator

LED illuminates to show that a GMAW or GMAW-P weld process is selected. Values appear on the Amperage/Wire Feed Speed/ Material Thickness display.

17 Material Thickness Indicator

This feature will be functional in a later software release.

18 Output ON Indicator

LED illuminates to show that open circuit voltage is present at the weld output terminals.

19 USB Port

Use this port for all USB related functions.

20 Parameter Lock Indicator

LED illuminates to show weld parameters are locked and active.

SECTION 8 – CONFIGURATION

8-1. Accessing Configuration Web Pages

To access the power source configuration webpages you will need the optional communication panel (See Section 5-8). Connect a PC directly to the jack on the communication panel with a CAT5 or CAT6 Ethernet cable.

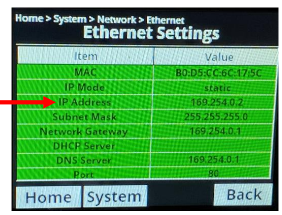
Enter the default IP address, 169.254.0.2, into a web browser and the welder configuration web pages will open to the Home screen.

1. Connect — Connect the computer to the Network Port on the back of the welding power source

(Auto-Continuum is shown)

2. Navigate — Navigate to the System menu and then the Network Settings menu in the User Interface

3. Choose — Choose Ethernet Settings to confirm the IP Address of the Network Port, in this example the IP Address is 169.254.0.2



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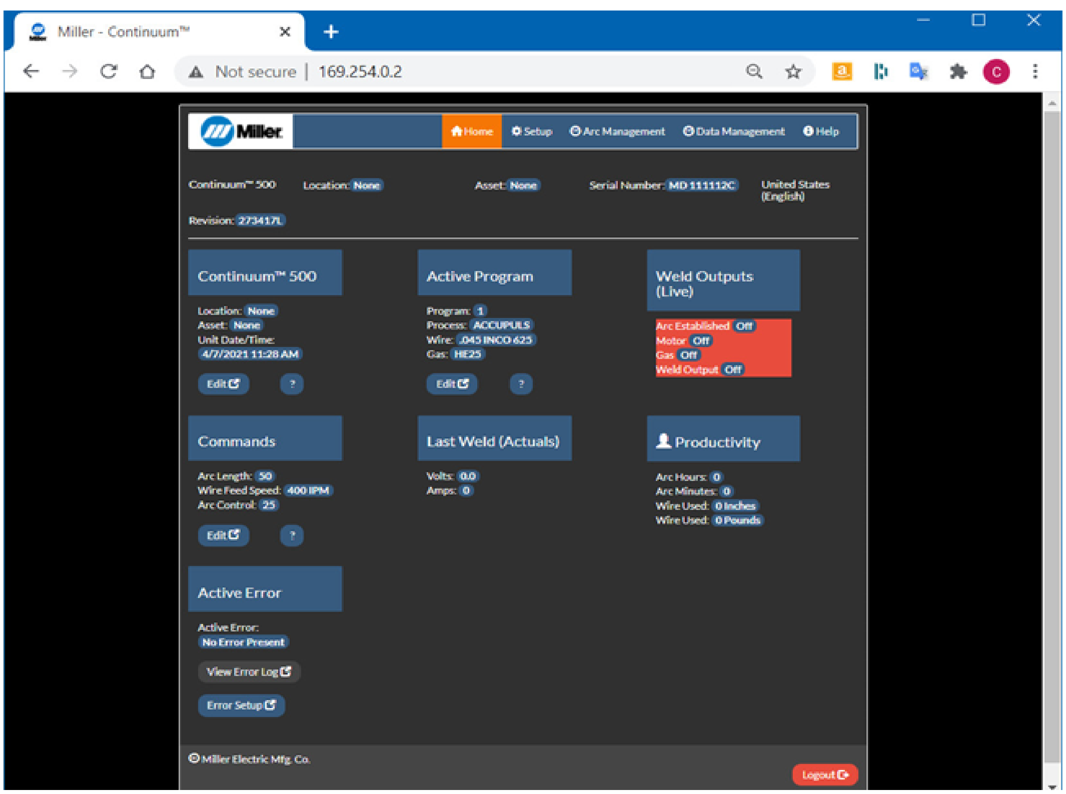


8-2. Home Screen

1 Navigation Bar

Select Home, Setup, Arc Management, Data Management, or Help screens.

2 Information Bar

Displays general information on power source, location, asset number, serial number, and display language.

3 Power Source Information

Displays power source information. Edit button allows these parameters to be changed.

4 Active Program

Displays the program number, process, wire size and alloy, and gas. Edit button allows these parameters to be changed.

5 Weld Outputs (Live)

Displays the weld output conditions of the power source and wire feeder in real time.

6 Commands

Displays current user changeable commands. Depending on the word process being used these commands will vary. Edit button allows parameters to be changed.

7 Last Weld

Displays the voltage and amperage used during the last weld made.

8 Productivity

Displays arc time, the amount of wire used and the amount of wire remaining.

9 Active Error

Displays any active error messages. View Error Log button will display a list of previous errors and the time at which they occurred.

8-3. Set Up Screen

1 Information Bar

Displays general information on power source, location, asset number, serial number, and display language.

2 Purge Time

Displays the length of time gas will flow when the purge button is pressed. Edit button allows time length to be changed.

3 Enabled Errors

Displays which system errors are enabled. Click Edit to enable/disable system errors.

4 Jog Speed

Displays the minimum and maximum Jog speeds in inches per minute (IPM). Edit button allows minimum and maximum jog speed setting to be changed for forward and reverse jog.

5 Ethernet Settings

Edit button allows user to change connectivity settings.

6 Data Connections

Displays which data connections are enabled or disabled.

7 Region

Displays current region. Edit button allows region to be changed.

8 Status

Displays the status of various system selections: English/Metric display values, Slope Adjustment enable/disable, Synergic MIG Enable/Disable, CE/English weld library availability, Arc length command scaling.

9 WiFi Settings

Edit button allows changes to WiFi settings

10 Auto Thread

Displays Load Rate in inches per minute (IPM) and Torch Length (Inches) to determine length of time for Auto Threading. Edit button allows these parameters to be changed.

11 Wire Spool

Displays wire type, spool size, and amount of wire left on the spool. Edit button allows wire information to be changed or reset.

12 Weld Cable Set-Up

Displays length and size of positive and negative weld leads, torch, and voltage feedback (VFB) method that are selected. Edit button allows these parameters to be changed.

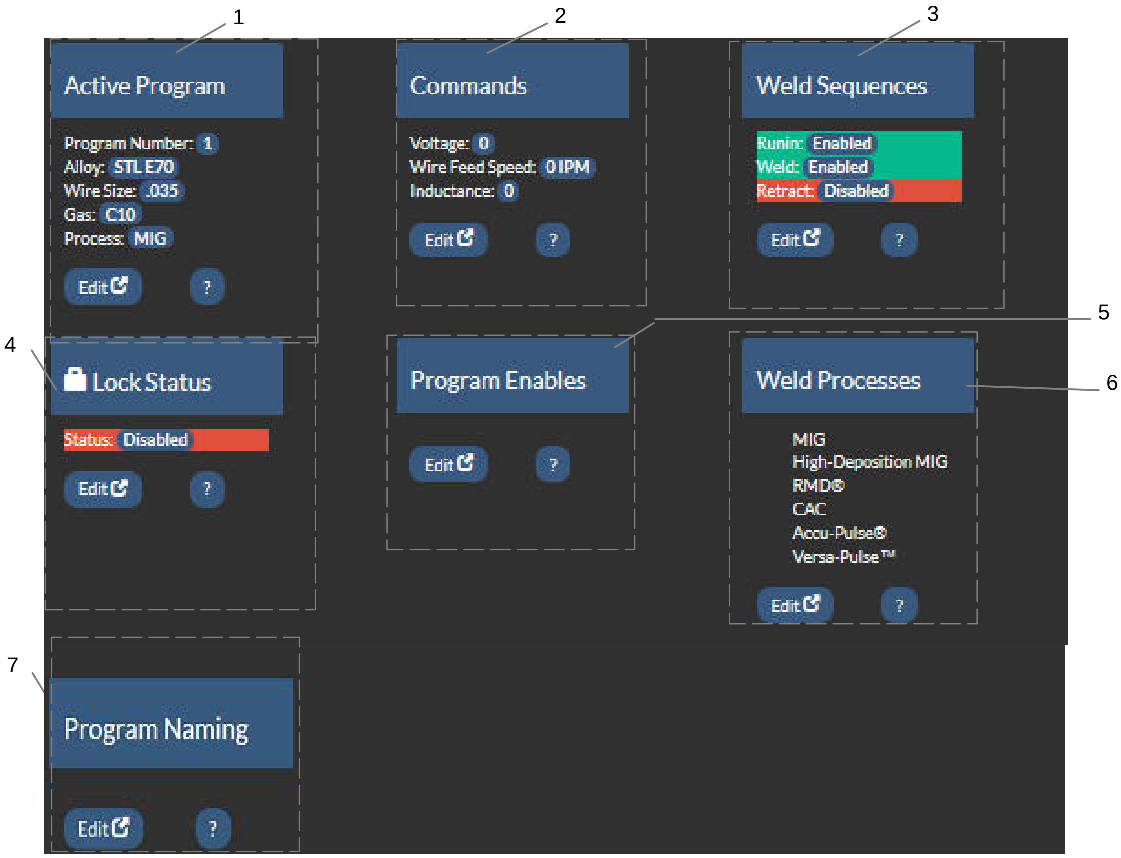
13 Robot Options

Displays current robot connectivity options. Edit button allows options to be changed.

14 Dry Run

Allows an automated weld cell to cycle without striking an arc by simulating the arc detected signal back to the weld cell

8-4. Arc Management Screen



1 Active Program

Displays the program number, process, wire size and alloy, and gas. Edit button allows these parameters to be changed.

2 Commands

Displays current user changeable commands. Depending on the weld process being used these commands will vary. Edit button allows parameters to be changed.

3 Weld Sequence

Displays active weld sequence parameters. Edit button allows changes to these parameters.

4 Lock Status

Displays the current status of system locks. Edit button allows the lock configuration to be changed.

5 Program Enable

Edit button will allow user to select which programs are available for use.

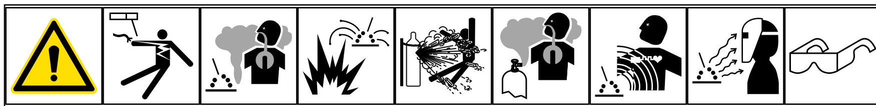
6 Weld Processes

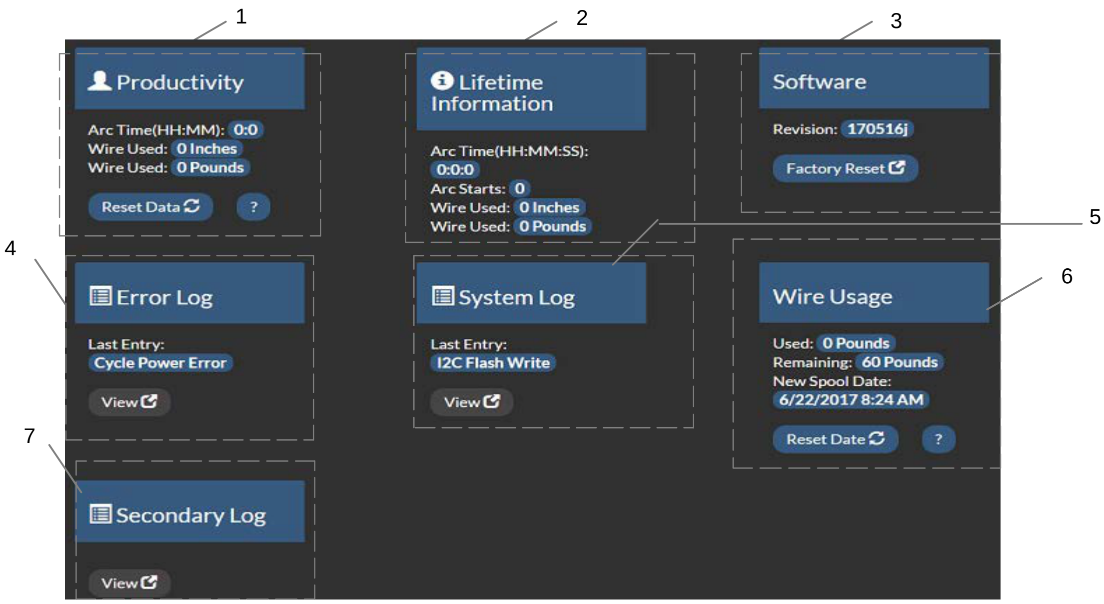
Displays available weld processes. Edit button allows processes to be enabled or disabled.

7 Program Naming

Edit button allows user to enter a short personalized name for a weld program and a longer description of what the program is for. Both fields must have data entered. The short name will then be viewable on the LCD screen on the User Interface of the Continuum.

8-5. Data Management Screen





1 Productivity

Displays arc time and wire used in inches and pounds. Click Reset Data to reset productivity data.

2 Lifetime Information

Displays total arc time, arc starts, and wire used over lifetime of unit.

3 Software Information

Displays the software revision.

Factory Reset button will take user to a Factory Reset screen. Read and follow all instructions on Factory Reset screen to perform a Factory Reset.

4 Error Log

Displays the last error experienced by the system.

View button displays a list of previous errors and the time at which they occurred.

5 System Log

Displays the last entry into the system log.

View button displays a list of the entries made to the system log and the time at which they occurred.

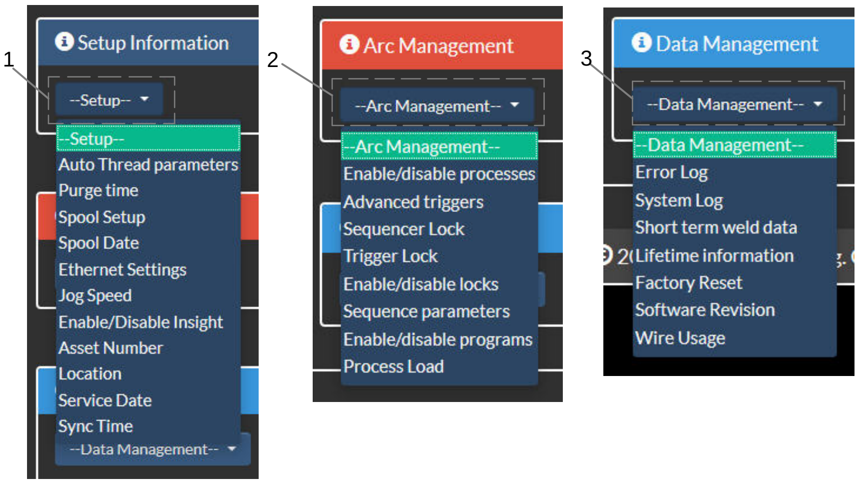
6 Wire Usage

Displays the amount of wire used, the amount remaining and the date the spool was started. Click reset data when starting a new spool.

7 Secondary Log

View button displays a list of secondary compensation tests that were run, listing Pass/Fail status, date/time test was run, torch resistance, torch voltage, weld cable resistance, weld cable voltage, and weld cable current.

8-6. Help Screen



☞ *. The Help Screen contains three drop down menus which contain the features on the Setup, Arc Management and Data Management screens.*

1 Set-Up Information Drop-Down Menu

2 Arc Management Drop-Down Menu

3 Data Management Drop-Down Menu

8-7. Lock Setup

1. Access the Locks Setup from the Arc Management page by clicking on the Edit button under Lock Status.

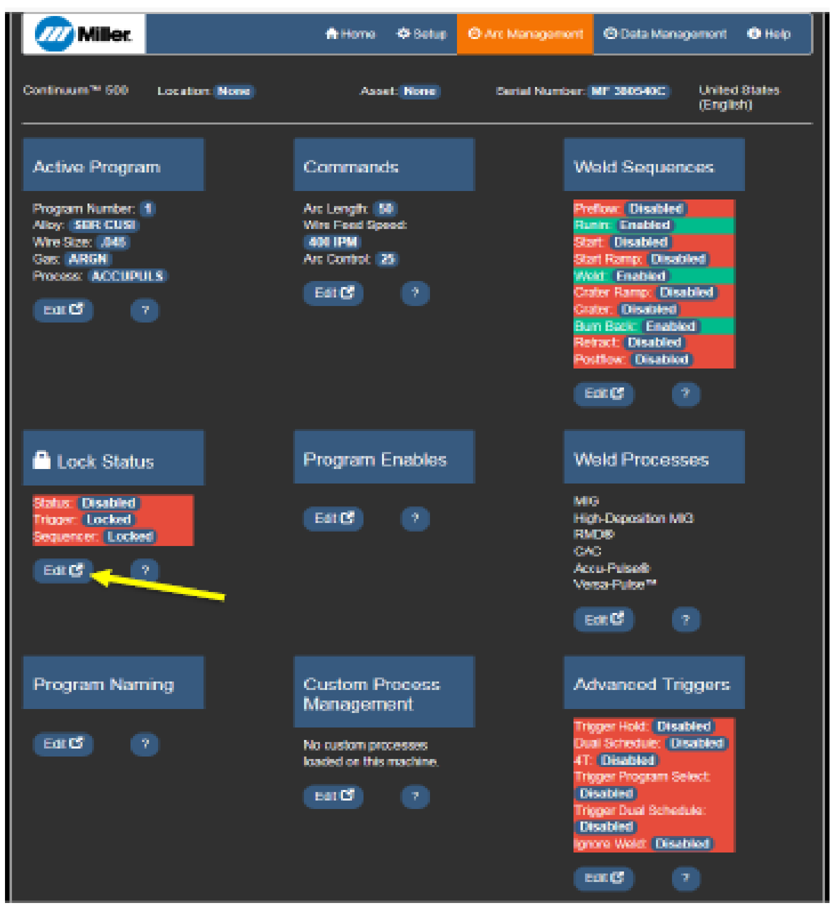


Figure 8-1. Home Page for Lock Status

2. The Weld Program section of the Lock Status page allows the individual programs to be Enabled — accessible to the operator or Disabled — inaccessible to the operator through the Use Interface.
3. In the Lock Setup section, the Locks Enable/Disable is the global lock setting. There are two features configurable from this screen. Trigger Functions and Weld Sequencer can be enabled unlocked to the operator when the Locks are Enabled. The other features made inaccessible by the Lock Setup are Load Process, Feedback Configuration, and Retract Start.

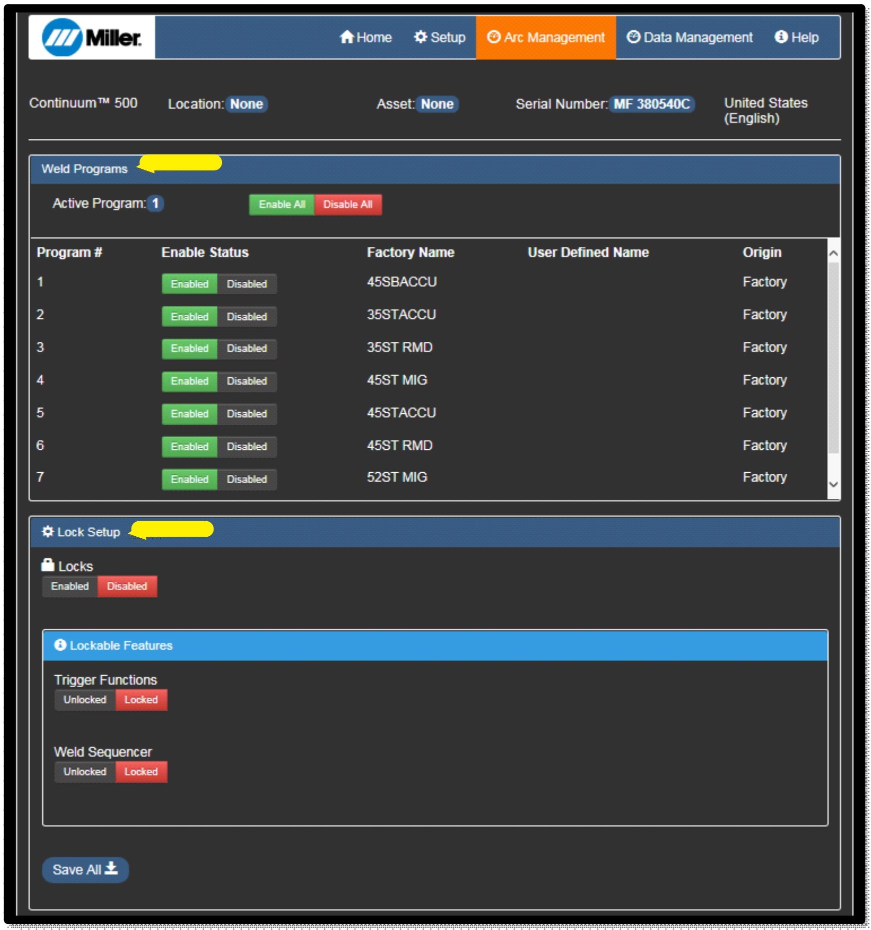


Figure 8-2. Enable the Locks and Lockable Features

8-8. Programming The Parameter Limits

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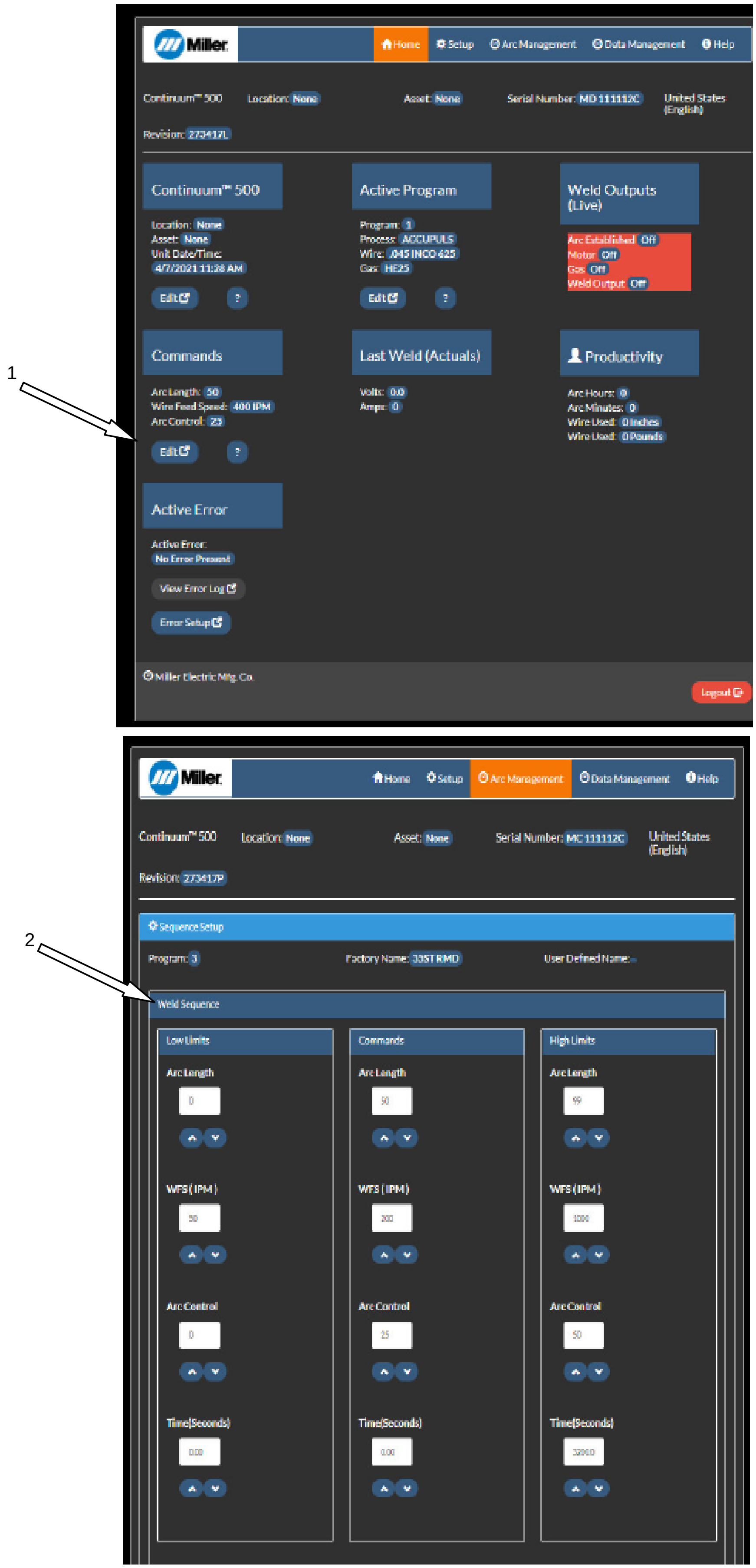

Figure 8-3. Home Page for Parameter Limits

1. On the Home page, under commands, click on the Edit button, which will take you to the Arc Management Program Set-Up screen for the Weld Sequence.
2. Choose the values required for your application for the Low Limit and High Limit for each welding parameter. Keep in mind, you must do this for every active program number.

The Parameter Limits do not take effect until the Global Locks are enabled. This will be explained on the following pages.

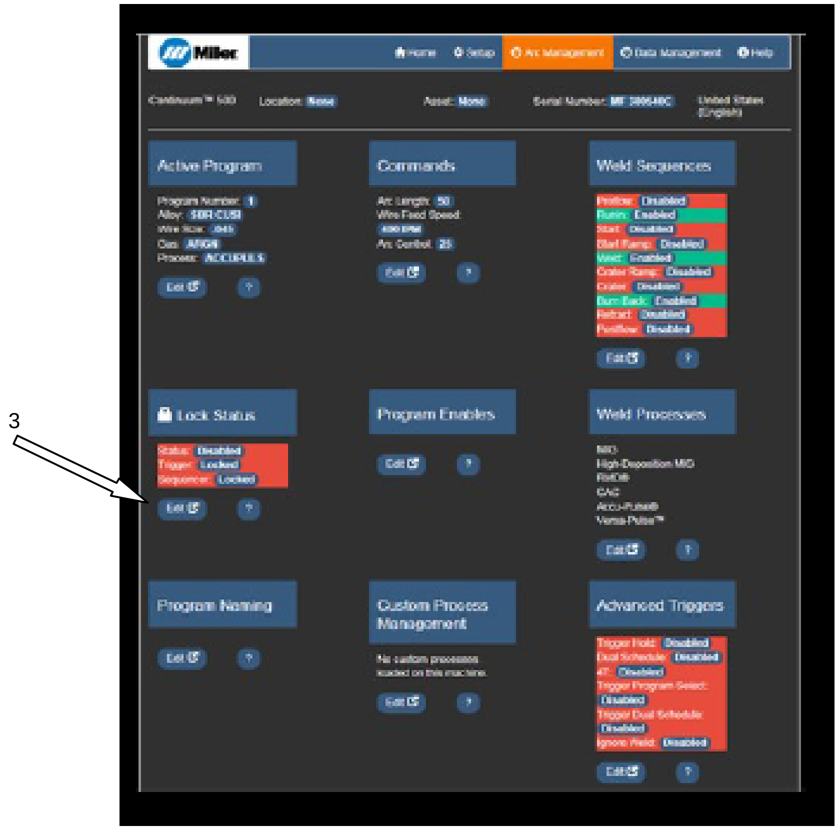

Figure 8-4. Enable the Locks and Lockable Features

3. Enabling the Locks will prevent changes to the Program and Process made from the wire feeder User Interface.

4

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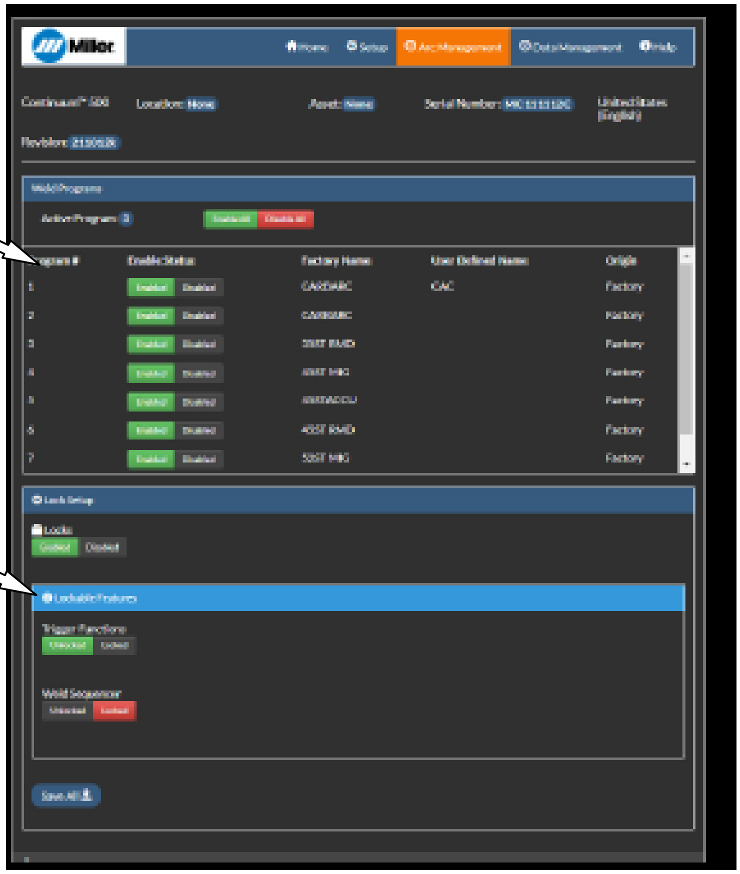
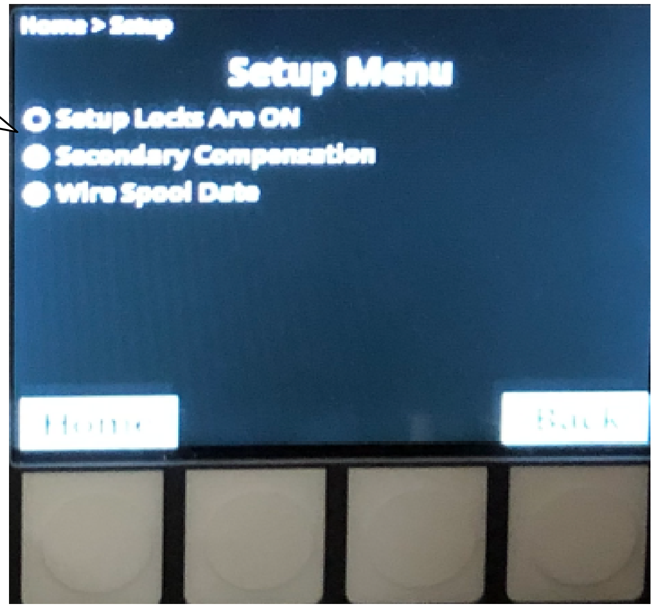


4. You can also disable entire Program numbers, turning them off in the User Interface.
5. Navigate to the Arc Management page, and click on the Edit button under Lock Status. There will be two panels on the Lock Status page, Weld Programs and Lock Set-Up.

6



7

Figure 8-5. Setup Menu

6. User Interface Set-Up Menu with Locks Enabled.

7. Locks disabled

8-9. Creating 98 Programs

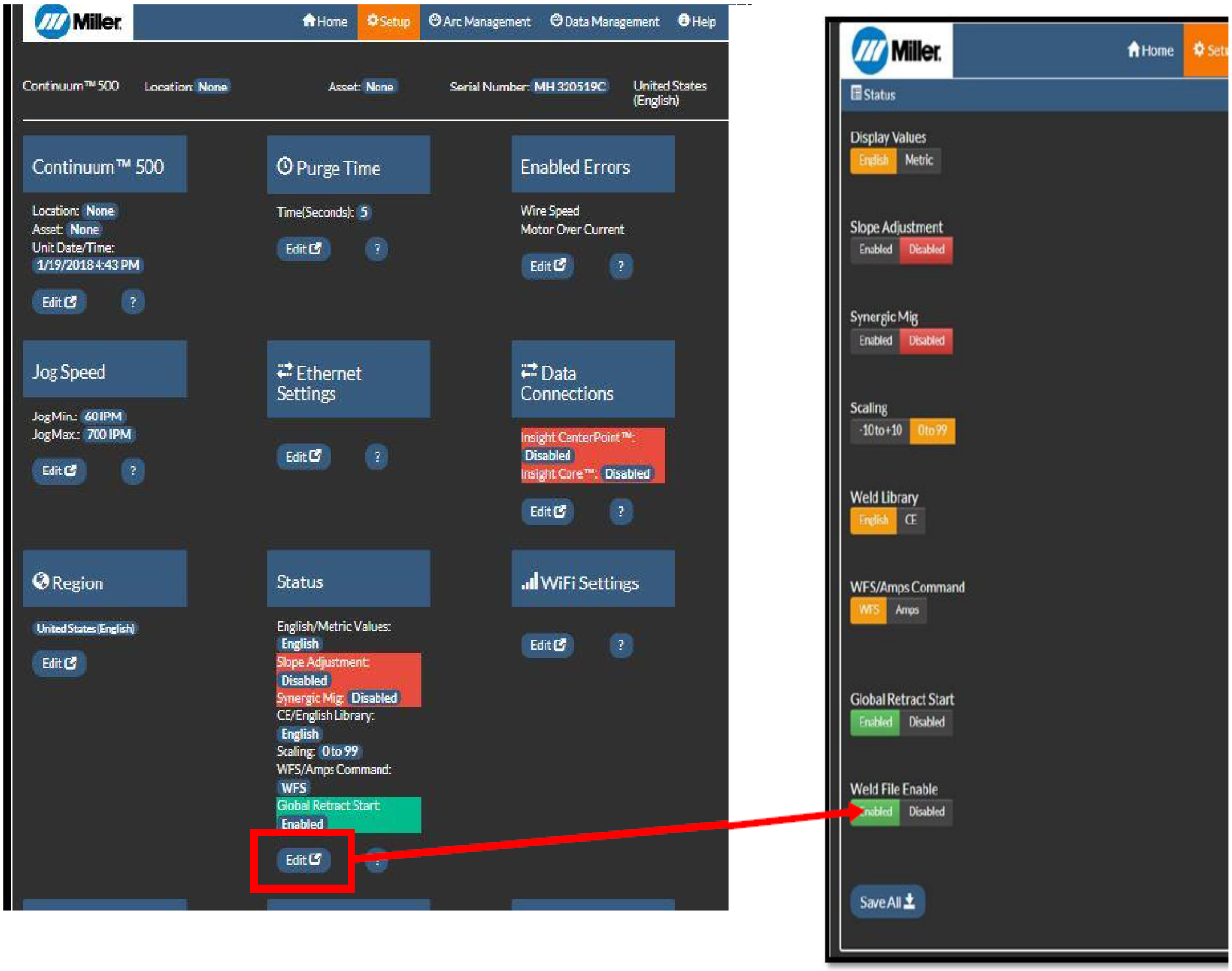


Figure 8-6. Creating 98 Programs

The ability to create 98 programs started with software revision 273417–J.

1. Enter the Status menu and click on the Weld File Enable button and Save All.

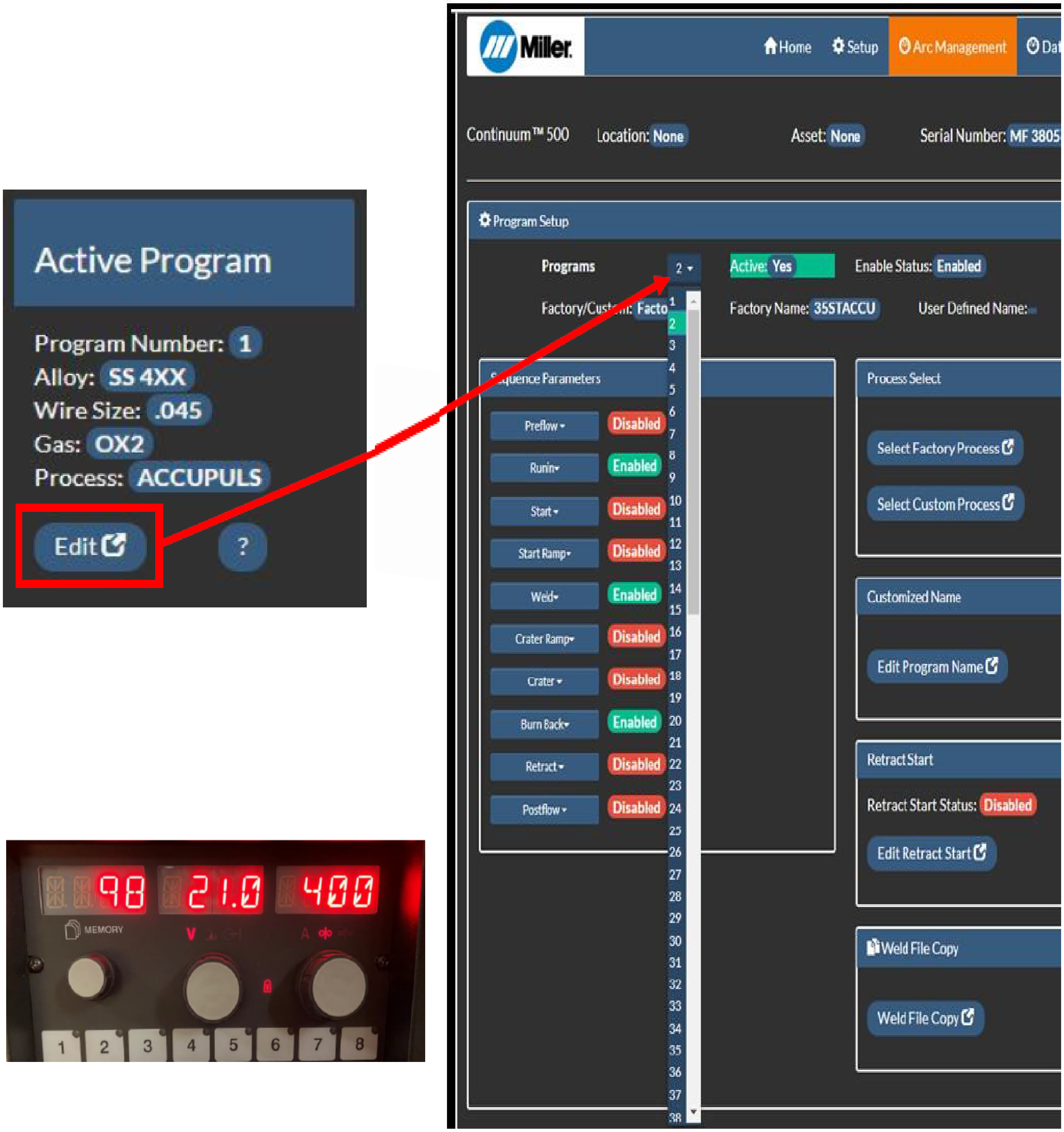


Figure 8-7. Program Set-Up.

Cycle the power on the system, if prompted.

2. To verify on the configuration webpage, navigate to Active Program from the Home Page or Arc Management.
3. Click on Programs and you will see the number of programs expand to 98 possible memory positions on the Program Setup page as well as the Memory positions on the User Interface.

8-10. Custom Naming

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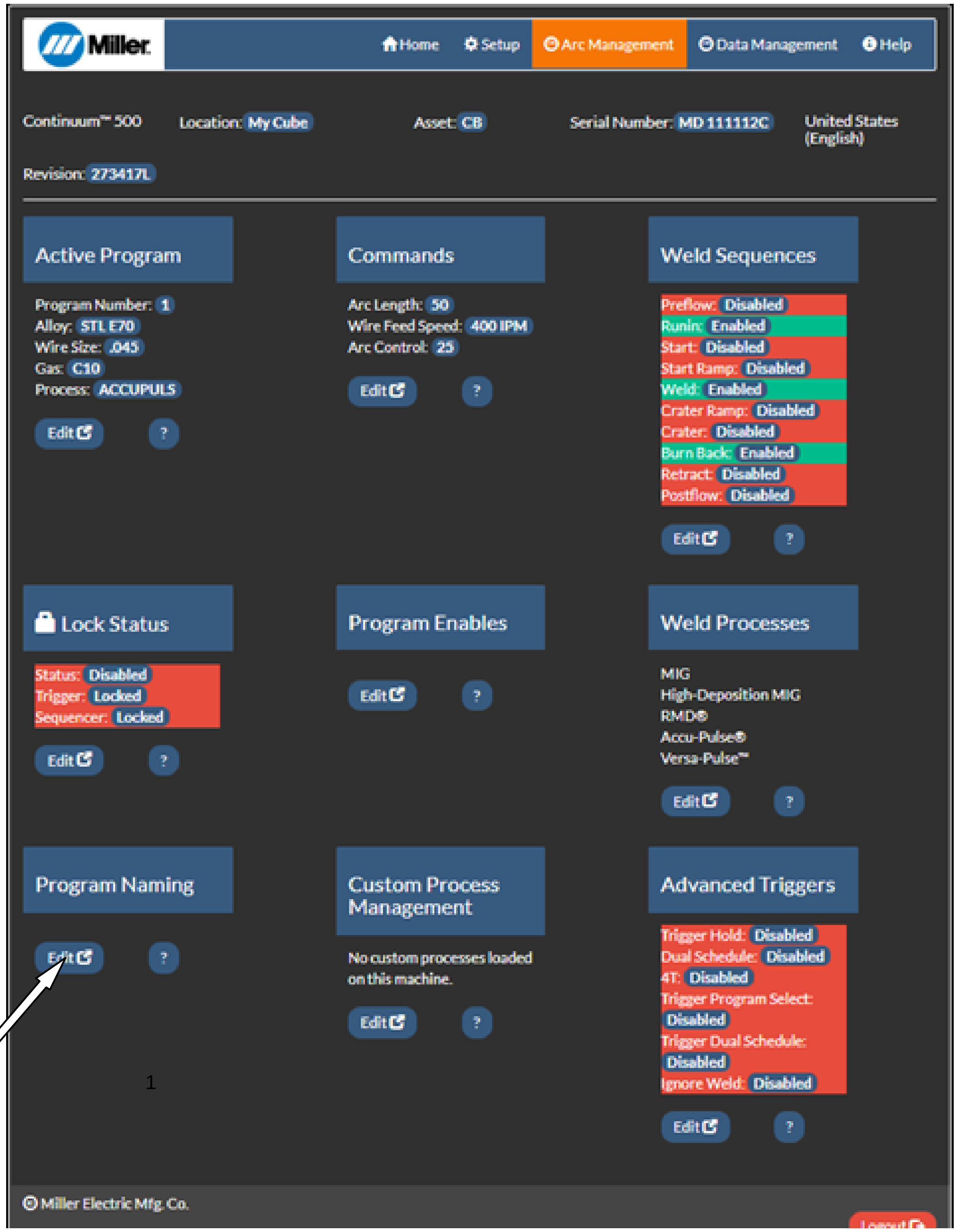

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Figure 8-8. Home Page for Program Naming

1. On the Home page for Arc Management, click on the Edit button located under Program Naming, which will take you to the Program Information screen for the Weld Sequence.

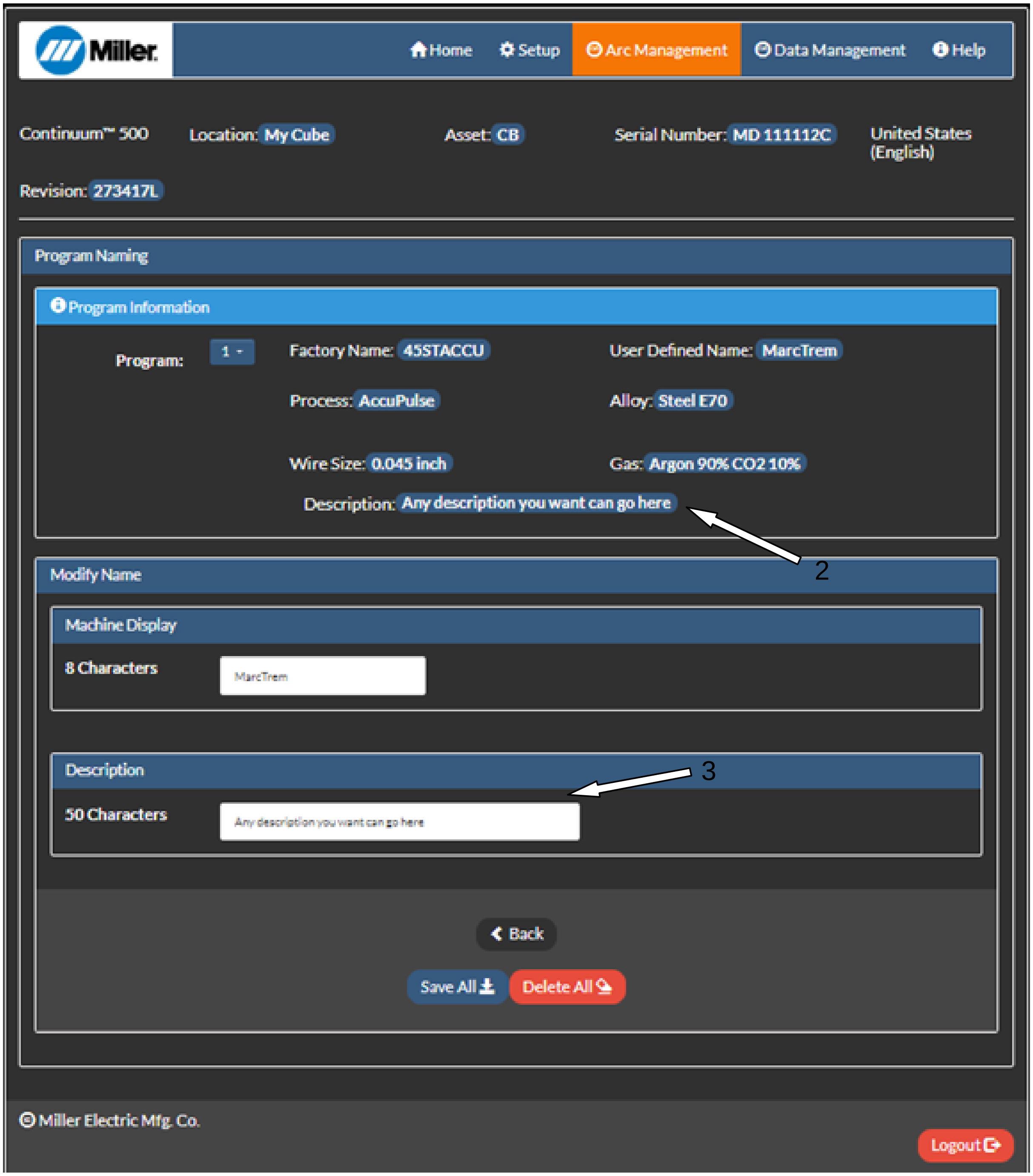


Figure 8-9. Programming name and description.

2. Enter any description that you like can be entered, up to 50 characters.

Figure 8-10. User Interface LCD Display

3. Custom naming located on the LCD display of the User Interface.

8-11. Enabling Cooler To Continuous Mode

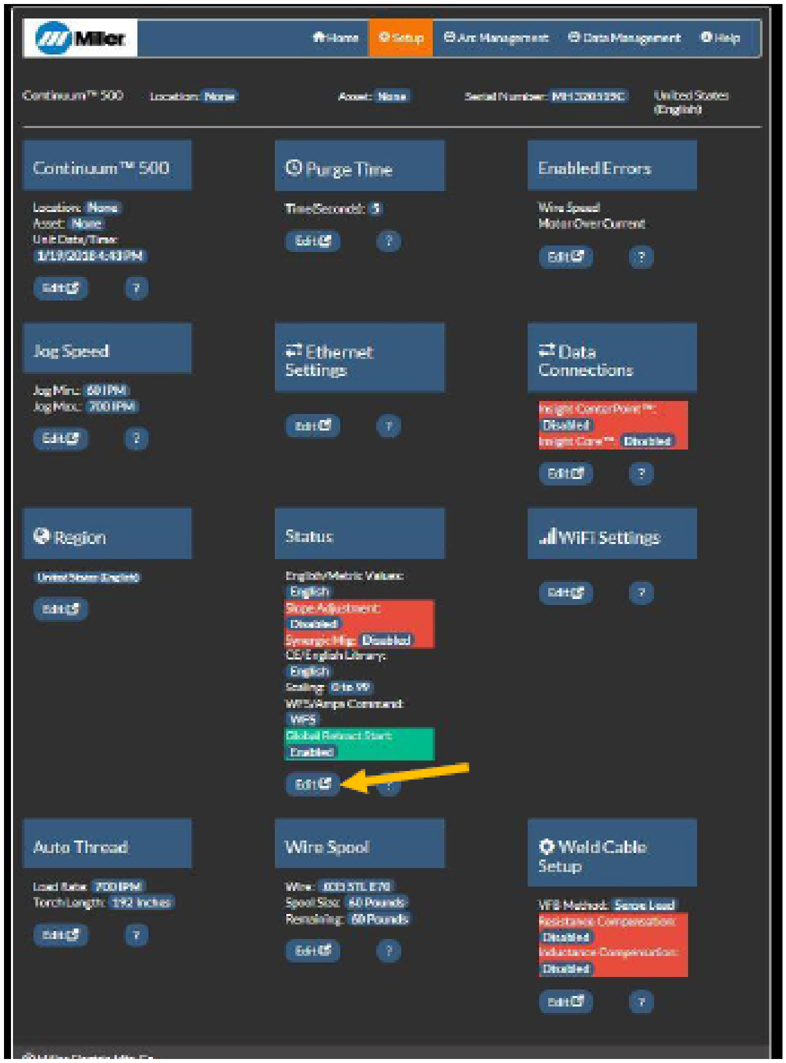


Figure 8-11. Accessing Cooler Status

1. Connect the computer to the Network Port on the back of the welding power source (Auto Continuum is shown.)
2. Navigate to the System menu and then the Network Settings menu in the User Interface

The Parameter Limits do not take effect until the Global Locks are enabled. This will be explained on the following pages.

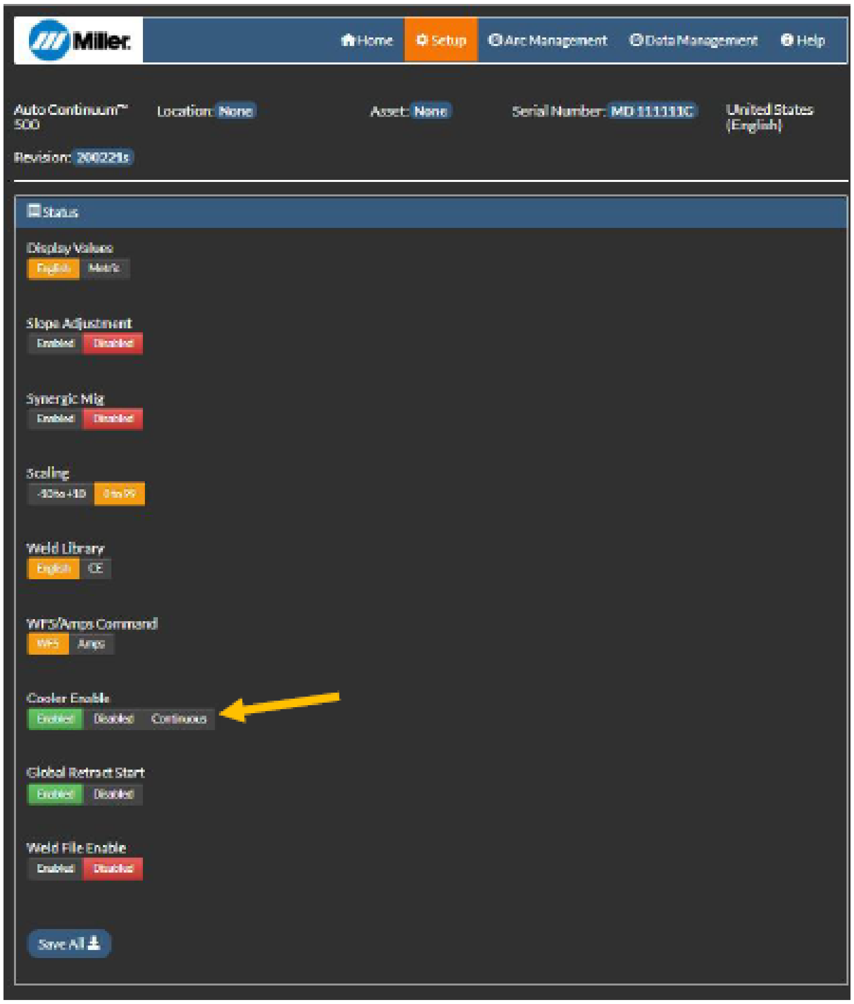
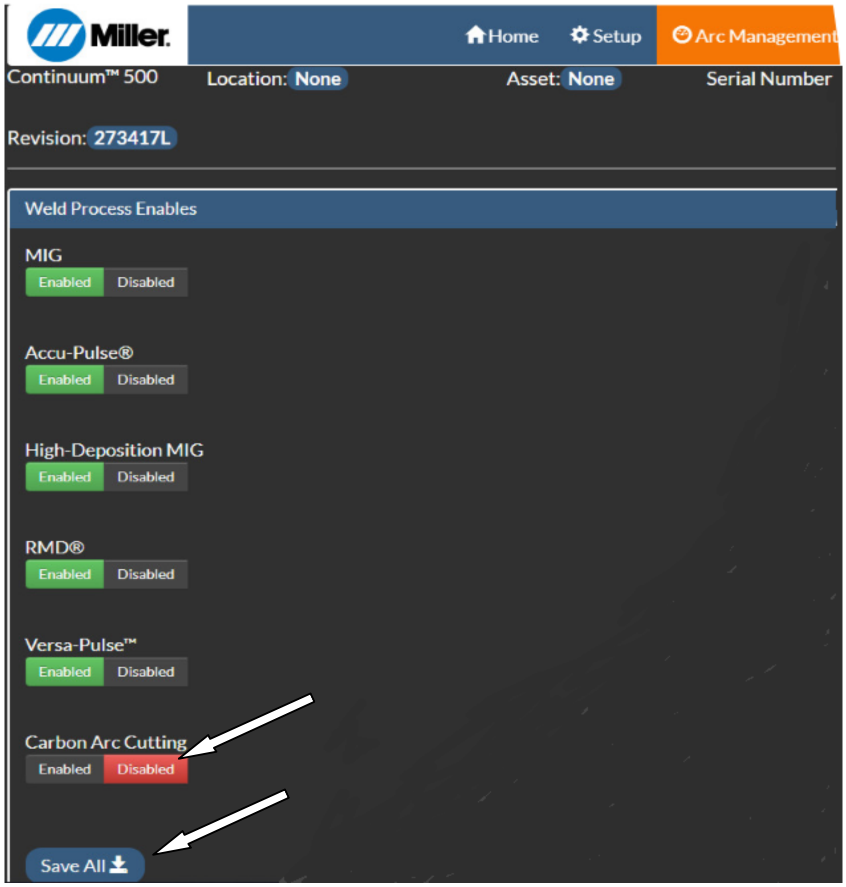


Figure 8-12. Enter the Address and Navigating the Menu.

3. Enter the Ethernet Port IP Address from the Ethernet settings into a web browser.

8-12. Enabling And Loading Carbon Arc Gouging



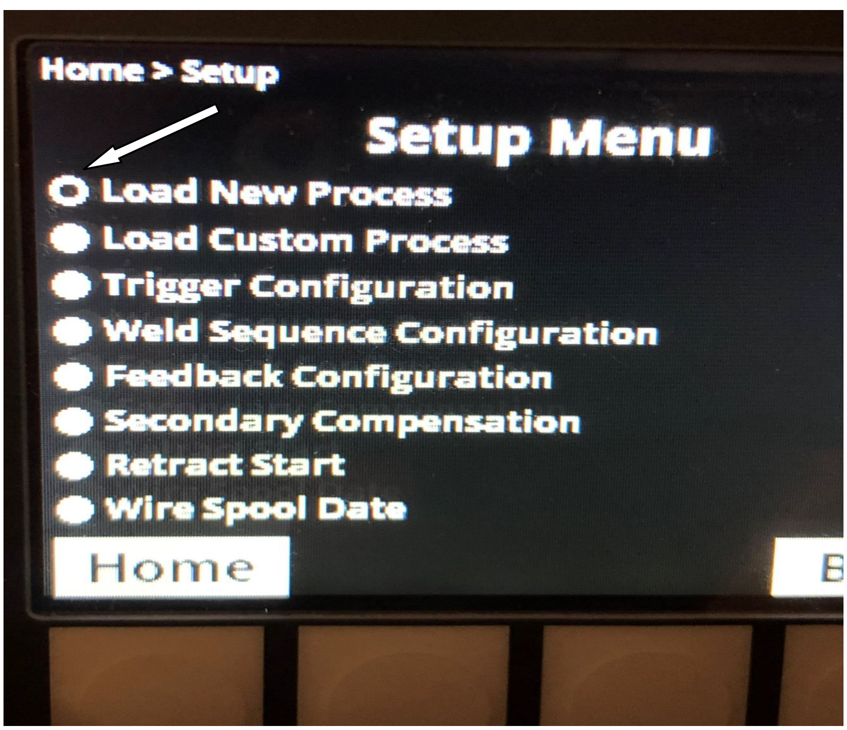
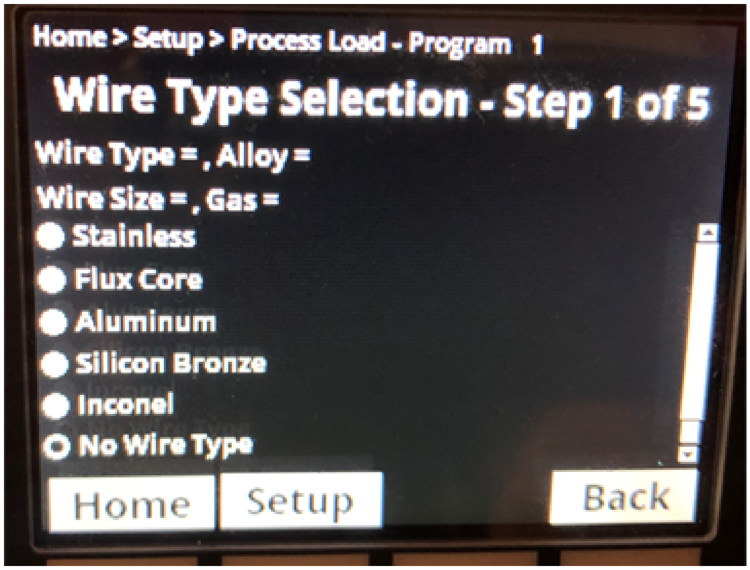


Figure 8-13. Accessing Carbon Arc Gouging Menu

1. Access the configuration pages of the welder and navigate to Weld Processes on the Arc Management page.
2. Enable Carbon Arc Cutting and Save All.
3. On the User Interface of the wire feeder, choose the soft key Setup and choose Load New Process.



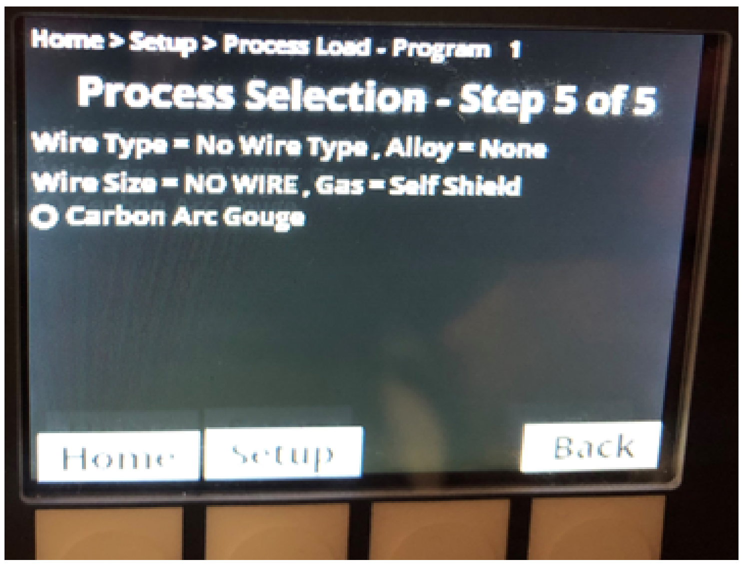


Figure 8-14. Programming for Carbon Arc Cutting.

4. In Step 1 of Process Load, choose No Wire Type

   In Step 2 through Step 4, select the following option.

   Step 2: None

   Step 3: No Wire

   Step 4: Self Shield

   Step 5: Carbon Arc Gouge

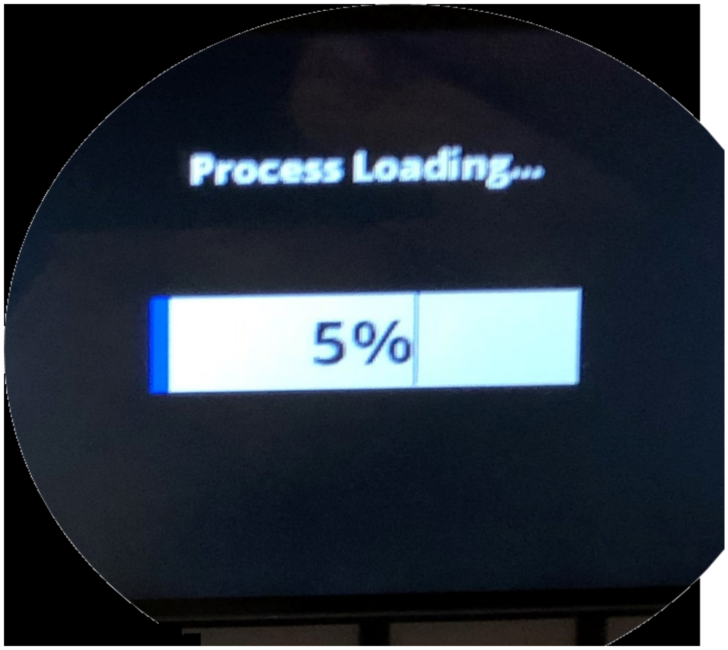


Figure 8-15. Loading The Carbon Arc Gouge Program

5. The Carbon Arc Gouge process will load to the program position chosen.
6. After the Carbon Arc Gouge Program is loaded, the Carbon Arc Gouge output is turned on and off through the User Interface by pressing the selection knob to the right of the LCD screen. The arrows on the display point to this.
7. Adjust the Amperage using the Amperage Adjustment knob under the Amp display.

8-13. Auto-Continuum Secondary Compensation

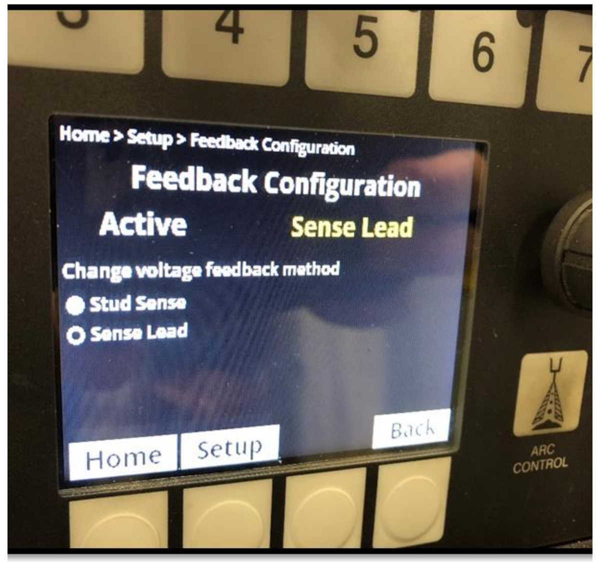
Auto-Continuum Secondary Compensation

- The Secondary Compensation can be used to check the integrity of the weld circuit, condition of the weld cables, quality of the air or liquid cooled MIG torch, or poor fit up in the fixturing.

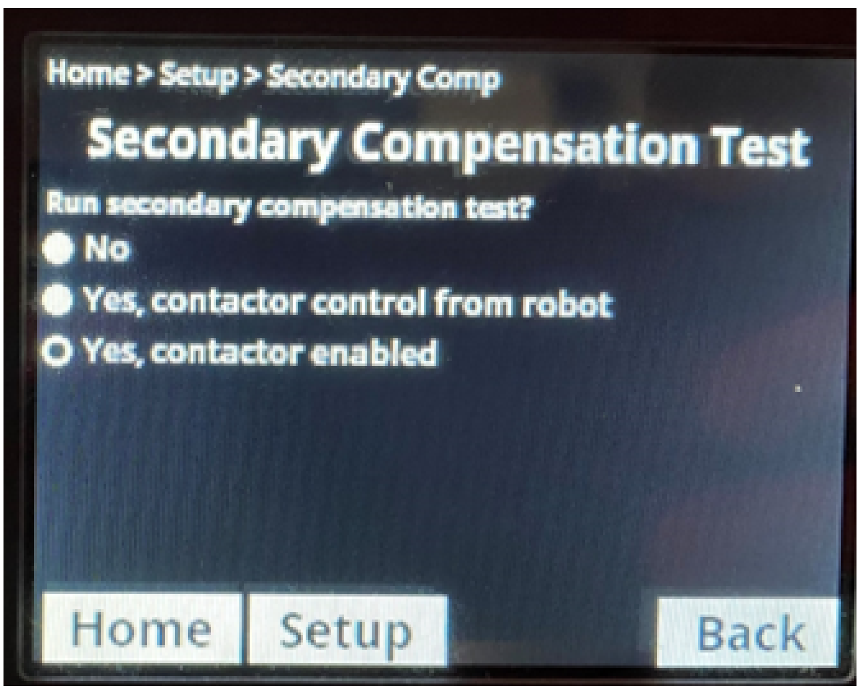
- Extra resistance or added inductance may result in a loss of weld voltage.

Secondary Compensation Steps

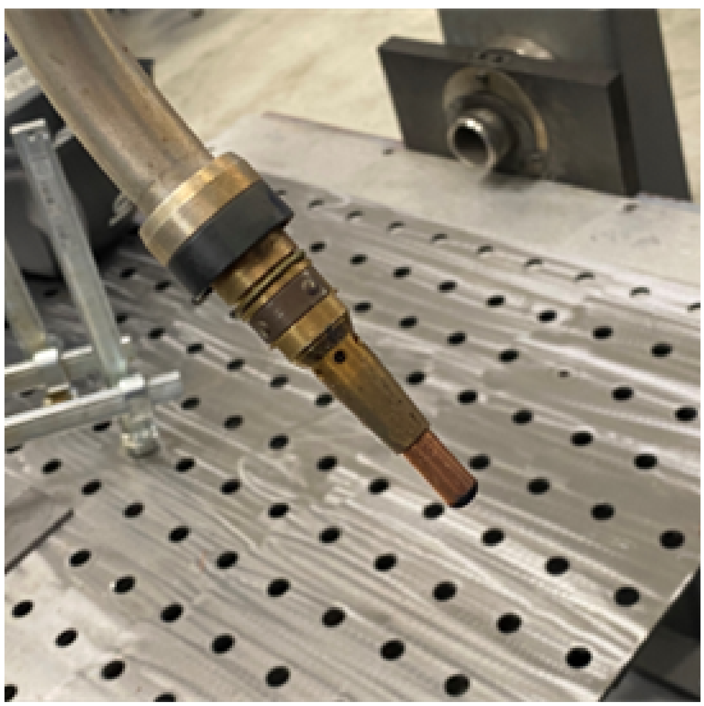
1. Enable the Feedback Configuration for V-Sense. You MUST attach the Volt-Sense cable from the Auto-Continuum to the work. This will calculate the cable resistance and torch resistance.

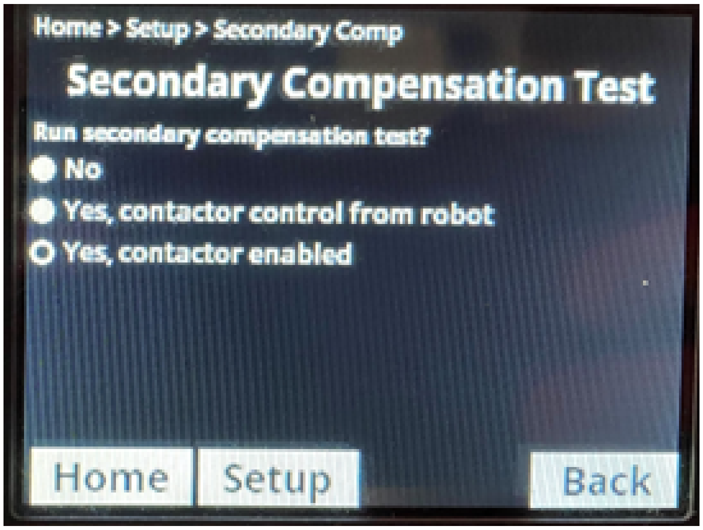


1

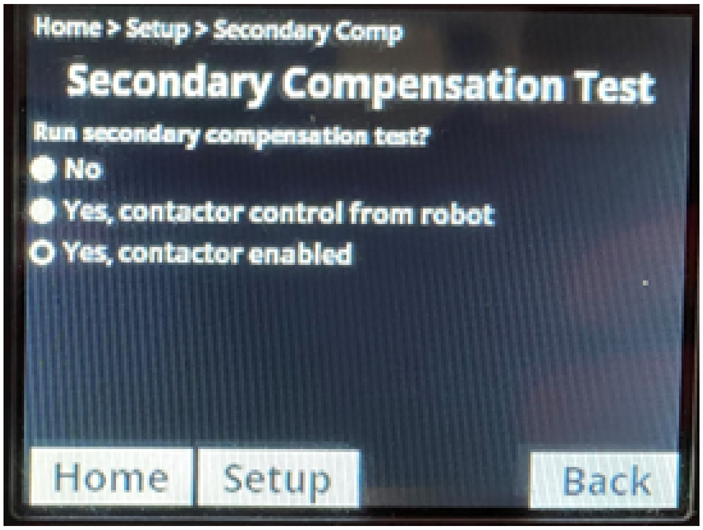



Home > Setup
Setup Menu
Load New Process
Trigger Configuration
Weld Sequence Configuration
Feedback Configuration
Secondary Compensation
Retract Start
Home
Back

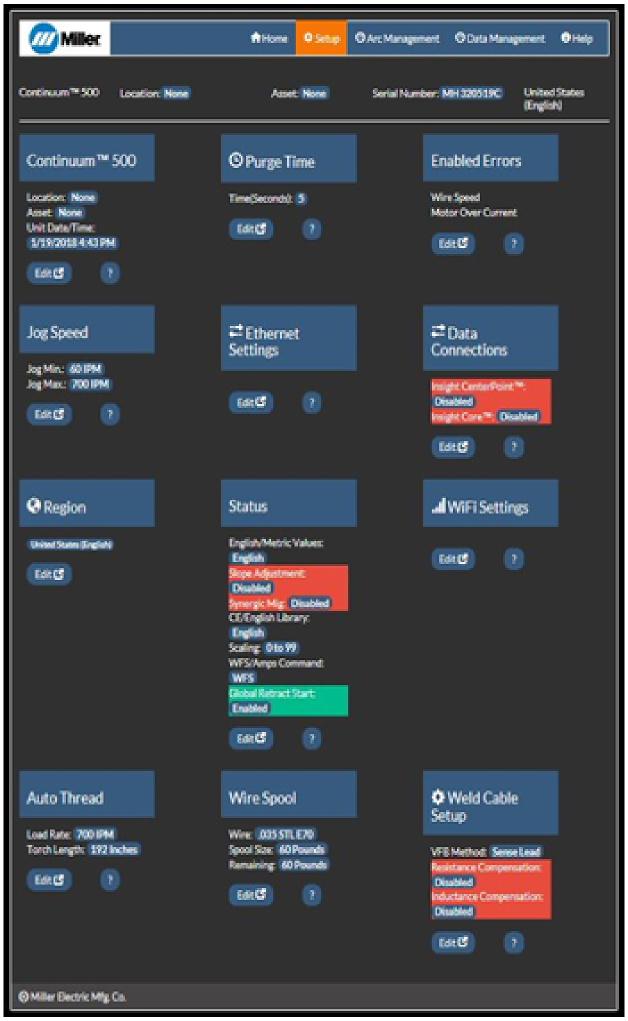


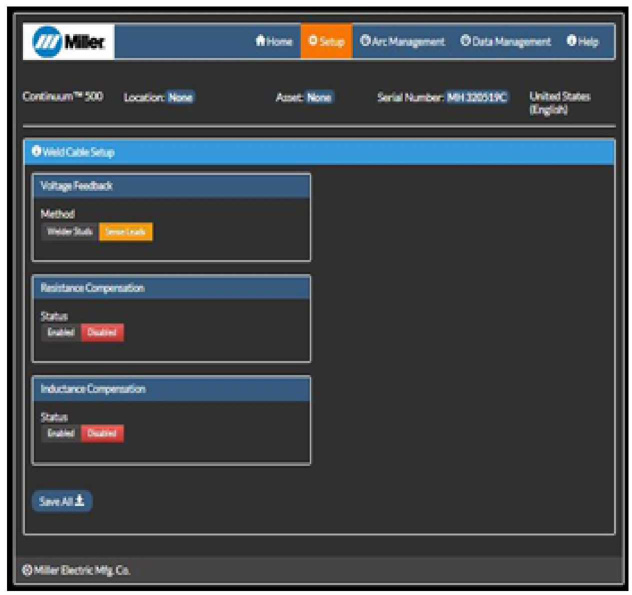

Home > Setup > Secondary Comp
Secondary Compensation Test
Run secondary compensation test?
No
Yes, contactor control from robot
Yes, contactor enabled
Home
Setup
Back


TEST COMPLETE


Home > Setup > Secondary Comp
Secondary Compensation Test
Run secondary compensation test?
No
Yes, contactor control from robot
Yes, contactor enabled
Home
Setup
Back

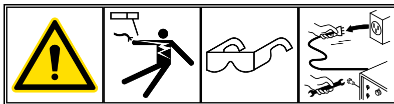

TEST COMPLETE


Miller
Home
Setup
Arc Management
Data Management
Help
Continuum™ 500
Location: None
Asset: None
Serial Number: MH 320519C
United States (English)
Continuum™ 500
Location: None
Asset: None
Unit Date/Time:
1/19/2018 4:43 PM
Edit
Purge Time
Time(Seconds): 5
Edit
Enabled Errors
Wire Speed
Motor Over Current
Edit
Jog Speed
Jog Min.: 60 IPM
Jog Max.: 700 IPM
Edit
Ethernet Settings
Edit
Data Connections
Insight CenterPoint™: Disabled
Insight Core™: Disabled
Edit
Region
United States (English)
Edit
Status
English/Metric Values: English
Slope Adjustment: Disabled
Synergic Mig: Disabled
CE/English Library: English
Scaling: 0 to 99
WFS/Amps Command: WFS
Global Retract Start: Enabled
Edit
WiFi Settings
Edit
Auto Thread
Load Rate: 700 IPM
Torch Length: 192 Inches
Edit
Wire Spool
Wire: .035 STL E70
Spool Size: 60 Pounds
Remaining: 60 Pounds
Edit
Weld Cable Setup
VFB Method: Sense Lead
Resistance Compensation: Disabled
Inductance Compensation: Disabled
Edit
Miller Electric Mfg. Co.


Miller
Home
Setup
Arc Management
Data Management
Help
Continuum™ 500
Location: None
Asset: None
Serial Number: MH 320519C
United States (English)
Weld Cable Setup
Voltage Feedback
Method
Welder Studs
Sense Leads
Resistance Compensation
Status
Enabled
Disabled
Inductance Compensation
Status
Enabled
Disabled
Save All
Miller Electric Mfg. Co.

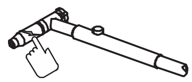
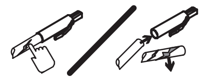
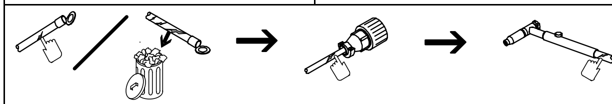
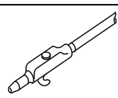
SECTION 9 – MAINTENANCE AND TROUBLESHOOTING

9-1. Routine Maintenance

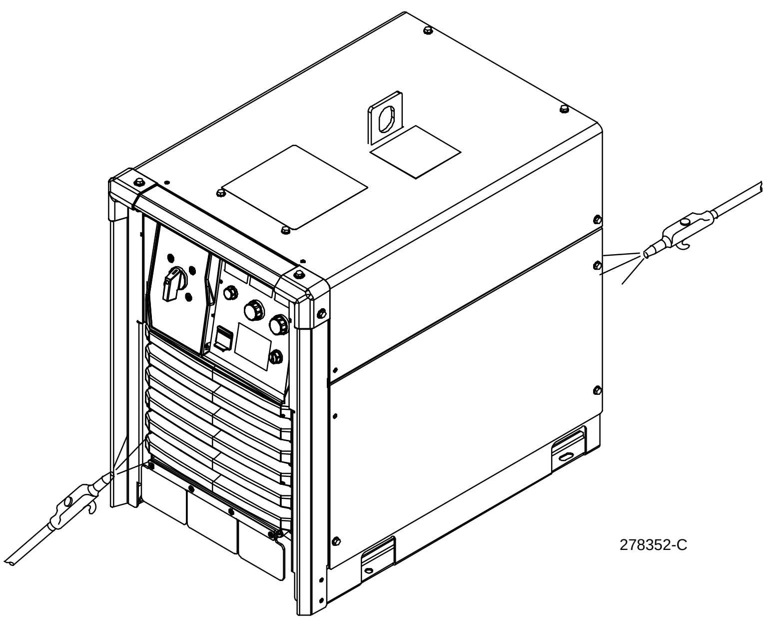


Disconnect power before maintaining.

Maintain more often during severe conditions.

	✓ = Check	◇ = Change	❍ = Clean	✰ = Replace
Every **3** Months	 ✓ ◇ ✰Damaged Or Unreadable Labels	 ✓ ✰ Cracked Torch Body	 ✓ ✰ Cracked Cables	
	 ✓ ✰ Cracked Cables And Cords		 ✓ ❍ Clean And Tighten Weld Connections	
Every **6** Months	 ✓ ❍ Blow Out Inside			

9-2. Blowing Out Inside Of Unit



Do not remove case when blowing out inside of unit.

To blow out unit, direct airflow through front and rear louvers.

9-3. Setting Up Load Bank Mode

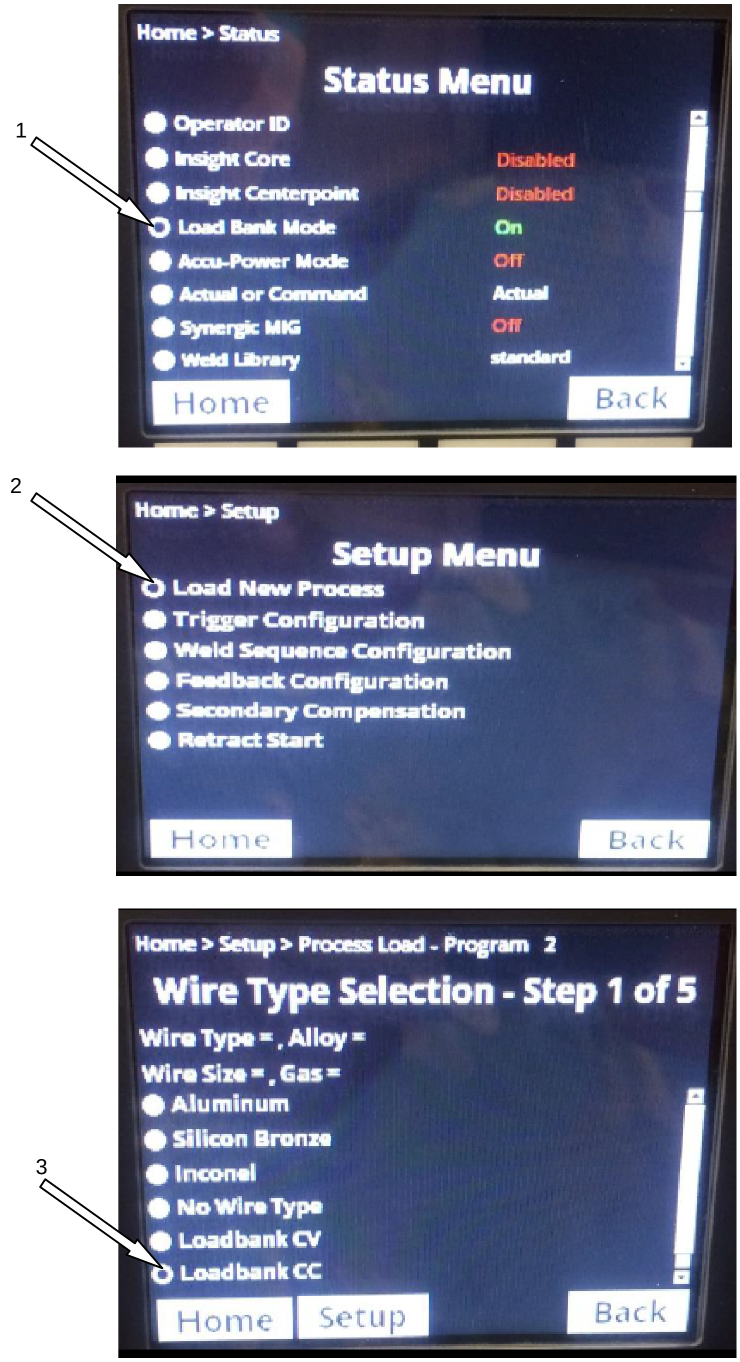
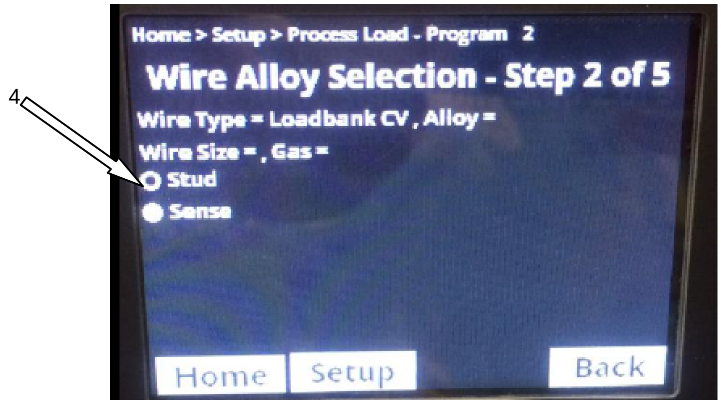
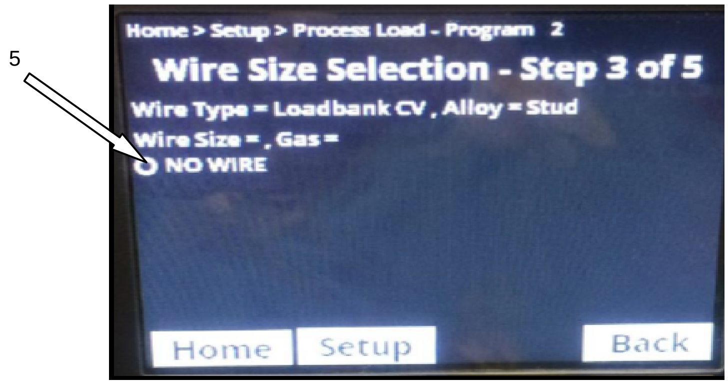


Figure 9-1. Selecting Location for various menus

1. Enable Load Bank Mode in the Status Menu.
2. Choose a memory location and load new process.
3. Select the type of output: **CV-Constant Voltage** or **CC-Constant Current**





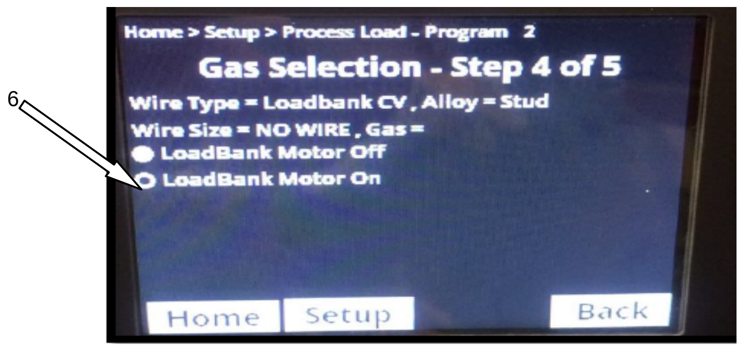


Figure 9-2. Continue with Menu selections for Load Bank Mode.

4. Choose the type of voltage feedback, Stud is preferred for test purposes.
5. Select No Wire for this screen.
6. Determine if you would like the Wire Feed Motor to operate while the unit is under load.

7

8

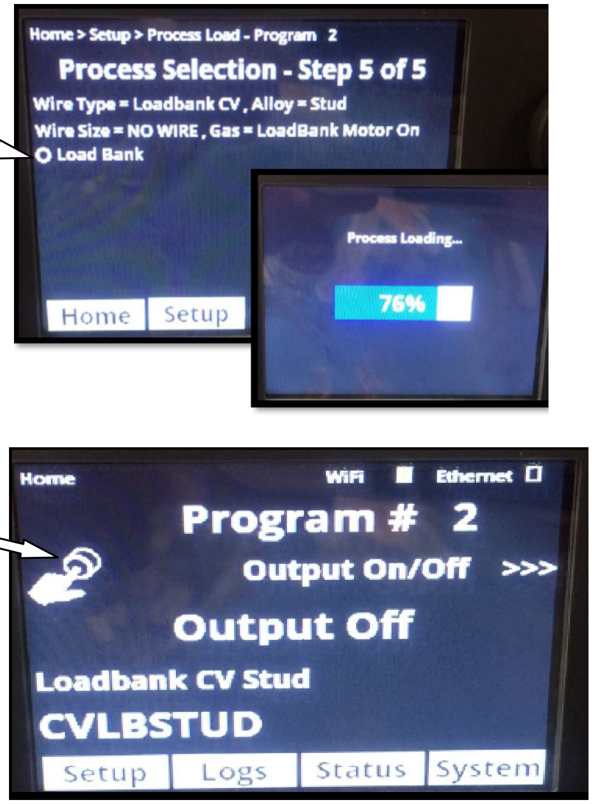
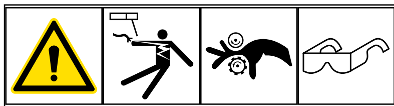
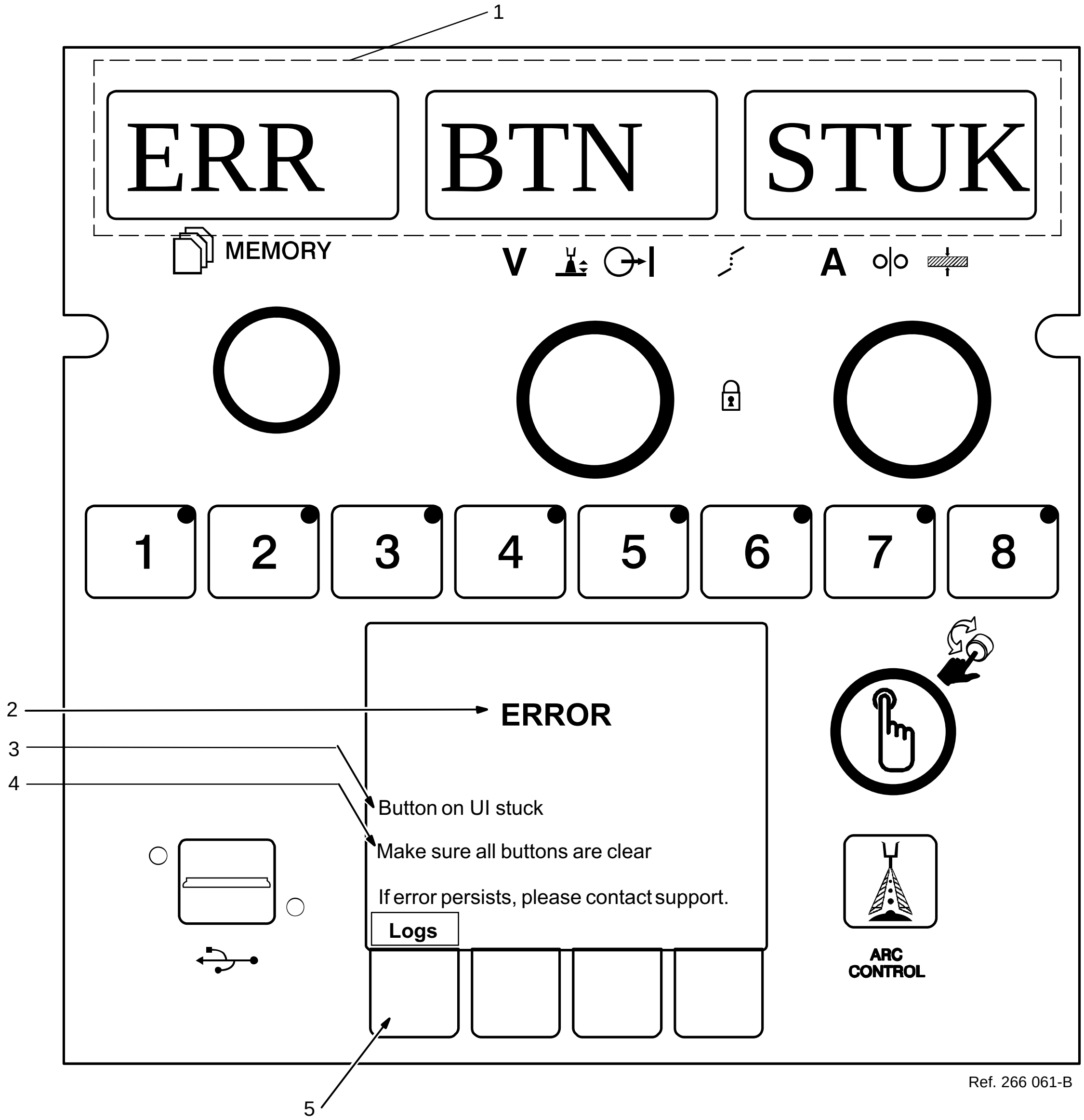


Figure 9-3. Continuation of loading Load Bank Program

7. Select Load Bank for this screen.
8. After the Load Bank program is loaded, the output is turned on and off through the User Interface by pressing the selection knob to the right of the LCD screen. The arrows on the display point to this.

9-4. User Interface Error Information





Ref. 266 061-B

1 LED Display Message

Displays a short message describing the error or where the error is occurring. Some messages may represent multiple different errors. For more specific error information see the LCD Display Message.

2 Message Type

There are three types of messages that will be displayed.

ERROR- The system has experienced a fault that must be addressed before continuing use.

MESSAGE- No fault has occurred, but some action must be taken before resuming use of the welding system. For instance, input power may need to be cycled after loading new software.

WARNING- There is something that should be addressed before continuing. For instance, the volt sense lead connection has been lost.

3 LCD Display Message

Provides more information about the message being displayed.

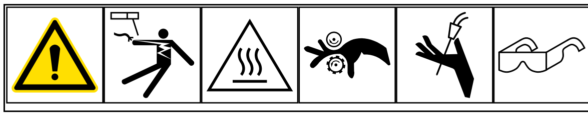
4 LCD Solution Message

Describes the action needed to clear the message or fault being experienced.

5 Logs Soft Key

Pressing the Logs soft key will take the user to the Logs Menu. From the Logs Menu it is possible to access the Error Log. The LCD Error Log Messages are stored in the Error Log.

9-5. Error Code Troubleshooting Description And Tables



Red LED Display	LCD Display Message	Description	Probable Cause(s)	Potential Solution(s)
BOOT ERROR or BOOT FAIL	System Booting Status bar stopped at 1%, 2%, 3%, or 4%	One of the components in the system is not connected to the network or the rest of the system.	Wires in the interconnecting cable are disconnected. There is a pin or socket in one of the communication receptacles that is broken. Internal circuit board failure.	Replace the interconnecting cable from the power source to the wire feeder, or between the ROI and wire drive, if it is a Swingarc system. Replace the power source, ROI, wire drive, or wire feeder to have serviced.
No Display	LCD is blank	Feeder is not powering up, white power light is illuminated and power source contactor is energized.	Bad communication cable Internal circuit board failure.	Replace the communication cable between power source and wire feeder. Replace the power source, ROI, wire drive, or wire feeder to have serviced
ERR THERM or ERR OVERTEMP	ERROR Thermal System Problem Welding Power Source has overheated Allow unit to cool Cycle Power on the Power Source	The power source is over heating. The overtemp light is on constant.	The fan louvers/wind tunnel openings are blocked The fan motors are not turning on or cannot rotate. The heat sinks are dirty.	Ensure that there is a minimum of 18 inches distance between power source and other objects. The power source must be serviced/repaired before proper operation. Blow out inside of wind tunnel per the preventative maintenance section.
CYCLE POWER	The unit requires a power cycle. Cycle power on power source	This message may display if the settings changed in the configuration of the system.		Cycle power on the power source
ERR COMMS	ERROR System Communication loss Cycle power on the power source	User Interface is not connected to the rest of the system.	Bad communication cable between the feeder and power source.	Replace interconnecting cable.
ERR COMMS	ERROR Internal comms loss:	There are many communication errors in the system indicating where the problem may be. The communication could be lost from the power source, wire feeder, ROI, or wire drive. The detailed message in the LCD screen will give the proper details for the servicing technician.	Poor connection in the communication cable between the wire feeder and the power source. Open connection in the interconnecting cable on the back of the power source, wire feeder, ROI, or wire drive. Database in the system may be corrupted.	Replace the communication cable. The power source, ROI, wire drive, or wire feeder must be serviced/repaired before proper operation. Reinstall the system software.
ERR WIRSTUK	ERROR Wire Stuck Error	Wire stuck to work piece at the end of the weld.		.Break/Cut away from the work piece.

Red LED Display	LCD Display Message	Description	Probable Cause(s)	Potential Solution(s)
ERR ARC	ERROR Arc error occurred. Check wire feeder and power source.	The arc produced was not a valid arc, weld voltage was too high or too low for the parameters set on the User Interface.	Broken or disconnected weld cable. Wire delivery is too slow, arc length too long.	Repair or replace weld cable. Repair or replace work clamp or cable connection. Check/Replace gun liner. Check drive roll pressure is properly set for the application.
ERR TRG STUK	ERROR Trigger held during power up. Release trigger and press button to clear error.	Trigger Stuck error occurs if the trigger is held when the feeder is powered up. The error may be cleared by releasing the trigger.	Trigger or trigger wires in the MIG gun may be shorted together.	Repair or replace the MIG gun.
WRN SOFTWARE	WARNING Incompatible software detected. Sys SW Rev Mismatch Error	The software in another component in the system is creating a software mismatch. The software in the wire feeder and power source are at different revisions.	A component in the system is newer than the rest of the components. A different wire feeder and power source were placed together for the first time.	Update system software
ERR INPUT PWR	ERROR Must use three phase primary power.	The system is experiencing a single phase condition, missing a primary leg.	There is a poor connection in the three phase plug or disconnect. There is an open fuse or circuit breaker.	Contact a qualified electrician to verify the primary power and possible solutions.
ERR INPUT PWR	ERROR The primary input voltage is too high.	Customer supplied primary voltage is too high.		Contact a qualified electrician to verify the primary power and possible solutions.
ERR INPUT PWR	ERROR The primary input voltage is too low. Check the primary connections.	Customer supplied primary voltage is too low.	Poor connection in the three phase plug or disconnect.	Contact a qualified electrician to verify the primary power and possible solutions.
ERR BTN STUK	ERROR Button on UI stuck Make sure all buttons are clear.		Button on UI was held too long. Membrane button on the overlay is damaged.	The wire feeder or ROI must be serviced/repaired before proper operation.
ERR ARC STRT	ERROR Trigger held too long without arc.	The weld output was active too long before striking an arc.	The drive rolls were open and the wire did not move. There is no weld cable between the feeder and power source.	Ensure proper drive roll pressure is applied. Ensure the lower drive roll carrier support plate is installed. Install weld cable.
ERR REL TRIG	ERROR Trigger was held too long. Release trigger and press button to clear error.	Release Trigger error occurs if the user has the trigger held for more than two minutes without striking an arc or if the user holds the trigger past the postflow phase in a timed weld. This may also happen if the Jog button is depressed too long.	The trigger in the MIG gun is shorted or stuck in a closed position.	Repair or replace the MIG gun.

Red LED Display	LCD Display Message	Description	Probable Cause(s)	Potential Solution(s)
WRN OVERAVER	WARNING Duty cycle exceeded System will be ready to weld shortly.	The duty cycle of the power source has been exceeded. The power source shut down to allow itself to cool down. The error will be active for a period of time.	The welding output was too high for too long of a period of time.	Allow the welder to be powered on and sitting idle until the error clears, typically for a period of 15 minutes.
ERR AUX PWR	ERROR Aux power problem Cycle power on the power source.	The 115 VAC aux power circuit board is experiencing a problem.	A device plugged into the 115 VAC receptacle is too much load. Auxiliary Power failure	Remove the device from the 115 VAC receptacle. The power source must be serviced/repaired before proper operation.
ERR THERM	ERROR Thermal System Problem Cycle power on the power source.	The system is experiencing overheating on the primary or secondary heat sinks. Also see ERR THERM 1,2, or 3 messages.	The fans are not turning on. The fan enable signal from the Arc Control board (PC4) is not present.	Check the fan output from the Control Power Supply board (PC13). Ensure the Fan Enable wires are terminated properly. Replace (PC4)
ERR PWR SRC	ERROR Boost CS1, CS2, CS3, or CS4 current error Cycle power on power source.	Primary current transducer sensor feedback is reporting too high or too low amperage draw of the input primary power.	Primary AC input may be too high or too low on one or more legs.	Contact a qualified electrician to verify the primary power and possible solutions.
WRN VSNS LOS	WARNING V-sense Feedback Warning Lost volt sense lead feedback.	The power source is not receiving the voltage feedback input from the internal connections or the external volt sense lead.	The feedback configuration setting is configured for V-Sense and the cable is broken or disconnected. Inspect the volt sense lead for breaks. The interconnecting cable has an open wire carrying the voltage feedback input.	Replace the Volt Sense lead.. Replace the Volt Sense lead. Replace the Interconnecting cable.
ERR COOL TEMP	ERROR Cooler Over Temperature	The temperature sensor in the cooler is reporting the coolant temperature is too high and shuts down the system until the temperature is low enough.	There is blockage in the torch or in the torch coolant line. The welding application is exceeding the cooler capabilities.	Replace the torch or clear the blockage in the cooler lines. Reduce the welding output.
ERR COOL FLOW	ERROR Cooler Low Flow	The coolant lines do not have coolant flowing.	The air in the coolant lines have not yet been purged. The filter is clogged with debris. There is blockage in the torch or the torch coolant line.	Turn on the weld output by Load Bank mode several times and clear the error each time until the coolant gets into the flow indicator and activates the flow switch. Follow the coolant maintenance instructions in the Cooler OM. Replace the torch or clear the blockage in the coolant lines.

SECTION 10 – INSIGHT CORE INSTALLATION

10-1. License Agreement

You have acquired a device (Insight Core) which includes software licensed by Miller Electric Mfg. Co. Such software products, as well as associated media, printed materials, and "online" or electronic documentation for such software are protected by international intellectual property laws and treaties. The software is licensed, not sold. All rights reserved.

10-2. Downloading Connectivity Survey

In order to ensure proper connectivity, visit the Insight Core Download page at https://insight.millerwelds.com/download

Download and print the Network Connectivity Survey Form.

Ask your corporate IT department to complete the survey.

10-3. Obtaining System IP Address

☞ *The factory default IP address of all Continuum systems is 169.254.0.2.*

☞ *Ethernet and WIFI connections each have their own unique IP addresses.*

☞ *The power source must be on the same Subnet as the computer used to access the Cpntinuum web pages.*

1. Connect your Continuum welding system per the instructions in the power source owner's manual.
2. Connect an Ethernet cable between your computer and power source.
3. Turn the welding power source on.
4. From the Feeder (or Pendant) main menu screen, go to the Ethernet settings screen. Using the scroll knob, select **System,**then **Network Settings,** then **Ethernet Settings.** Note the IP address.

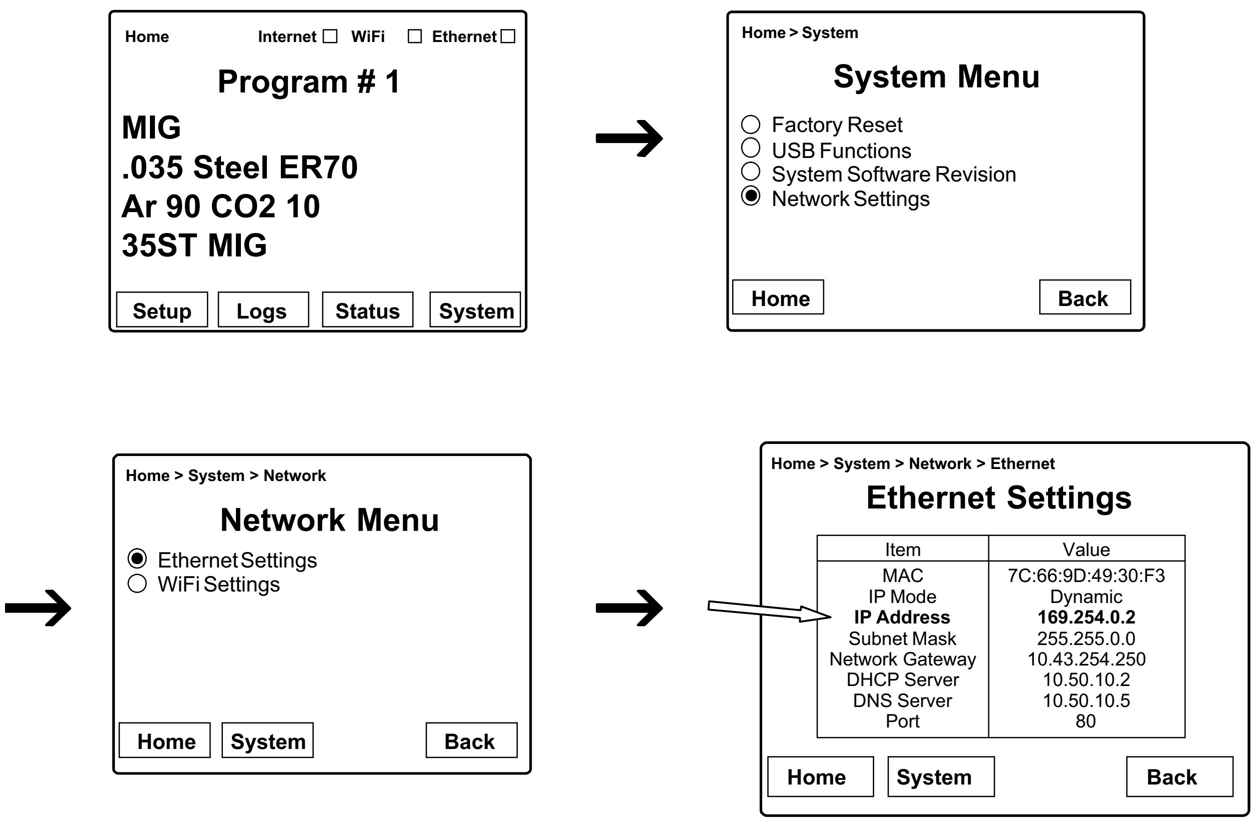


Figure 10-1. Obtaining IP Address

10-4. Enabling Insight Core Through The Continuum Web Pages

1. Connect a computer to the welding power source with an Ethernet cable and open a browser.
2. Enter the system IP address (see section10–3) into the address bar.The default IP address is 169.254.0.2.
3. Press **Enter**.

Figure 10-2. Connecting To Continuum Web Page

4. The Continuum system web page appears. On the top of the page, click **Setup.**

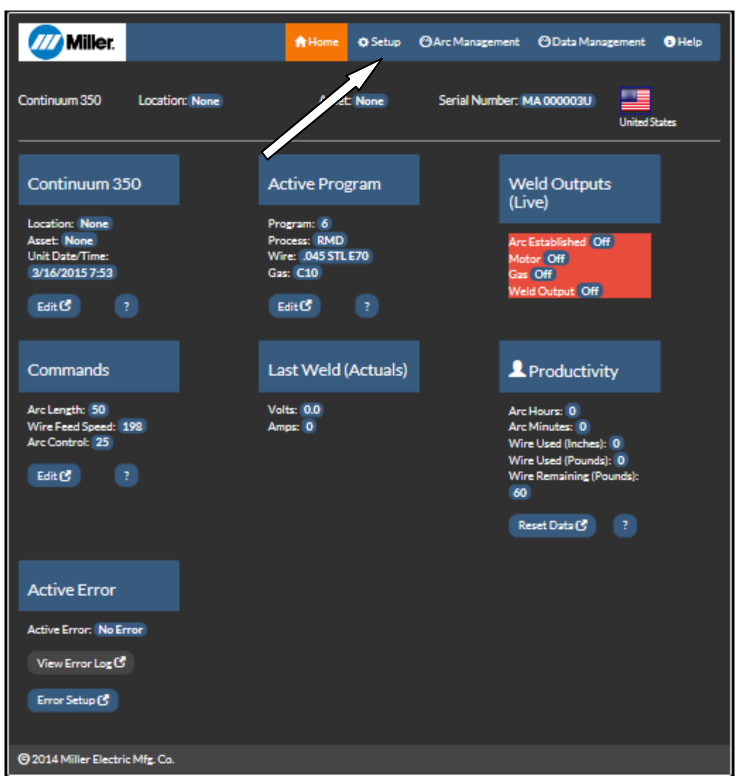


Figure 10-3. Continuum Web Page

5. Under **Data Connections,** click **Edit.**

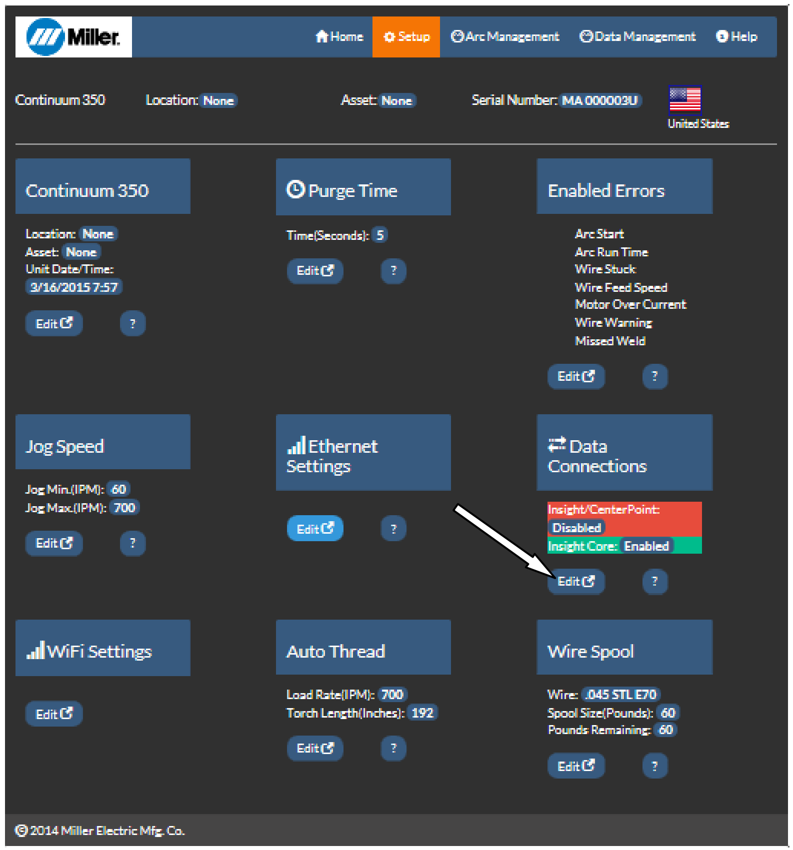


Figure 10-4. Setup Page

6. Enable or disable Insight Core by clicking your selection. After your selection is made, click **Save All.**
7. Cycle the machine power by turning the machine off, waiting until the Power On light goes off, and turning the power on again.

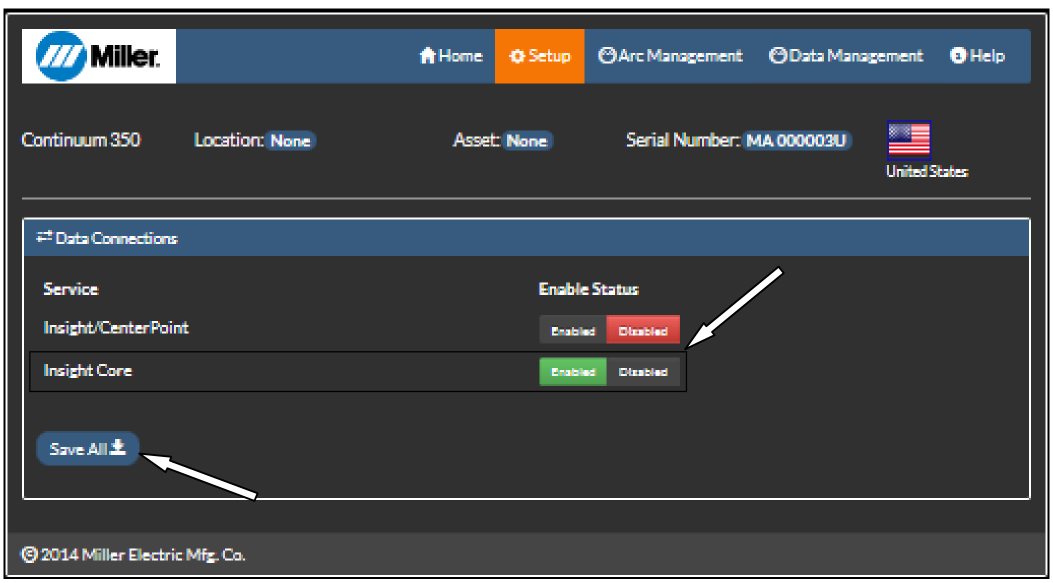


Figure 10-5. Data Connections Page

10-5. Configuring The Network Connection

☞ *If this is the initial installation of a unit, determine what type of installation will be required. Connection information is required for each new installation of the Insight Core product.*

☞ *To set up more than one Continuum system, or if a computer near the Continuum is not available, it is more efficient to use the Insight Core Configuration Utility available at the Insight Core Downloads web page: https://insight.millerwelds.com/download*

1 Connect a computer to the power source with an Ethernet cable and open a browser.

2 Enter the IP address (see Section 10-3) into the address bar. The default IP address is 169.254.0.2.

3 Press **Enter**.

4 The Continuum system's web page appears. On the top of the page, click **Setup**.

- To configure a WIFI connection using DHCP (dynamic IP addresses), go to Section A.
- To configure a WiFi connection using static WiFi addressing, go to Section B.
- To configure a wired Ethernet connection using DHCP (dynamic IP addresses) go to Section C.*
- To configure a wired Ethernet connection using static IP addressing, go to Section D*.

* For static address connections, a computer must be connected directly to the module with an Ethernet cable.

A. WiFi Connection Using DHCP (Dynamic Addressing)

1 On the Setup page, under **WiFi Settings**, click **Edit**.

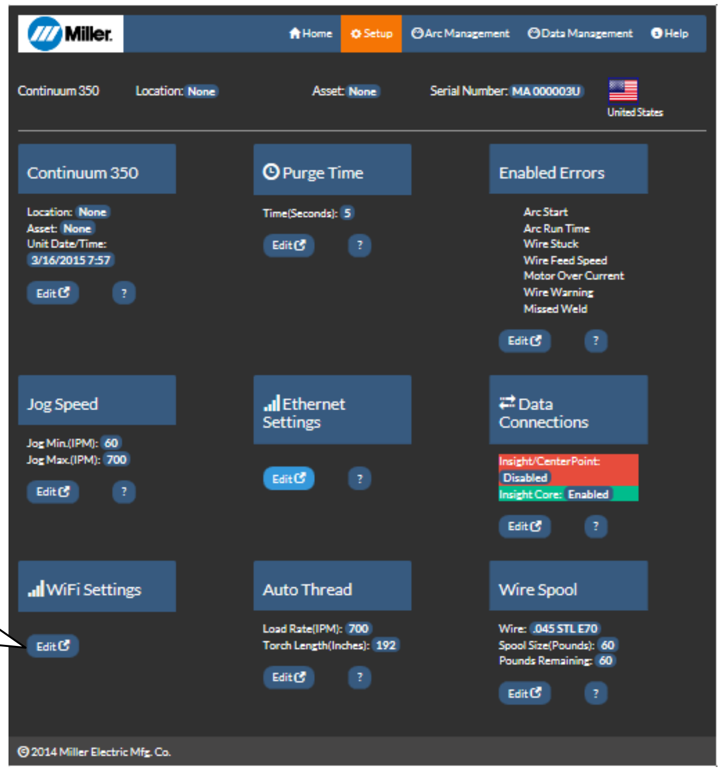


Figure 10-6. Editing WiFi Settings

2 On the screen that appears, under **Security Setup**, enter your network name (SSID), security type, and security key. This information is found on the Network Connectivity Survey completed by your IT Department (see Section 10-2).

3 Click **Save Security Settings** to save your wireless network settings.

4 Under **WiFi Address**, select **Dynamic**.

5 Under **DNS Address**, select **DHCP Automatic**.

6 When finished, click **Save All**.

7 Cycle the machine power by turning the machine off, waiting until the Power On light goes off, and turning power on again.

Figure 10-7. Configuring WiFi Connection Using DHCP

B. WiFi Connection Using Static Addressing

1 On the Setup page, under **WiFi Settings**, click **Edit** (see Editing WIFI Settings).

2 On the screen that appears, under **Security Setup**, enter your network name (SSID), security type, and security key. This information is found on the Network Connectivity Survey completed by your IT Department (see Section Connecting To Continuum Web Page).

3 Click **Save Security Settings** to save your wireless network settings.

4 Under **WiFi Address**, select **Static**.

5 Under **Network Information**, enter your IP address, subnet mask, and network gateway. This information is found on the Network Connectivity Survey completed by your IT Department (see Section Connecting To Continuum Web Page).

6 Under **DNS Address**, select **DNS Server**. Enter DNS address from the Network Connectivity Survey completed by your IT Department (see Section Connecting To Continuum Web Page).

7 When finished, click **Save All**.

8 Cycle the machine power by turning the machine off, waiting until the Power On light goes off, and turning power on again.

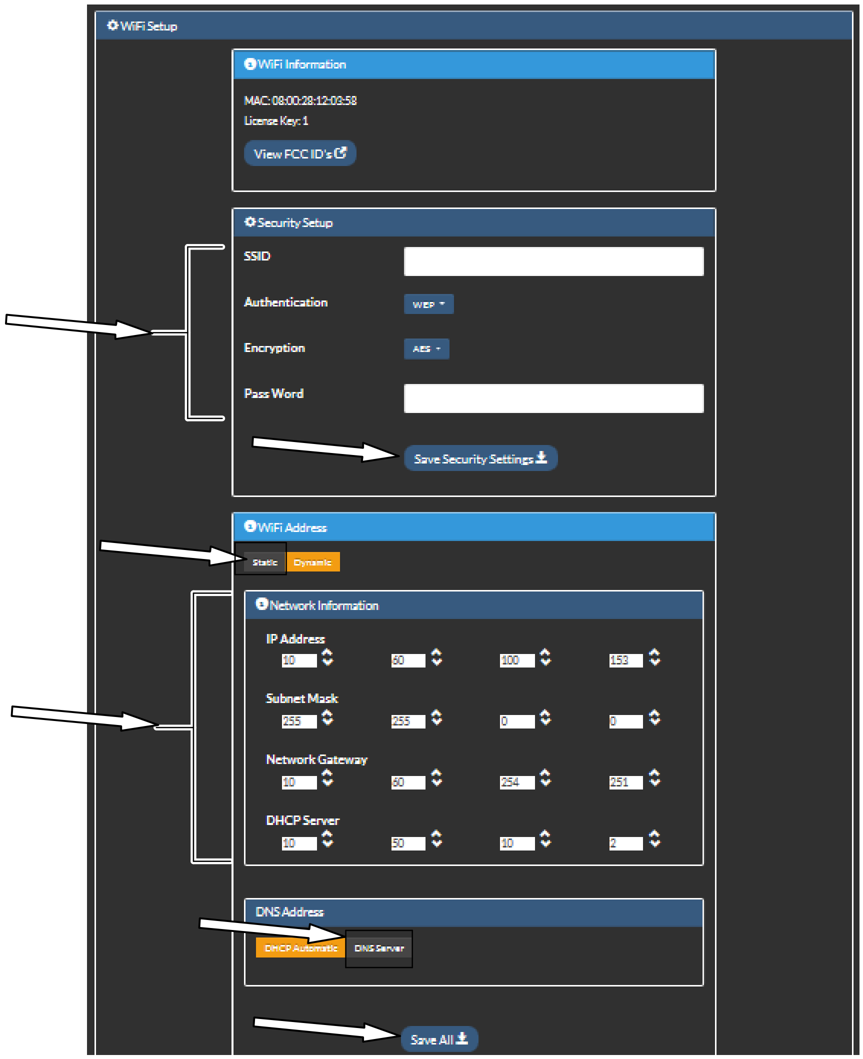


Figure 10-8. Configuring WiFi Connection Using Static Addressing

C. Wired (Ethernet) Connection Using DHCP (Dynamic Addressing)

1 On the Setup page, under **Ethernet Settings**, click **Edit**.

2 At the top of the screen that appears, select **Dynamic** underneath the IP address. The Network Information below will automatically fill in.

3 Under **DNS Address**, select **DHCP Automatic**.

4 When finished, click **Save All**.

5 Cycle the machine power by turning the machine off, waiting until the Power On light goes off, and turning power on again.

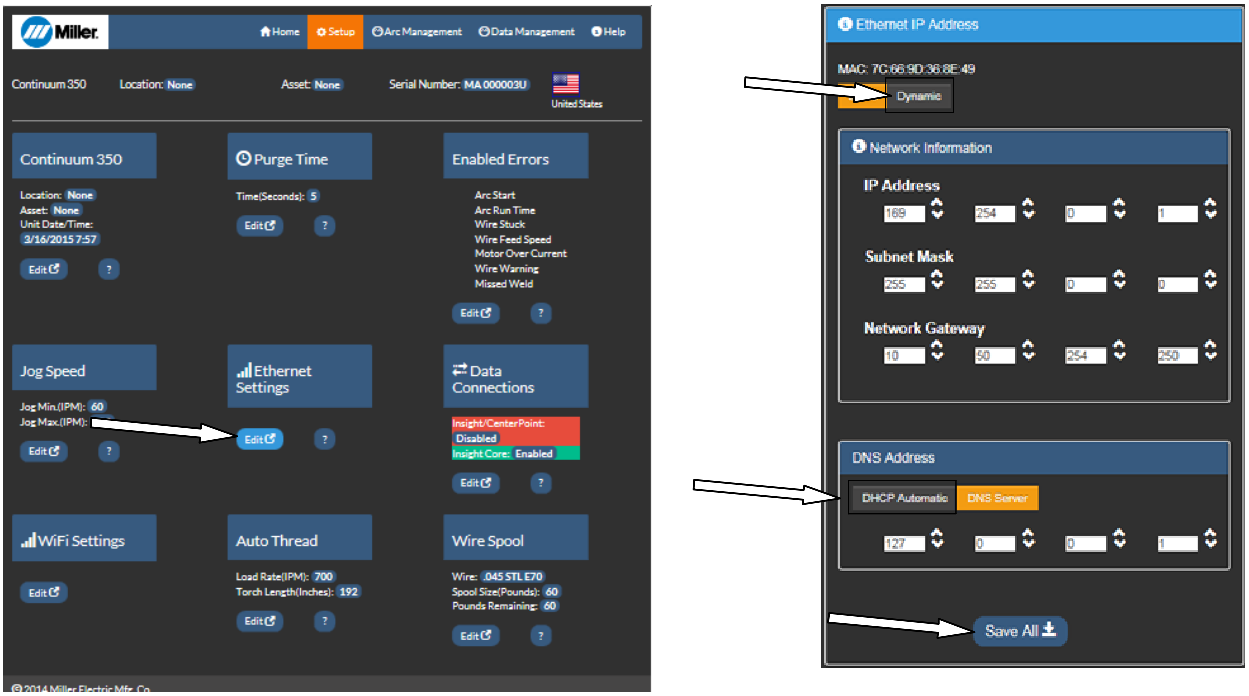


Figure 10-9. Configuring Ethernet Connection Using DHCP

D. Wired (Ethernet) Connection Using Static Addressing

1 On the Setup page, under **Ethernet Settings**, click **Edit**.

2 At the top of the screen that appears, select **Static** under the IP address.

3 Under **Network Information**, enter your IP address, subnet mask, and network gateway. This information is found on the Network Connectivity Survey completed by your IT Department (see Section Connecting To Continuum Web Page).

4 Under **DNS Address**, select **DNS Server**. Enter DNS address from the Network Connectivity Survey completed by your IT Department (see Section Connecting To Continuum Web Page).

Double-check the numbers before saving. A wrong number may be hard to correct.

5 When finished, click **Save All**.

6 Cycle the machine power by turning the machine off, waiting until the Power On light goes off, and turning power on again.

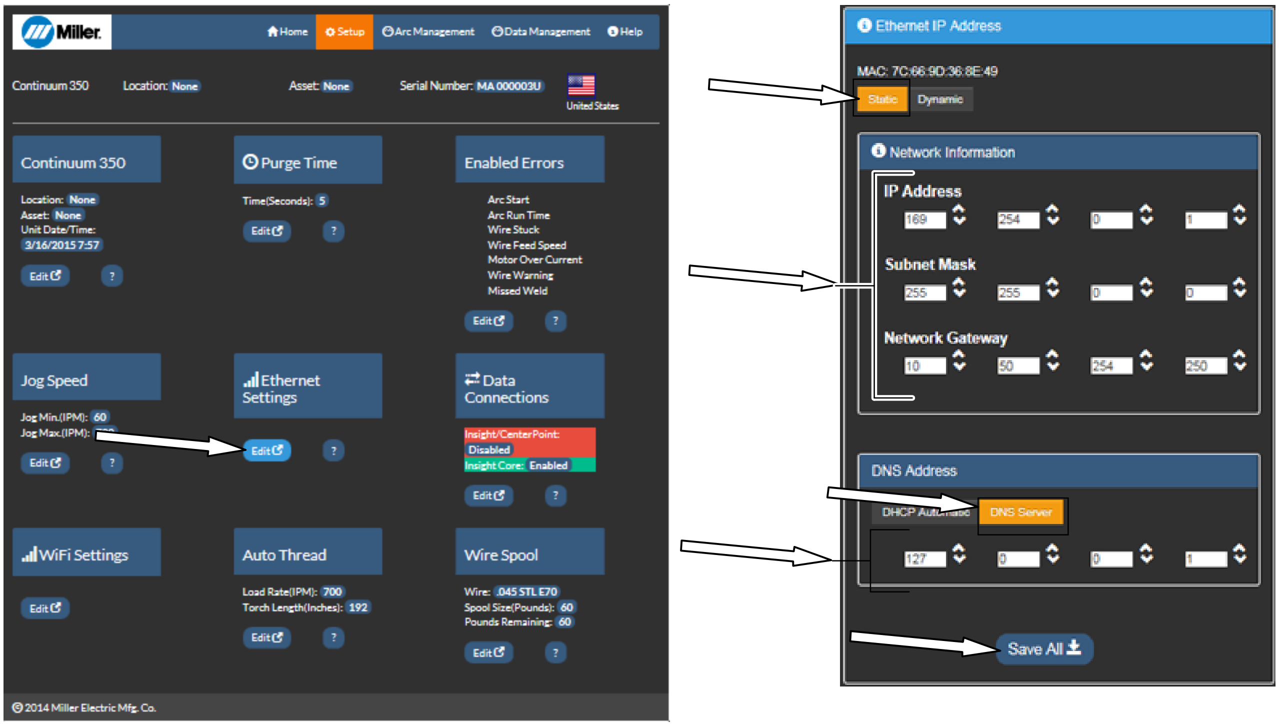


Figure 10-10. Configuring Ethernet Connection Using Static Addressing

10-6. Verifying Network Connection

It can take up to five minutes for a WiFi network to connect the first time.

To verify the network connection, reference the feeder (or pendant) display. If a WiFi connection is being used, the box next to **WiFi** should be yellow. If an Ethernet connection is being used, the box next to **Ethernet** should be yellow. Only one (Ethernet or Wifi) is required to establish a network connection. Once the network is identified, the box next to **Internet** should be green.

1. Internal Connection Indicator (Green When Connected)
2. WiFi Connection Indicator (Yellow When Connected)
3. Ethernet Connection Indicator (Yellow When Connected)

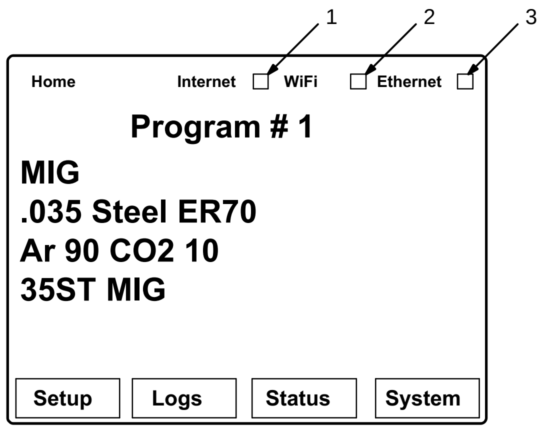


Figure 10-11. Verifying Network Connection

To review settings from the feeder, go to the Ethernet Settings screen. Using the scroll knob, select **System**, then **Network Settings**, then either **Ethernet Settings** or **Wifi Settings**.

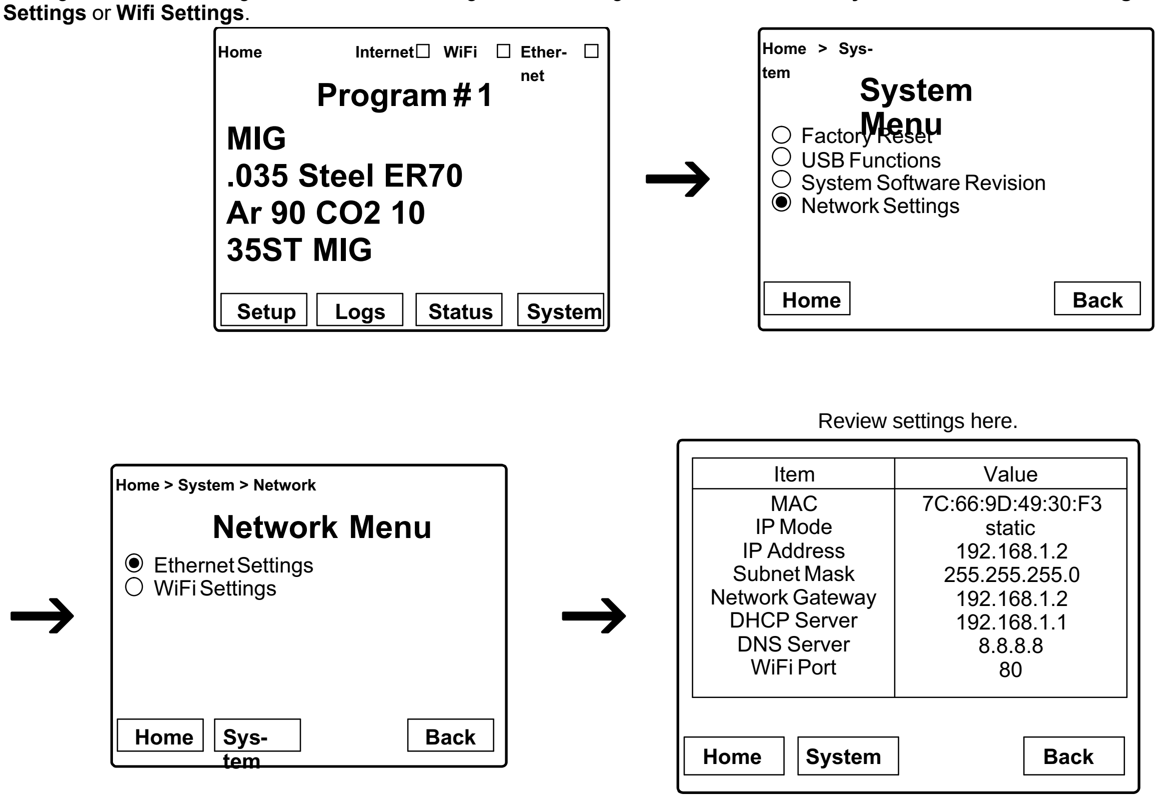


Figure 10-12. Verifying Network Connection

10-7. Determining Device Serial Number And License Key

☞ *Obtaining the device serial number and license key is necessary for product registration. Registration will activate your Insight Core account and give access to all Insight Core data.*

1. Insert a USB drive into the front port of the Continuum Feeder

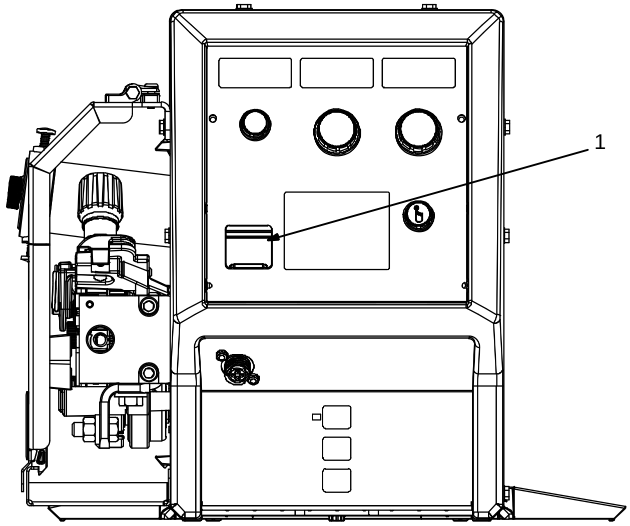


Figure 10-13. Continuum Feeder USB Location

2. From the feeder (or pendant) main menu, select **System** to bring up the System menu. Using the scroll knob, select **USB Functions**, then **Write to USB**, then **License Key**, then **Yes** to save the license key and serial number data. This process will load an IP_MAC file onto the USB drive, which contains the unique license key for your system

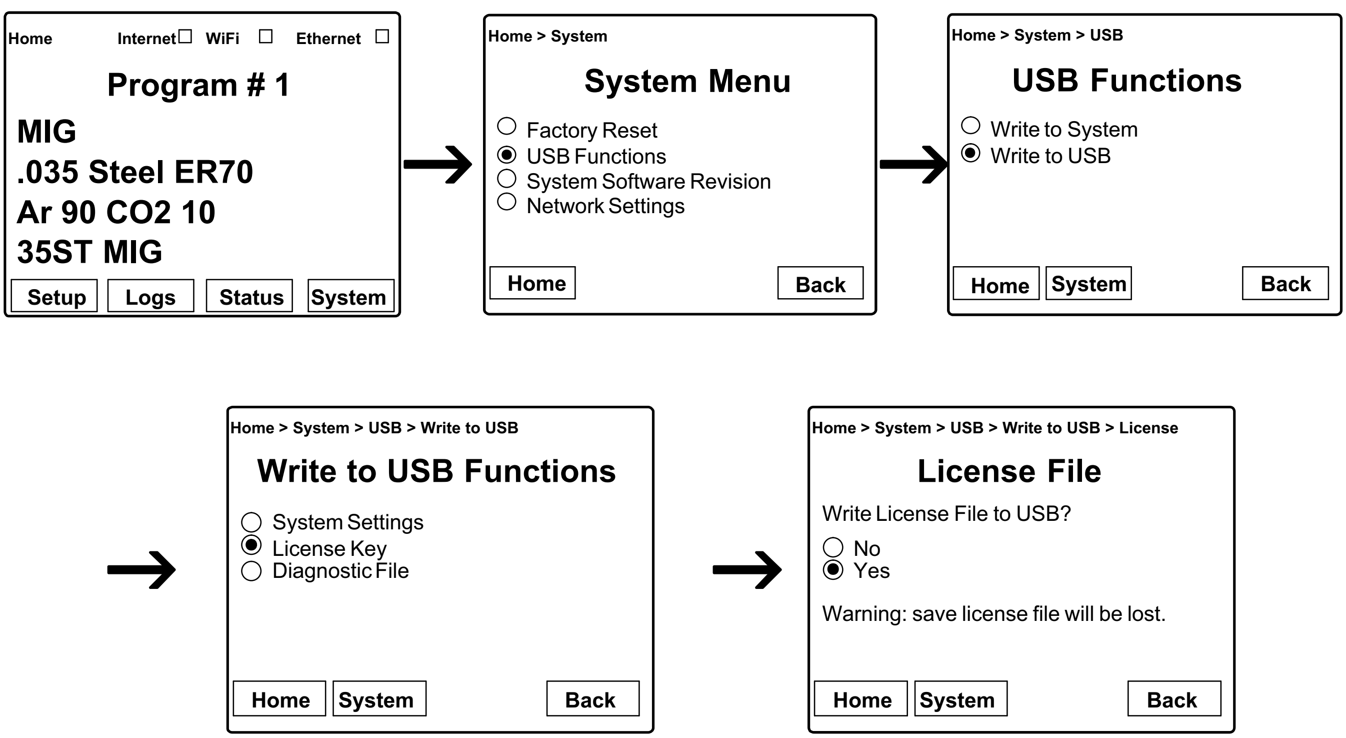


Figure 10-14. Obtaining Serial Number And License Key

3. Remove the USB drive from the feeder and insert it into your computer. Allow time for the computer to recognize the USB drive.
4. Find and open the folder named Device License Keys on the USB drive.

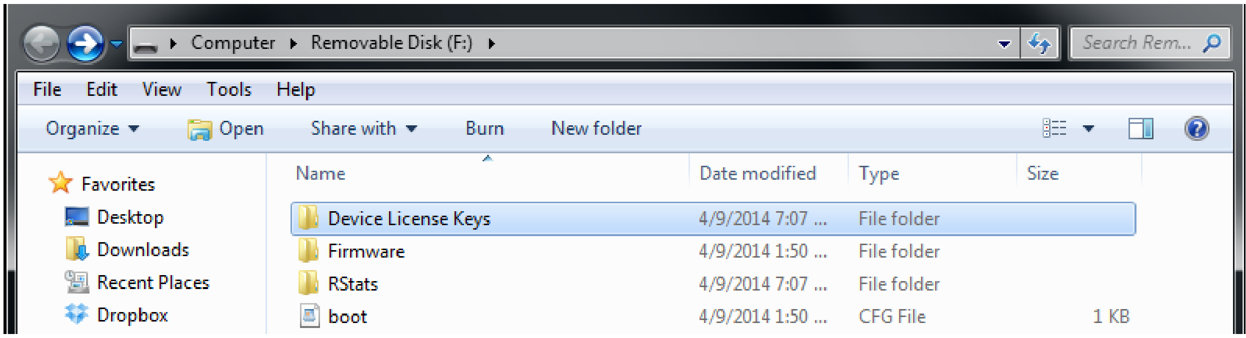


Figure 10-15. Device License Keys Folder

5. Open the .txt file called "IP_MAC_ADDR_LIC…" with a text editor (Notepad) to find the module/device serial number and license key needed to register the device.

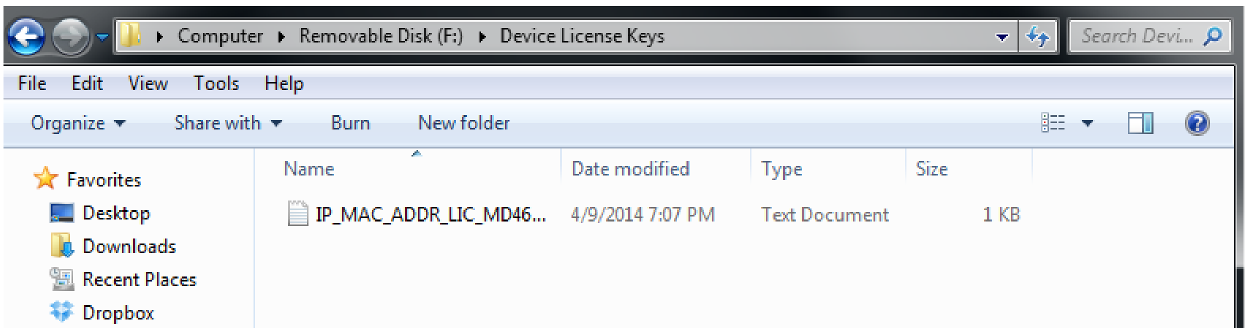


Figure 10-16. Device License Key File

6. Write down or copy and paste the serial number and license key for registration.

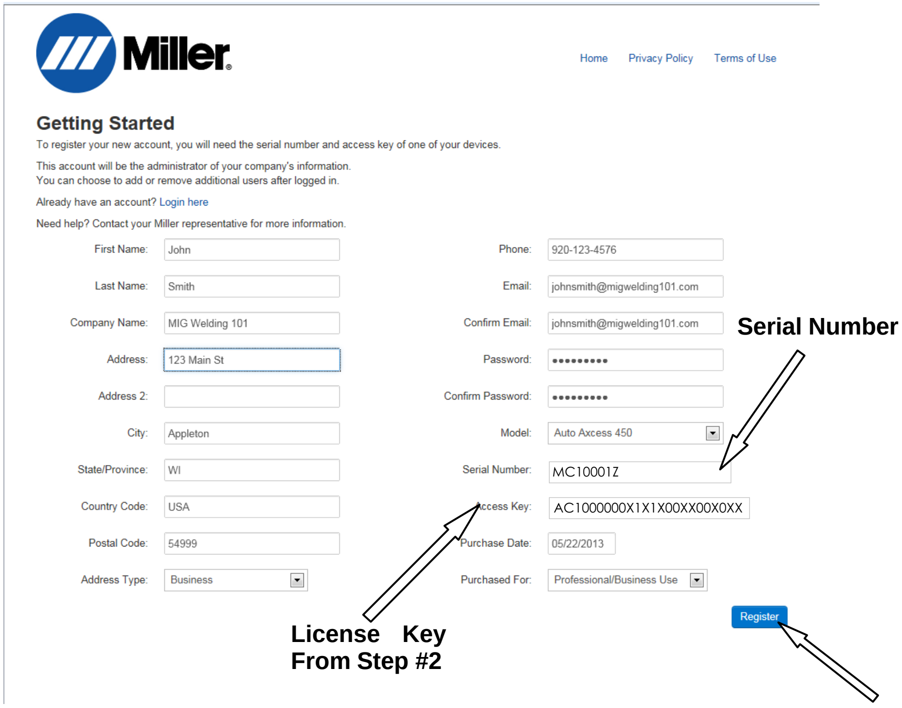
```
Firmware: --(Insight v1.0.0 Continuum)
Ethernet Status: Connected
WiFi Status: --
Ethernet Device IP: 0.0.0.0
Ethernet Subnet: XXX.000.X.0
Ethernet Gateway: 00.XX.000.XXX
Ethernet DHCP: ENABLED
WiFi Device IP: --
WiFi Subnet: --
WiFi Gateway: --
WiFi DHCP: --
Ethernet MAC: 00:XX:00:XX:00:XX
WiFi MAC: 00:XX:00:XX:00:XX
Ethernet Port: 80
WiFi Port: --
Power Source Serial: MF000003U
License Key: A0000000XXXX00X00000XXX0
```

Figure 10-17. Device License Key Information

☞ *This file also verifies the connections information. Exact information in the file will vary based on connection type and device.*

☞ *To continue device registration, proceed to Section NO TAG.*

10-8. Registering Initial Device And Creating An Account



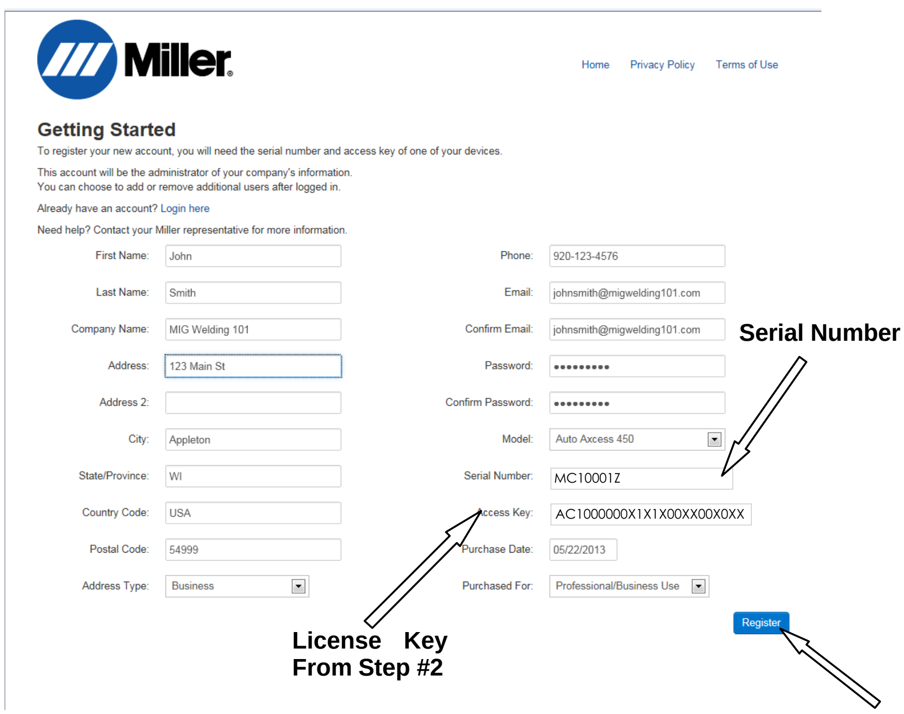
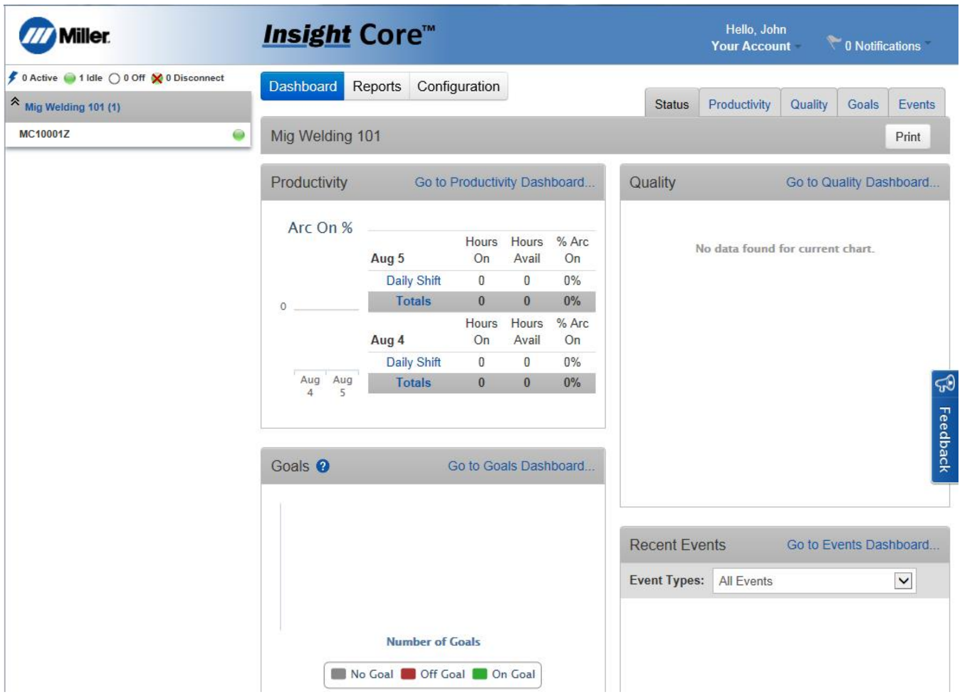


Figure 10-18. Registering A New Account

1. After clicking **Register**, the homepage for the company that was just created will appear (see Figure 10-19).



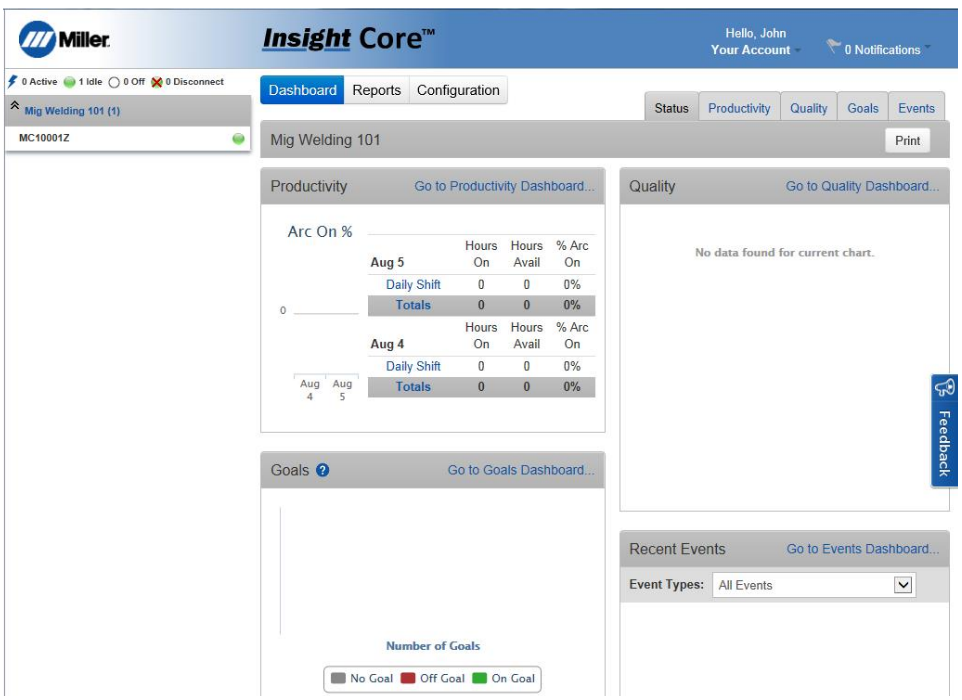


Figure 10-19. Insight Core Company Dashboard

☞ *To register additional devices, proceed to Section 10-9.*

10-9. Registering Additional Devices

1. Log in to Insight Core website at https://insight.millerwelds.com.
2. Click the **Configuration** tab.
3. Click **Register New Device**.

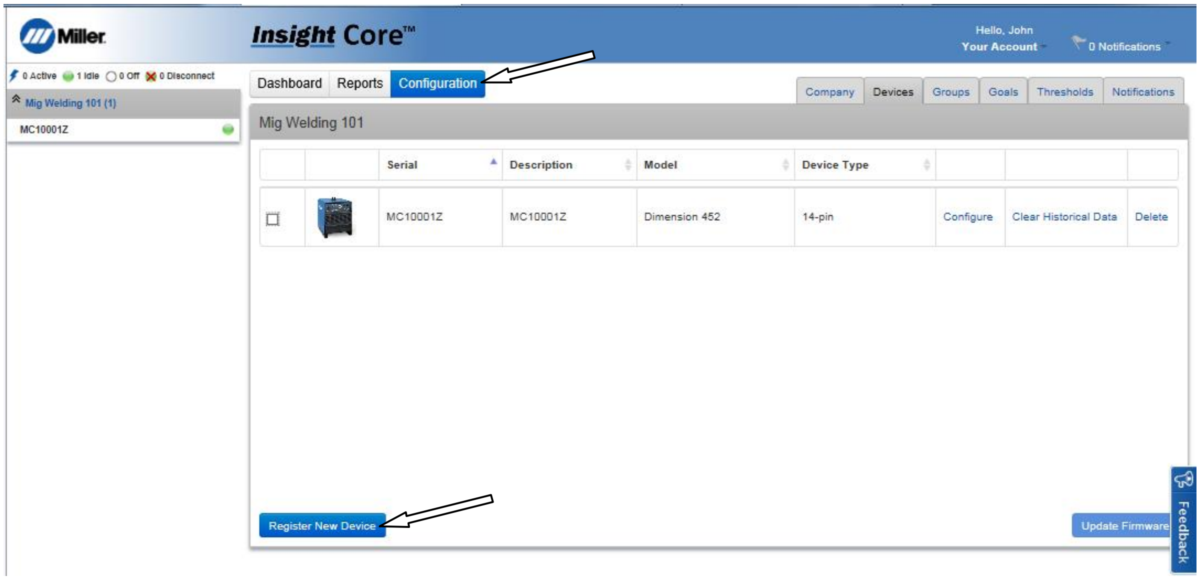


Figure 10-20. Configuration Screen

4. Enter the information for the new device you intend to register.
5. A confirmation screen appears. If the information is correct, click **Save**.

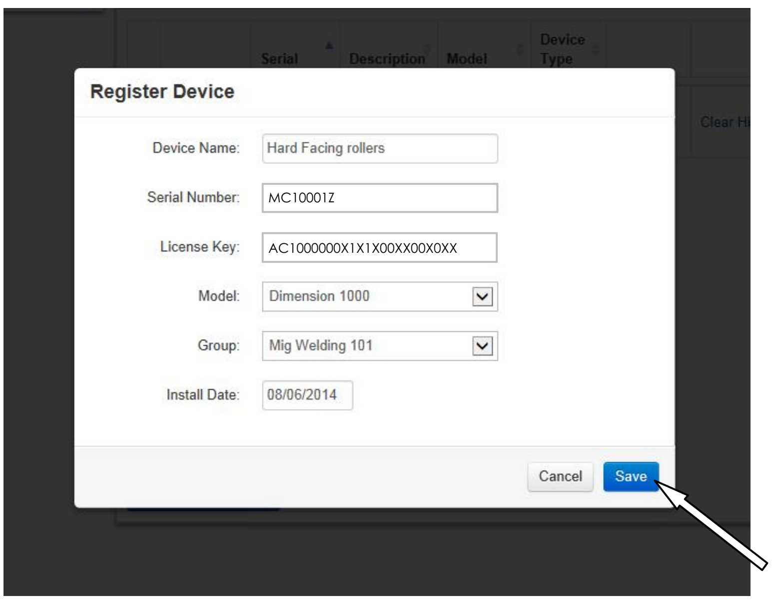


Figure 10-21. Register Device Confirmation Screen

6. After registration, the device will appear inside the corporate asset tree and begin displaying activity as operators begin welding with the power source.

Notes

Notes

Notes

SECTION 11 – ELECTRICAL DIAGRAMS

⚠ WARNING	• Do not touch live electrical parts. • Disconnect input power or stop engine before servicing. • Do not operate with covers removed. • Have only qualified persons install, use, or service this unit.
ELECTRIC SHOCK HAZARD	

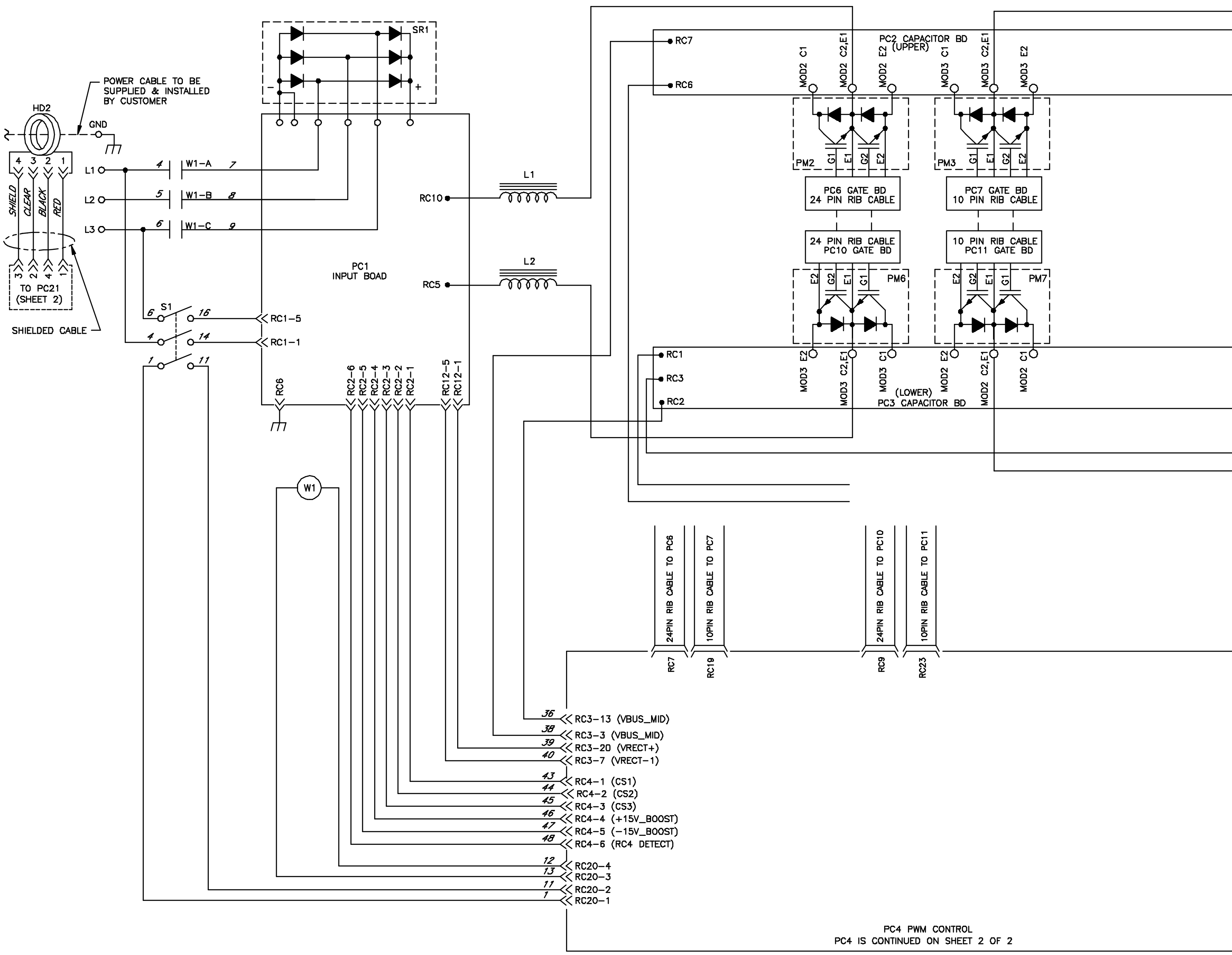
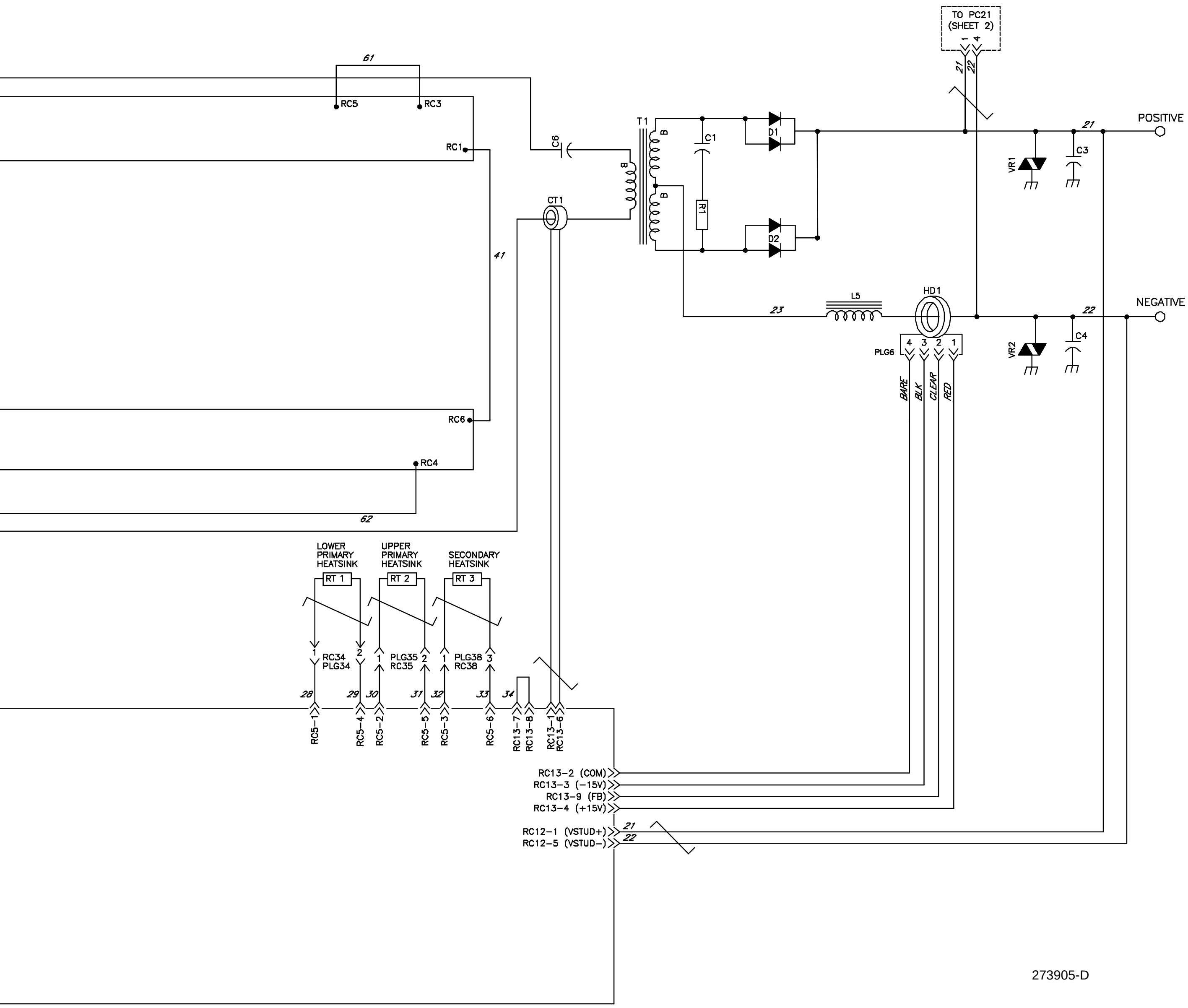


Figure 11-1. Circuit Diagram For Auto-Continuum 350 Model (Page 1 of 2)


TO PC21
(SHEET 2)
61
RC5
RC3
RC1
C6
T1
C1
D1
D2
R1
CT1
41
POSITIVE
NEGATIVE
VR1
VR2
C3
C4
L5
HD1
PLG6
BARE
BLK
CLEAR
RED
23
21
22
RC6
RC4
62
LOWER PRIMARY HEATSINK
UPPER PRIMARY HEATSINK
SECONDARY HEATSINK
RT 1
RT 2
RT 3
RC34
PLG34
PLG35
RC35
PLG38
RC38
28
29
30
31
32
33
34
RC5-1
RC5-4
RC5-2
RC5-5
RC5-3
RC5-6
RC13-7
RC13-8
RC13-1
RC13-6
RC13-2 (COM)
RC13-3 (-15V)
RC13-9 (FB)
RC13-4 (+15V)
RC12-1 (VSTUD+)
RC12-5 (VSTUD-)
273905-D

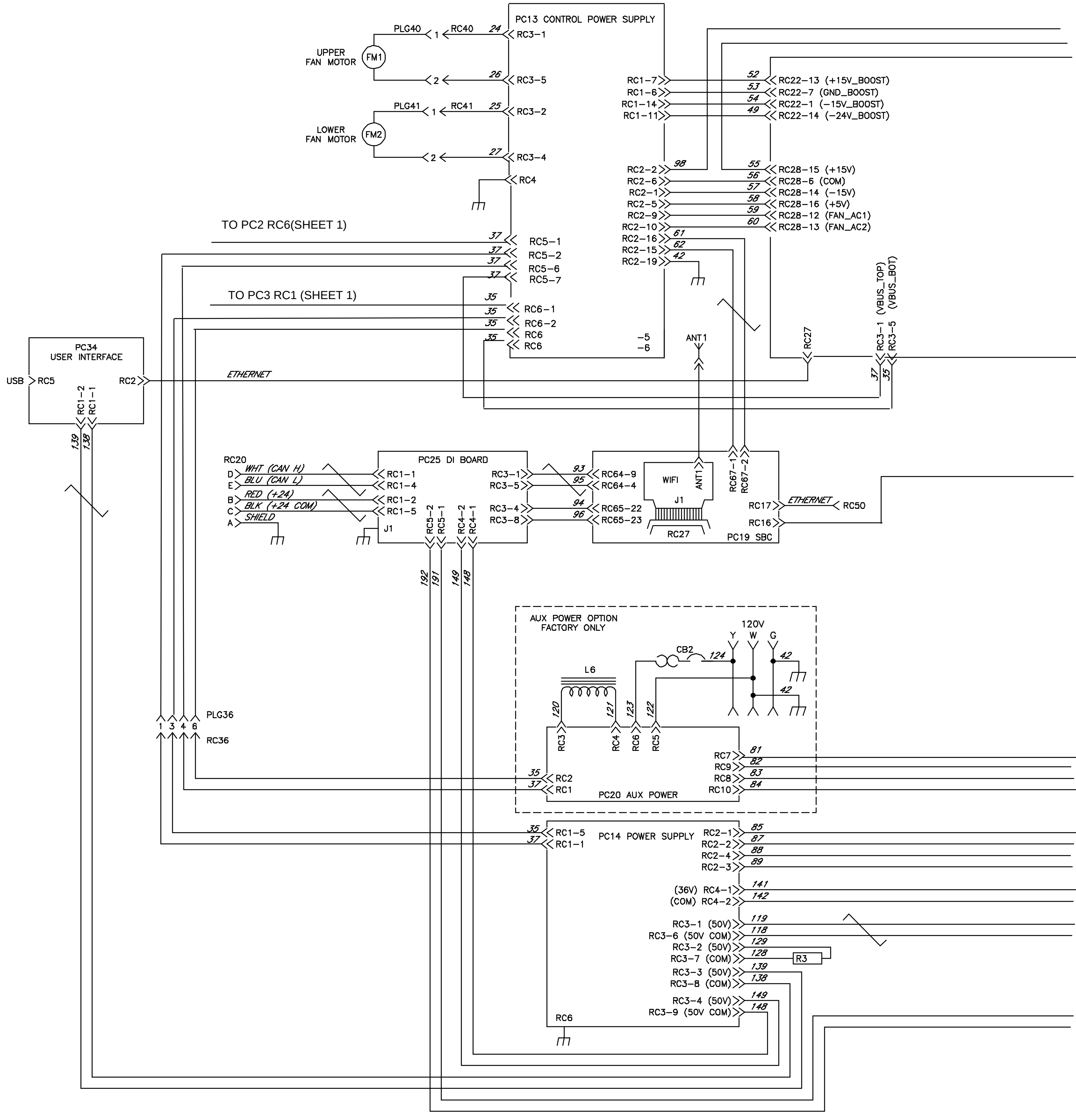
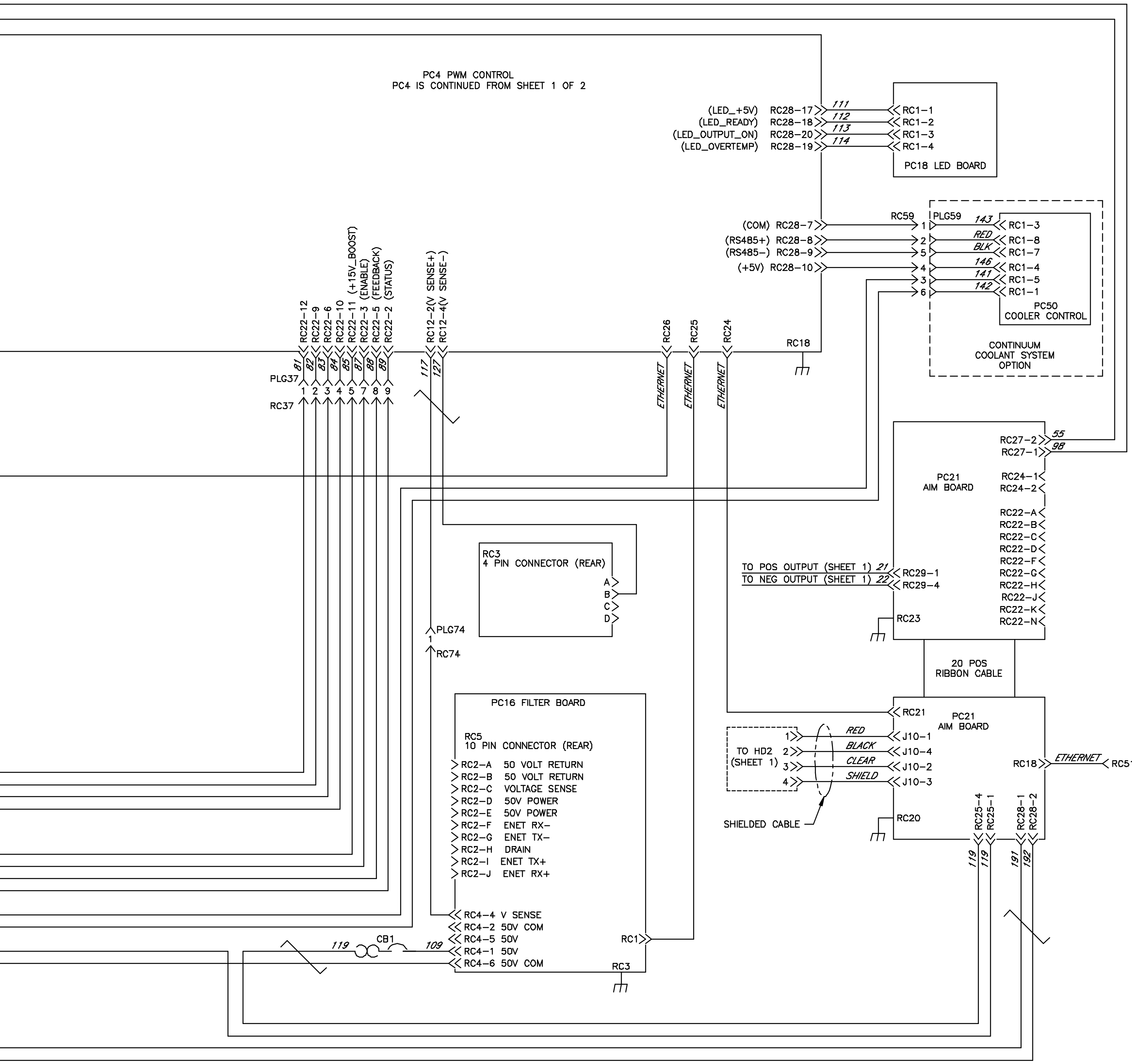
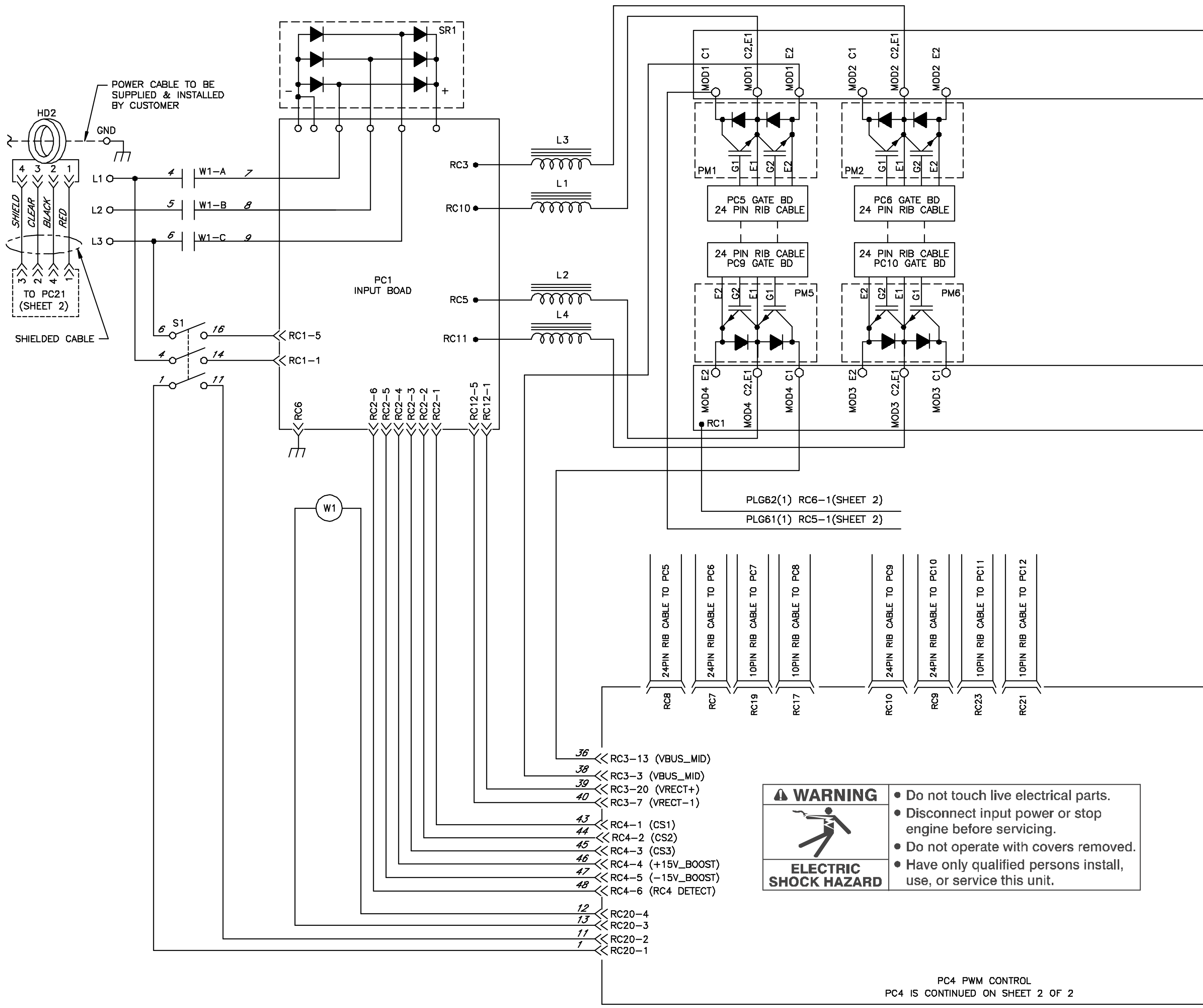


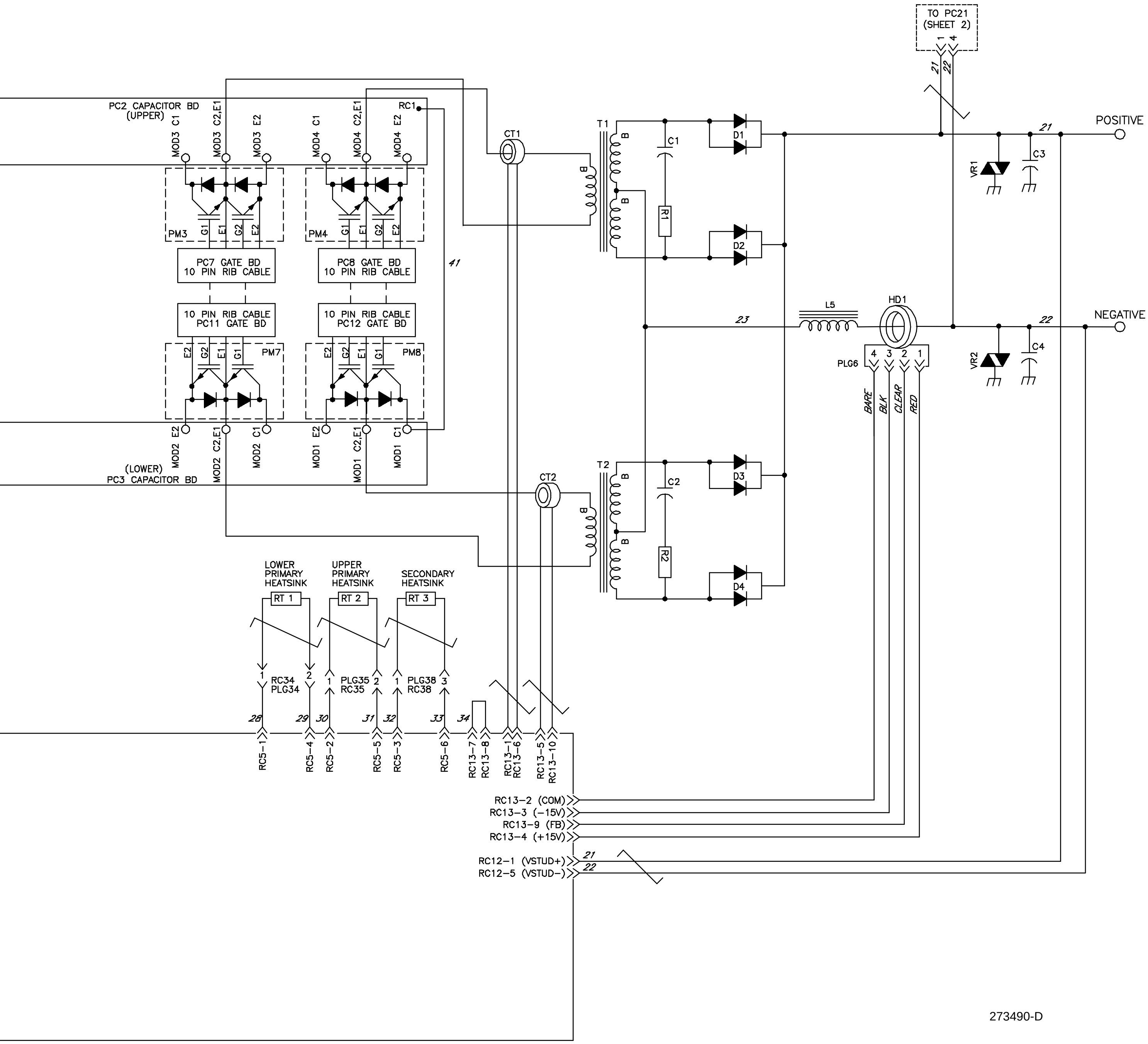
Figure 11-2. Circuit Diagram For Auto-Continuum 350 Model (Page 2 of 2)


PC4 PWM CONTROL
PC4 IS CONTINUED FROM SHEET 1 OF 2
(LED_+5V) RC28-17
(LED_READY) RC28-18
(LED_OUTPUT_ON) RC28-20
(LED_OVERTEMP) RC28-19
111
112
113
114
RC1-1
RC1-2
RC1-3
RC1-4
PC18 LED BOARD
(COM) RC28-7
(RS485+) RC28-8
(RS485-) RC28-9
(+5V) RC28-10
RC59
PLG59
143
RED
BLK
146
141
142
RC1-3
RC1-8
RC1-7
RC1-4
RC1-5
RC1-1
PC50
COOLER CONTROL
CONTINUUM
COOLANT SYSTEM
OPTION
RC18
RC22-12
RC22-9
RC22-6
RC22-10
RC22-11 (+15V_BOOST)
RC22-3 (ENABLE)
RC22-5 (FEEDBACK)
RC22-2 (STATUS)
RC12-2(V SENSE+)
RC12-4(V SENSE-)
RC26
RC25
RC24
ETHERNET
PLG37
RC37
81 82 83 84 85 87 88 89
117 127
RC27-2
RC27-1
55
98
PC21
AIM BOARD
RC24-1
RC24-2
RC22-A
RC22-B
RC22-C
RC22-D
RC22-F
RC22-G
RC22-H
RC22-J
RC22-K
RC22-N
TO POS OUTPUT (SHEET 1) 21
TO NEG OUTPUT (SHEET 1) 22
RC29-1
RC29-4
RC23
RC3
4 PIN CONNECTOR (REAR)
PLG74
RC74
20 POS
RIBBON CABLE
PC21
AIM BOARD
RC21
TO HD2
(SHEET 1)
RED
BLACK
CLEAR
SHIELD
J10-1
J10-4
J10-2
J10-3
RC18
ETHERNET
RC51
SHIELDED CABLE
RC20
RC25-4
RC25-1
RC28-1
RC28-2
119
119
191
192
PC16 FILTER BOARD
RC5
10 PIN CONNECTOR (REAR)
RC2-A 50 VOLT RETURN
RC2-B 50 VOLT RETURN
RC2-C VOLTAGE SENSE
RC2-D 50V POWER
RC2-E 50V POWER
RC2-F ENET RX-
RC2-G ENET TX-
RC2-H DRAIN
RC2-I ENET TX+
RC2-J ENET RX+
RC4-4 V SENSE
RC4-2 50V COM
RC4-5 50V
RC4-1 50V
RC4-6 50V COM
RC1
RC3
CB1
119
109



⚠ WARNING	
ELECTRIC SHOCK HAZARD	• Do not touch live electrical parts. • Disconnect input power or stop engine before servicing. • Do not operate with covers removed. • Have only qualified persons install, use, or service this unit.

Figure 11-3. Circuit Diagram For Auto-Continuum 500 Model (Page 1 of 2)



TO PC21
(SHEET 2)
PC2 CAPACITOR BD
(UPPER)
MOD3 C1
MOD3 C2,E1
MOD3 E2
MOD4 C1
MOD4 C2,E1
MOD4 E2
RC1
CT1
T1
C1
D1
R1
D2
POSITIVE
VR1
C3
PM3
PM4
G1
E1
G2
E2
PC7 GATE BD
10 PIN RIB CABLE
PC8 GATE BD
10 PIN RIB CABLE
10 PIN RIB CABLE
PC11 GATE BD
PC12 GATE BD
PM7
PM8
L5
HD1
NEGATIVE
VR2
C4
PLG6
BARE
BLK
CLEAR
RED
MOD2 E2
MOD2 C2,E1
MOD2 C1
MOD1 E2
MOD1 C2,E1
MOD1 C1
(LOWER)
PC3 CAPACITOR BD
CT2
T2
C2
D3
R2
D4
LOWER PRIMARY HEATSINK
UPPER PRIMARY HEATSINK
SECONDARY HEATSINK
RT 1
RT 2
RT 3
RC34
PLG34
PLG35
RC35
PLG38
RC38
RC5-1
RC5-4
RC5-2
RC5-5
RC5-3
RC5-6
RC13-7
RC13-8
RC13-1
RC13-6
RC13-5
RC13-10
RC13-2 (COM)
RC13-3 (-15V)
RC13-9 (FB)
RC13-4 (+15V)
RC12-1 (VSTUD+)
RC12-5 (VSTUD-)
273490-D

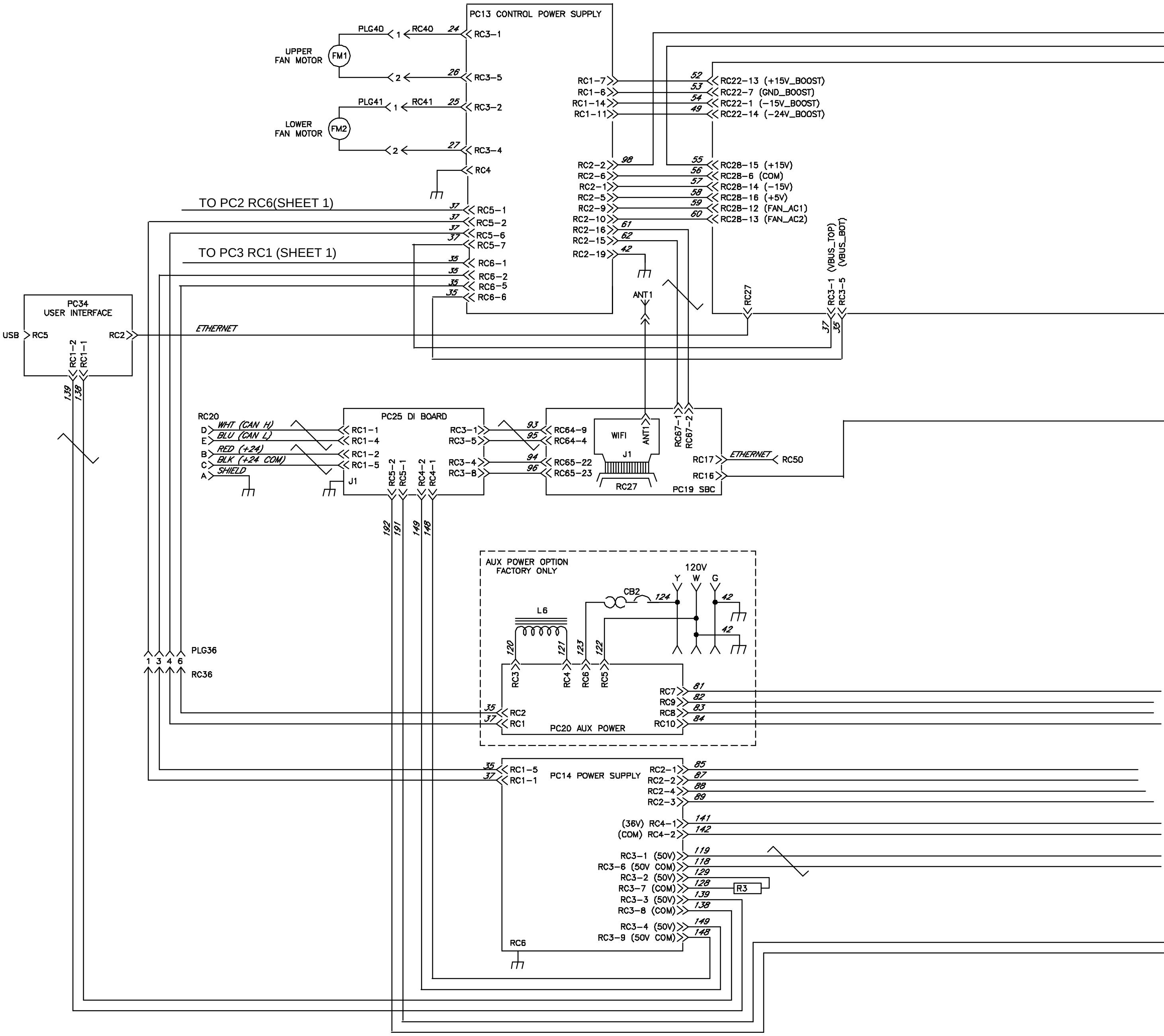
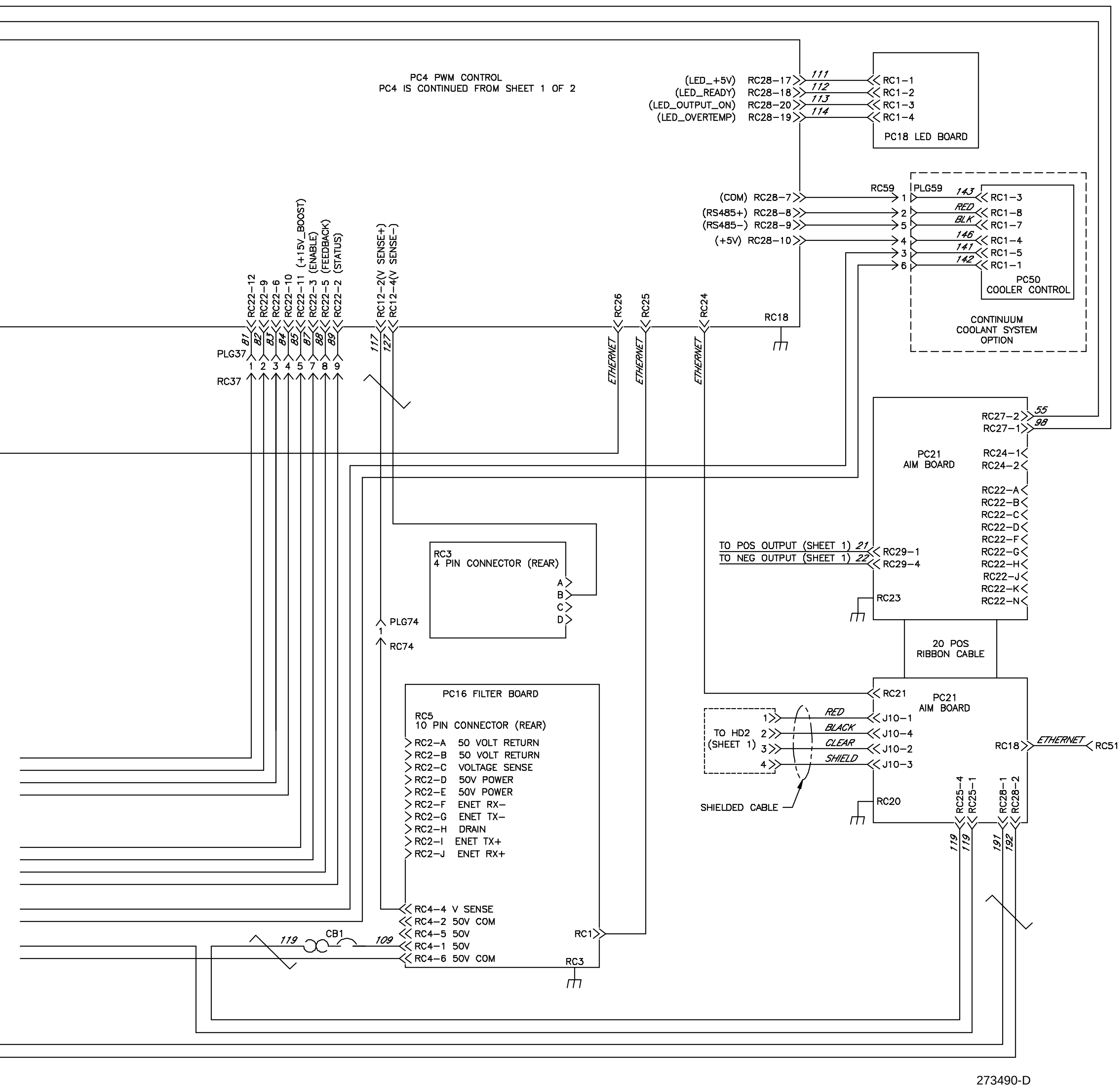


Figure 11-4. Circuit Diagram For Auto-Continuum 500 Model (Page 2 of 2)


PC4 PWM CONTROL
PC4 IS CONTINUED FROM SHEET 1 OF 2
(LED_+5V) RC28-17
(LED_READY) RC28-18
(LED_OUTPUT_ON) RC28-20
(LED_OVERTEMP) RC28-19
PC18 LED BOARD
(COM) RC28-7
(RS485+) RC28-8
(RS485-) RC28-9
(+5V) RC28-10
RC59
PLG59
PC50
COOLER CONTROL
CONTINUUM
COOLANT SYSTEM
OPTION
RC22-12 RC22-9 RC22-6 RC22-10 RC22-11 (+15V_BOOST) RC22-3 (ENABLE) RC22-5 (FEEDBACK) RC22-2 (STATUS)
RC12-2(V SENSE+) RC12-4(V SENSE-)
PLG37
RC37
ETHERNET
RC18
PC21
AIM BOARD
TO POS OUTPUT (SHEET 1)
TO NEG OUTPUT (SHEET 1)
RC3
4 PIN CONNECTOR (REAR)
PLG74
RC74
20 POS
RIBBON CABLE
PC16 FILTER BOARD
RC5
10 PIN CONNECTOR (REAR)
RC2-A 50 VOLT RETURN
RC2-B 50 VOLT RETURN
RC2-C VOLTAGE SENSE
RC2-D 50V POWER
RC2-E 50V POWER
RC2-F ENET RX-
RC2-G ENET TX-
RC2-H DRAIN
RC2-I ENET TX+
RC2-J ENET RX+
RC4-4 V SENSE
RC4-2 50V COM
RC4-5 50V
RC4-1 50V
RC4-6 50V COM
CB1
TO HD2
(SHEET 1)
RED
BLACK
CLEAR
SHIELD
SHIELDED CABLE
RC51
273490-D

TRUE BLUE® WARRANTY

Effective January 1, 2024 (Equipment with a serial number preface of NE or newer)
This limited warranty supersedes all previous Miller warranties and is exclusive with no other guarantees or warranties expressed or implied.

LIMITED WARRANTY - Subject to the terms and conditions below, Miller Electric Mfg. LLC, Appleton, Wisconsin, warrants to authorized distributors that new Miller equipment sold after the effective date of this limited warranty is free of defects in material and workmanship at the time it is shipped by Miller. THIS WARRANTY IS EXPRESSLY IN LIEU OF ALL OTHER WARRANTIES, EXPRESS OR IMPLIED, INCLUDING THE WARRANTIES OF MERCHANTABILITY AND FITNESS.

Within the warranty periods listed below, Miller will repair or replace any warranted parts or components that fail due to such defects in material or workmanship. Notifications submitted as online warranty claims must provide detailed descriptions of the fault and troubleshooting steps taken to diagnose failed parts. Warranty claims that lack the required information as defined in the Miller Service Operation Guide (SOG) may be denied by Miller.

Miller shall honor warranty claims on warranted equipment listed below in the event of a defect within the warranty coverage time periods listed below. Warranty time periods start on the delivery date of the equipment to the end-user purchaser.

1 5 Years Parts — 3 Years Labor

- Original Main Power Rectifiers Only to Include SCRs, Diodes, and Discrete Rectifier Modules in non-inverter products

2 4 Years Parts (No Labor)

- Auto-Darkening ClearLight 2.0 Helmet Lenses

3 3 Years — Parts and Labor Unless Specified

- Auto-Darkening Helmet Lenses (No Labor)
- Copilot Collaborative Welding Systems
- Engine Driven Welder/Generators (Including EnPak) **(NOTE: Engines are Warranted Separately by the Engine Manufacturer.)**
- Handheld Laser Power Sources
- Insight Welding Intelligence Products (Except External Sensors)
- Inverter Power Sources
- Plasma Arc Cutting Power Sources
- Process Controllers
- Semi-Automatic and Automatic Wire Feeders
- Transformer/Rectifier Power Sources

4 2 Years — Parts and Labor

- Auto-Darkening Weld Masks (No Labor)
- Fume Extractors - Filtair 215, Capture 5, and Industrial Collector Series

5 1 Year — Parts and Labor Unless Specified

- ArcReach Heater
- AugmentedArc, LiveArc, and MobileArc Welding Systems
- Automatic Motion Devices
- Bernard BTB Air-Cooled MIG Guns (No Labor)
- CoolBelt, PAPR Blower, and PAPR Face Shield (No Labor)
- Desiccant Air Dryer System
- Field Options **(NOTE: Field options are covered for the remaining warranty period of the product they are installed in, or for a minimum of one year — whichever is greater.)**
- RFCS Foot Controls (Except RFCS-RJ45)
- Fume Extractors - Filtair 130, MWX and SWX Series, Standard, Telescopic, and ZoneFlow Extraction Arms and Motor Control Box
- Handheld Laser Torches (No Labor)
- HF Units
- Induction Heating Power Sources, Coolers **(NOTE: Digital Recorders are Warranted Separately by the Manufacturer.)**
- Insight Sensors
- Laser Welding Helmets (No Labor)
- Load Banks
- MDX Series MIG Guns (No Labor)
- Motor-Driven Guns
- Positioners and Controllers
- Racks (For Housing Multiple Power Sources)
- Running Gear/Trailers
- Spoolmate Spoolguns (No Labor)
- Subarc Wire Drive Assemblies
- Supplied Air Respirator (SAR) Boxes and Panels
- TIG Torches (No Labor)
- Tregaskiss Guns (No Labor)
- Water Cooling Systems
- Wireless Remote Foot/Hand Controls and Receivers
- Work Stations/Weld Tables (No Labor)
- XT Plasma Cutting Torches (No Labor)

6 6 Months — Parts

- 12 Volt Automotive-Style Batteries

7 90 Days — Parts

- Accessories (Kits)
- ArcReach Heater Quick Wrap and Air Cooled Cables
- Canvas Covers
- Induction Heating Coils and Blankets, Cables, and Non-Electronic Controls
- M-Guns
- MIG Guns, Subarc (SAW) Torches, and External Cladding Heads
- Remote Controls and RFCS-RJ45
- Replacement Parts (No labor)

Miller's True Blue® Limited Warranty shall not apply to:

1. **Consumable components; such as contact tips, cutting nozzles, contactors, brushes, relays, work station table tops and welding curtains, or parts that fail due to normal wear. (Exception: brushes and relays are covered on all engine-driven products.)**
2. Items furnished by Miller, but manufactured by others, such as engines or trade accessories. These items are covered by the manufacturer's warranty, if any.
3. Equipment that has been modified by any party other than Miller, or equipment that has been improperly installed, improperly operated or misused based upon industry standards, or equipment which has not had reasonable and necessary maintenance, or equipment which has been used for operation outside of the specifications for the equipment.
4. Defects caused by accident, unauthorized repair, or improper testing.

MILLER PRODUCTS ARE INTENDED FOR COMMERCIAL AND INDUSTRIAL USERS TRAINED AND EXPERIENCED IN THE USE AND MAINTENANCE OF WELDING EQUIPMENT.

The exclusive remedies for warranty claims are, at Miller's option, either: (1) repair; or (2) replacement; or, if approved in writing by Miller, (3) the pre-approved cost of repair or replacement at an authorized Miller service station; or (4) payment of or credit for the purchase price (less reasonable depreciation based upon use). Products may not be returned without Miller's written approval. Return shipment shall be at customer's risk and expense.

The above remedies are F.O.B. Appleton, WI, or Miller's authorized service facility. Transportation and freight are the customer's responsibility. TO THE EXTENT PERMITTED BY LAW, THE REMEDIES HEREIN ARE THE SOLE AND EXCLUSIVE REMEDIES REGARDLESS OF THE LEGAL THEORY. IN NO EVENT SHALL MILLER BE LIABLE FOR DIRECT, INDIRECT, SPECIAL, INCIDENTAL OR CONSEQUENTIAL DAMAGES (INCLUDING LOSS OF PROFIT) REGARDLESS OF THE LEGAL THEORY. ANY WARRANTY NOT PROVIDED HEREIN AND ANY IMPLIED WARRANTY, GUARANTY, OR REPRESENTATION, INCLUDING ANY IMPLIED WARRANTY OF MERCHANTABILITY OR FITNESS FOR PARTICULAR PURPOSE, ARE EXCLUDED AND DISCLAIMED BY MILLER.

Some US states do not allow limiting the duration of an implied warranty or the exclusion of certain damages, so the above limitations may not apply to you. This warranty provides specific legal rights, and other rights may be available depending on your state. In Canada, some provinces provide additional warranties or remedies, and to the extent the law prohibits their waiver, the limitations set out above may not apply. This Limited Warranty provides specific legal rights, and other rights may be available, but may vary by province.

Warranty Questions?

Call 1-800-4-A-MILLER for your local Miller distributor.

Your distributor also gives you...

Service

You always get the fast, reliable response you need. Most replacement parts can be in your hands in 24 hours.

Support

Need fast answers to the tough welding questions? The expertise of the distributor and Miller is there to help you, every step of the way.

Owner's Record

Please complete and retain with your personal records.

Model Name	Serial/Style Number
Purchase Date	(Date which equipment was delivered to original customer.)
Distributor	
Address	
City	
State	Zip

For Service

Contact a DISTRIBUTOR or SERVICE AGENCY near you.

Always provide Model Name and Serial/Style Number.

Contact your Distributor for:	Welding Supplies and Consumables Options and Accessories Personal Protective Equipment (PPE) Service and Repair Replacement Parts Training (Schools, Videos, Books) Welding Process Handbooks To locate a Distributor or Service Agency visit www.millerwelds.com or call 1-800-4-A-Miller
Contact the Delivering Carrier to:	File a claim for loss or damage during shipment. For assistance in filing or settling claims, contact your distributor and/or equipment manufacturer's Transportation Department.

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For International Locations Visit
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